



Actividad 4.1 — Detección de Señales de Tránsito con CNN

Maestría en Inteligencia Artificial Aplicada · *Navegación Autónoma*

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EQUIPO 24

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Contenido del notebook

Este notebook documenta **paso a paso** el diseño, entrenamiento y evaluación de una **Red Neuronal Convolucional** capaz de clasificar las 43 categorías del *German Traffic Sign Recognition Benchmark* (GTSRB) y su posterior integración en un controlador de Webots.

1. Setup y verificación del entorno (TensorFlow + Apple GPU vía Metal).
2. Carga del dataset GTSRB y exploración visual de las 43 clases.
3. Construcción de los tensores X, y (resize 32×32, normalización).
4. División train / val / test y *one-hot encoding*.

5. Diseño de la arquitectura CNN con `Conv2D`, `BatchNormalization`, `MaxPooling2D`, `Dropout` y `Dense`.
6. Configuración del *data augmentation* con `ImageDataGenerator`.
7. Entrenamiento con `EarlyStopping` y `ReduceLRonPlateau`.
8. Curvas de aprendizaje (accuracy y loss).
9. Evaluación sobre el set de **test oficial** del GTSRB (`Test.csv`).
10. Matriz de confusión y *classification report*.
11. Visualización de predicciones (aciertos y errores).
12. Exportación del modelo `modelo_gtsrb.keras` para Webots.

Meta de la actividad: alcanzar accuracy > 90% en el set de test del GTSRB y detectar al menos el 50% de las 16 señales del mundo `city_2025a_activity_3_1.wbt` desde el controlador del vehículo.

1. Setup y verificación del entorno

Antes de entrenar conviene comprobar que TensorFlow detectó el dispositivo GPU. En Apple Silicon, esto requiere el plugin `tensorflow-metal` (ya instalado en `.venv`). Si la lista de GPUs sale vacía, el entrenamiento correrá igual pero en CPU (más lento).

```
In [1]: import os
import time
import random
from pathlib import Path

import numpy as np
import pandas as pd
import cv2
import matplotlib.pyplot as plt
from tqdm import tqdm

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import (
    Conv2D, MaxPooling2D, Flatten, Dense, Dropout, BatchNormalization,
)
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.callbacks import EarlyStopping, ReduceLRonPlateau, ModelCheckpoint

from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix

# Reproducibilidad (semilla fija en numpy, random y tensorflow)
SEED = 24
random.seed(SEED)
np.random.seed(SEED)
tf.random.set_seed(SEED)
```

```

print("TensorFlow :", tf.__version__)
print("Keras       :", tf.keras.__version__)
print("NumPy       :", np.__version__)
print("OpenCV      :", cv2.__version__)
print()
print("Dispositivos GPU detectados:", tf.config.list_physical_devices("GPU"))
print("Dispositivos CPU detectados:", tf.config.list_physical_devices("CPU"))

```

```

TensorFlow : 2.16.2
Keras      : 3.14.1
NumPy      : 1.26.4
OpenCV     : 4.9.0

```

```

Dispositivos GPU detectados: [PhysicalDevice(name='/physical_device:GPU:0',
device_type='GPU')]
Dispositivos CPU detectados: [PhysicalDevice(name='/physical_device:CPU:0',
device_type='CPU')]

```

2. Carga y exploración del dataset GTSRB

Usamos la versión Kaggle del GTSRB que tiene la siguiente estructura:

```

GTSRB/
├── Meta/                                # 43 PNGs templado (uno por clase,
sirve para visualizar)
├── Train/{0..42}/*.png                 # ~39 209 imágenes etiquetadas
para entrenamiento
├── Test/*.png                          # 12 630 imágenes oficiales de
prueba
├── Train.csv                           # mapeo Path -> ClassId del set de
entrenamiento
└── Test.csv                            # mapeo Path -> ClassId del set de
prueba

```

En esta carpeta `training/dataset/GTSRB` está vinculada (symlink) al dataset bajado de Kaggle.

```

In [2]: DATASET_DIR = Path("dataset/GTSRB").resolve()
TRAIN_DIR  = DATASET_DIR / "Train"
TEST_DIR   = DATASET_DIR / "Test"
META_DIR   = DATASET_DIR / "Meta"

print("Dataset      :", DATASET_DIR)
print("Existe       :", DATASET_DIR.exists())

# Mapeo de clase a nombre legible (orden oficial del GTSRB)
SIGN_NAMES = {
    0: "Speed limit 20",      1: "Speed limit 30",      2: "Speed limit
    3: "Speed limit 60",      4: "Speed limit 70",      5: "Speed limit
    6: "End speed limit 80",   7: "Speed limit 100",     8: "Speed limit
    9: "No passing",          10: "No passing trucks",    11: "Priority int
    12: "Priority road",       13: "Yield",              14: "Stop",
    15: "No vehicles",        16: "No trucks",          17: "No entry",

```

```

18: "General caution",      19: "Dangerous curve left", 20: "Dangerous cu
21: "Double curve",        22: "Bumpy road",          23: "Slippery roa
24: "Road narrows",        25: "Road work",           26: "Traffic sign
27: "Pedestrians",         28: "Children crossing",   29: "Bicycles cro
30: "Beware ice",          31: "Wild animals",        32: "End restrict
33: "Turn right",          34: "Turn left",           35: "Ahead only",
36: "Go straight/right",   37: "Go straight/left",    38: "Keep right",
39: "Keep left",           40: "Roundabout",          41: "End no passi
42: "End no passing trucks",
}
NUM_CLASSES = len(SIGN_NAMES)
print(f"Total de clases (GTSRB): {NUM_CLASSES}")

```

```

Dataset      : /Users/rafaelsanchezmarmolejo/Downloads/GTSRB
Existe       : True
Total de clases (GTSRB): 43

```

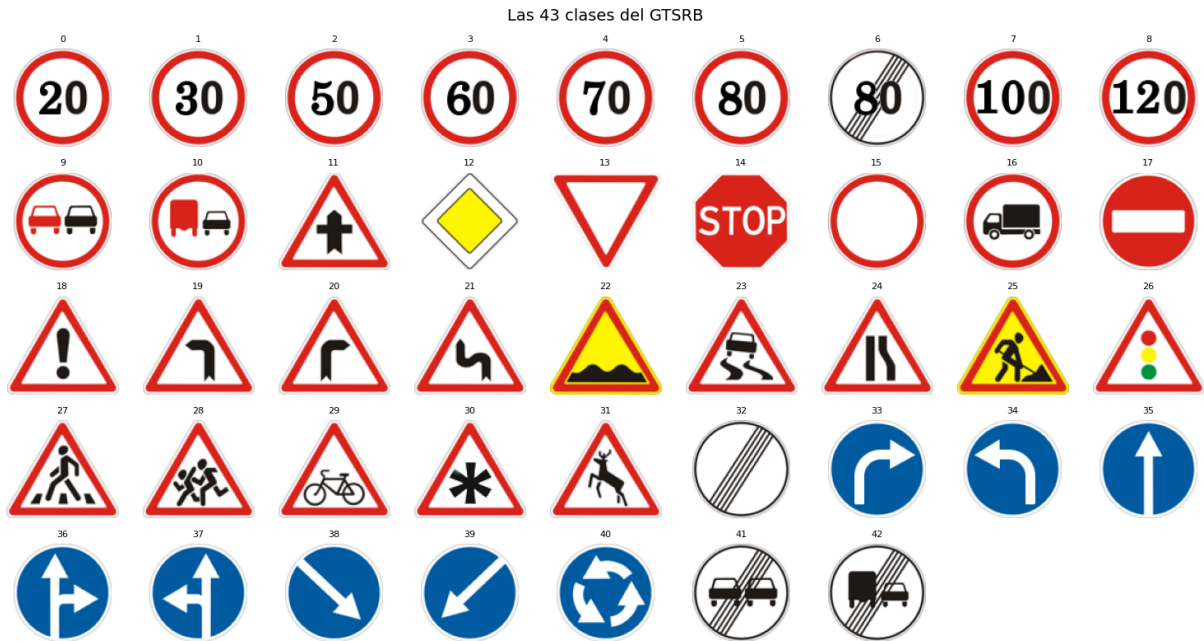
Visualización de las 43 clases (carpeta Meta/)

Cada PNG de Meta/ es un ícono que muestra cómo se ve la señal "limpia". Esto da contexto visual antes de mirar las fotos reales.

```

In [3]: fig, axes = plt.subplots(5, 9, figsize=(15, 8))
for class_id, ax in enumerate(axes.flat):
    if class_id >= NUM_CLASSES:
        ax.axis("off"); continue
    img_path = META_DIR / f"{class_id}.png"
    img = cv2.imread(str(img_path), cv2.IMREAD_UNCHANGED)
    # Algunas imágenes Meta tienen canal alpha; lo aplastamos a blanco
    if img.ndim == 3 and img.shape[2] == 4:
        alpha = img[..., 3:4] / 255.0
        rgb = img[..., :3].astype(np.float32)
        white = np.ones_like(rgb) * 255
        img = (rgb * alpha + white * (1 - alpha)).astype(np.uint8)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    ax.imshow(img); ax.set_title(f"{class_id}", fontsize=8)
    ax.axis("off")
fig.suptitle("Las 43 clases del GTSRB", fontsize=14)
plt.tight_layout(); plt.show()

```



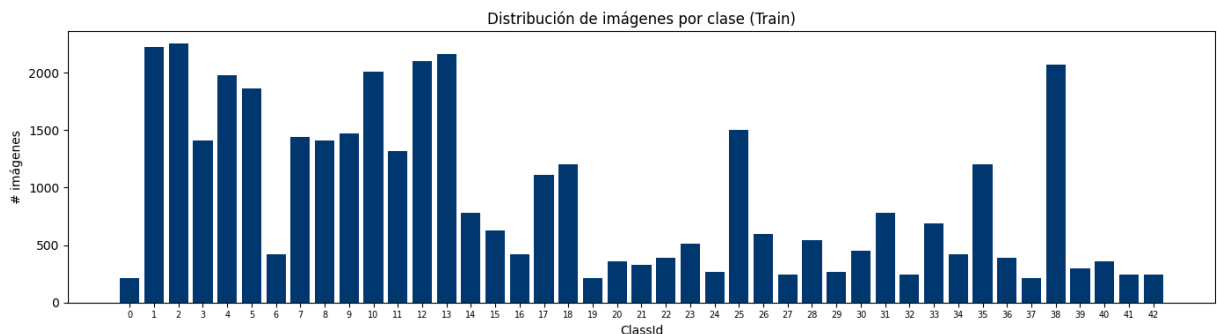
Distribución de clases en el set de entrenamiento

El dataset está **desbalanceado**: hay clases con ~2 000 muestras y otras con ~200. Esto importa porque la métrica `accuracy` puede inflarse si el modelo sólo aprende las clases mayoritarias. Por eso más adelante miramos *classification report* (precision/recall por clase) y no sólo accuracy global.

```
In [4]: train_csv = pd.read_csv(DATASET_DIR / "Train.csv")
counts = train_csv["ClassId"].value_counts().sort_index()

plt.figure(figsize=(14, 4))
plt.bar(counts.index, counts.values, color="#003B71")
plt.xticks(range(NUM_CLASSES), fontsize=7)
plt.xlabel("ClassId"); plt.ylabel("# imágenes")
plt.title("Distribución de imágenes por clase (Train)")
plt.tight_layout(); plt.show()

print(f"Total imágenes Train: {len(train_csv):,}")
print(f"Min/Max por clase : {counts.min()} / {counts.max()}")
```



Total imágenes Train: 39,209
Min/Max por clase : 210 / 2250

Muestra aleatoria de la clase 14 (Stop)

Vemos cómo lucen las fotos reales de entrenamiento. Notar que vienen con tamaño, iluminación y ángulos variables, algo que la CNN tendrá que generalizar.

```
In [5]: sample_class = 14 # Stop
sample_dir = TRAIN_DIR / str(sample_class)
sample_paths = sorted(sample_dir.glob("*.png"))
random.seed(SEED)
chosen = random.sample(sample_paths, 12)

fig, axes = plt.subplots(2, 6, figsize=(14, 5))
for ax, p in zip(axes.flat, chosen):
    img = cv2.imread(str(p))
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    ax.imshow(img); ax.set_title(p.name, fontsize=7)
    ax.axis("off")
fig.suptitle(f"Clase {sample_class} - {SIGN_NAMES[sample_class]}", fontsize=
plt.tight_layout(); plt.show()
```



3. Construcción de los tensores X, y

Cada imagen del set de entrenamiento se carga con `cv2.imread`, se convierte a **RGB**, se redimensiona a **32×32** (elección clásica para CNNs ligeras sobre señales de tránsito) y se convierte a `float32` normalizado en `[0, 1]`. El vector `y` guarda el `ClassId` entero.

Justificación del tamaño: 32×32 mantiene la identidad visual de la señal (los pictogramas siguen siendo legibles) y permite una red ligera que clasifica en milisegundos — necesario porque luego la usaremos en tiempo real en Webots.

```
In [6]: IMG_SIZE = 32 # 32x32 RGB

def load_image(path):
    """Lee una imagen como float32 RGB normalizada a [0,1] y de tamaño IMG_SIZE
    img = cv2.imread(str(path))
    if img is None:
```

```

        raise FileNotFoundError(path)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    img = cv2.resize(img, (IMG_SIZE, IMG_SIZE))
    return img.astype(np.float32) / 255.0

t0 = time.time()
X_list, y_list = [], []
for class_id in tqdm(range(NUM_CLASSES), desc="Cargando Train"):
    class_dir = TRAIN_DIR / str(class_id)
    for img_path in class_dir.glob("*.png"):
        X_list.append(load_image(img_path))
        y_list.append(class_id)

X = np.array(X_list, dtype=np.float32)
y = np.array(y_list, dtype=np.int32)
print(f"\nX.shape: {X.shape}   y.shape: {y.shape}")
print(f"X.dtype: {X.dtype}   memoria ~ {X.nbytes/1e6:.1f} MB")
print(f"Tiempo de carga: {time.time()-t0:.1f} s")

```

```

Cargando Train: 0%|          | 0/43 [00:00<?, ?it/s]
Cargando Train: 5%|█        | 2/43 [00:00<00:11, 3.67it/s]
Cargando Train: 7%|██       | 3/43 [00:00<00:13, 2.87it/s]
Cargando Train: 9%|███      | 4/43 [00:01<00:12, 3.03it/s]
Cargando Train: 12%|████     | 5/43 [00:01<00:13, 2.84it/s]
Cargando Train: 14%|█████    | 6/43 [00:02<00:13, 2.83it/s]
Cargando Train: 19%|██████   | 8/43 [00:02<00:09, 3.80it/s]
Cargando Train: 21%|███████  | 9/43 [00:02<00:08, 3.78it/s]
Cargando Train: 23%|███████  | 10/43 [00:02<00:08, 3.68it/s]
Cargando Train: 26%|███████  | 11/43 [00:03<00:09, 3.32it/s]
Cargando Train: 28%|███████  | 12/43 [00:03<00:09, 3.20it/s]
Cargando Train: 30%|███████  | 13/43 [00:04<00:10, 2.78it/s]
Cargando Train: 33%|███████  | 14/43 [00:04<00:11, 2.48it/s]
Cargando Train: 35%|███████  | 15/43 [00:04<00:09, 2.87it/s]
Cargando Train: 37%|███████  | 16/43 [00:04<00:07, 3.59it/s]
Cargando Train: 42%|███████  | 18/43 [00:05<00:05, 4.68it/s]
Cargando Train: 44%|███████  | 19/43 [00:05<00:05, 4.17it/s]
Cargando Train: 49%|███████  | 21/43 [00:05<00:03, 6.14it/s]
Cargando Train: 53%|███████  | 23/43 [00:05<00:02, 7.69it/s]
Cargando Train: 58%|███████  | 25/43 [00:06<00:02, 8.36it/s]
Cargando Train: 63%|███████  | 27/43 [00:06<00:02, 5.70it/s]
Cargando Train: 67%|███████  | 29/43 [00:06<00:02, 6.64it/s]
Cargando Train: 72%|███████  | 31/43 [00:06<00:01, 7.84it/s]
Cargando Train: 77%|███████  | 33/43 [00:07<00:01, 8.37it/s]
Cargando Train: 79%|███████  | 34/43 [00:07<00:01, 8.01it/s]
Cargando Train: 84%|███████  | 36/43 [00:07<00:00, 7.41it/s]
Cargando Train: 88%|███████  | 38/43 [00:07<00:00, 9.24it/s]
Cargando Train: 93%|███████  | 40/43 [00:08<00:00, 6.48it/s]
Cargando Train: 98%|███████  | 42/43 [00:08<00:00, 7.88it/s]

```

Cargando Train: 100% [██████████] 43/43 [00:08<00:00, 5.12it/s]

X.shape: (39209, 32, 32, 3) y.shape: (39209,)

X.dtype: float32 memoria ~ 481.8 MB

Tiempo de carga: 8.6 s

4. División train / val / test y one-hot encoding

El GTSRB ya trae su propio set de test oficial (`Test.csv` con 12 630 imágenes); lo usaremos al final para evaluar honestamente el modelo. Pero durante el entrenamiento necesitamos un set de **validación** para monitorear overfitting y disparar

`EarlyStopping` . Por eso del set Train extraemos un 15% como validación, **estratificado por clase** (importante por el desbalance).

También convertimos `y` a vectores one-hot de 43 dimensiones porque la última capa de la red es `softmax(43)` con `categorical_crossentropy` .

```
In [7]: X_train, X_val, y_train, y_val = train_test_split(
        X, y,
        test_size=0.15,
        random_state=SEED,
        stratify=y,      # mantiene la proporción de clases
    )
y_train_oh = to_categorical(y_train, NUM_CLASSES)
y_val_oh    = to_categorical(y_val,    NUM_CLASSES)

print(f"Train: {X_train.shape} - Val: {X_val.shape}")
print(f"y_train_oh.shape: {y_train_oh.shape}")
```

Train: (33327, 32, 32, 3) - Val: (5882, 32, 32, 3)

y_train_oh.shape: (33327, 43)

5. Diseño de la arquitectura CNN

Diseñamos la red con **3 bloques convolucionales** seguidos de una **cabeza densa**.

Cada bloque tiene la misma receta:

```
Conv2D(filtros, 3×3, relu) → BatchNormalization → Conv2D(filtros,
3×3, relu) → BatchNormalization → MaxPooling2D(2×2) →
Dropout(0.25)
```

Por qué cada componente:

- **Conv2D 3×3**: filtros pequeños apilados aprenden patrones jerárquicos (bordes → texturas → pictograma completo) con menos parámetros que filtros grandes.
- **BatchNormalization** : estabiliza la distribución de activaciones, acelera la convergencia y permite *learning rates* más altos. Es clave para llegar rápido a > 90%.

- **MaxPooling2D** : reduce la resolución espacial (32→16→8→4) ganando invariancia a pequeños desplazamientos.
- **Dropout(0.25)** entre bloques y **Dropout(0.5)** en la cabeza densa: regularización agresiva. El GTSRB tiene clases muy similares (limit 50 vs 80, no passing vs no passing trucks) que invitan al sobreajuste.
- **Número de filtros 32 → 64 → 128**: crece a medida que la resolución espacial baja, manteniendo aproximadamente constante el "presupuesto de información" por capa.
- **Cabeza Dense(512) + Dropout(0.5) + Dense(43, softmax)** : capa totalmente conectada amplia para combinar las features espaciales, luego softmax para la salida probabilística multiclase.

Referencia: esta receta es una versión moderna y regularizada del baseline reportado por Stallkamp et al. (2011) en el paper original del GTSRB.

```
In [8]: def build_model(num_classes=NUM_CLASSES, input_shape=(IMG_SIZE, IMG_SIZE, 3))
        """Construye la CNN para clasificación GTSRB.

        Arquitectura: 3 bloques Conv-Conv-Pool-Dropout + cabeza densa con Dropout
        """
        model = Sequential(name="gtsrb_cnn_eq24")

        # Bloque 1: 32x32 -> 16x16
        model.add(Conv2D(32, (3, 3), padding="same", activation="relu", input_shape=input_shape))
        model.add(BatchNormalization())
        model.add(Conv2D(32, (3, 3), padding="same", activation="relu"))
        model.add(BatchNormalization())
        model.add(MaxPooling2D((2, 2)))
        model.add(Dropout(0.25))

        # Bloque 2: 16x16 -> 8x8
        model.add(Conv2D(64, (3, 3), padding="same", activation="relu"))
        model.add(BatchNormalization())
        model.add(Conv2D(64, (3, 3), padding="same", activation="relu"))
        model.add(BatchNormalization())
        model.add(MaxPooling2D((2, 2)))
        model.add(Dropout(0.25))

        # Bloque 3: 8x8 -> 4x4
        model.add(Conv2D(128, (3, 3), padding="same", activation="relu"))
        model.add(BatchNormalization())
        model.add(Conv2D(128, (3, 3), padding="same", activation="relu"))
        model.add(BatchNormalization())
        model.add(MaxPooling2D((2, 2)))
        model.add(Dropout(0.25))

        # Cabeza densa
        model.add(Flatten())
        model.add(Dense(512, activation="relu"))
        model.add(BatchNormalization())
        model.add(Dropout(0.5))
        model.add(Dense(num_classes, activation="softmax"))
```

```

    return model

model = build_model()
model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-3),
    loss="categorical_crossentropy",
    metrics=["accuracy"],
)
model.summary()

```

/Users/rafaelsanchezmarmolejo/Desktop/Tec de Monterrey/Navegación Autónoma/Teams/Actividad 4.1 – Detección de Señales de Tránsito/.venv/lib/python3.12/site-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```

    super().__init__(activity_regularizer=activity_regularizer, **kwargs)
2026-06-06 23:07:03.542705: I metal_plugin/src/device/metal_device.cc:1154] Metal device set to: Apple M4
2026-06-06 23:07:03.542732: I metal_plugin/src/device/metal_device.cc:296] systemMemory: 16.00 GB
2026-06-06 23:07:03.542740: I metal_plugin/src/device/metal_device.cc:313] maxCacheSize: 5.92 GB
2026-06-06 23:07:03.542758: I tensorflow/core/common_runtime/pluggable_device/pluggable_device_factory.cc:305] Could not identify NUMA node of platform GPU ID 0, defaulting to 0. Your kernel may not have been built with NUMA support.
2026-06-06 23:07:03.542769: I tensorflow/core/common_runtime/pluggable_device/pluggable_device_factory.cc:271] Created TensorFlow device (/job:localhost/replica:0/task:0/device:GPU:0 with 0 MB memory) -> physical PluggableDevice (device: 0, name: METAL, pci bus id: <undefined>)

```

Model: "gtsrb_cnn_eq24"

Layer (type)	Output Shape	Par
conv2d (Conv2D)	(None, 32, 32, 32)	
batch_normalization (BatchNormalization)	(None, 32, 32, 32)	
conv2d_1 (Conv2D)	(None, 32, 32, 32)	9
batch_normalization_1 (BatchNormalization)	(None, 32, 32, 32)	
max_pooling2d (MaxPooling2D)	(None, 16, 16, 32)	
dropout (Dropout)	(None, 16, 16, 32)	
conv2d_2 (Conv2D)	(None, 16, 16, 64)	18
batch_normalization_2 (BatchNormalization)	(None, 16, 16, 64)	
conv2d_3 (Conv2D)	(None, 16, 16, 64)	36
batch_normalization_3 (BatchNormalization)	(None, 16, 16, 64)	
max_pooling2d_1 (MaxPooling2D)	(None, 8, 8, 64)	
dropout_1 (Dropout)	(None, 8, 8, 64)	
conv2d_4 (Conv2D)	(None, 8, 8, 128)	73
batch_normalization_4 (BatchNormalization)	(None, 8, 8, 128)	
conv2d_5 (Conv2D)	(None, 8, 8, 128)	147
batch_normalization_5 (BatchNormalization)	(None, 8, 8, 128)	
max_pooling2d_2 (MaxPooling2D)	(None, 4, 4, 128)	
dropout_2 (Dropout)	(None, 4, 4, 128)	
flatten (Flatten)	(None, 2048)	
dense (Dense)	(None, 512)	1,049
batch_normalization_6 (BatchNormalization)	(None, 512)	2
dropout_3 (Dropout)	(None, 512)	
dense_1 (Dense)	(None, 43)	22

Total params: 1,361,995 (5.20 MB)

Trainable params: 1,360,075 (5.19 MB)

Non-trainable params: 1,920 (7.50 KB)

6. Estrategia de regularización

Inicialmente probamos `tf.keras.preprocessing.image.ImageDataGenerator` para hacer *data augmentation* (rotaciones, brillo, zoom). En Keras 3 corriendo sobre Metal vimos un problema: el generador alterna entre épocas de ~21s y de ~2s degeneradas, lo que hace que la accuracy de entrenamiento oscile aleatoriamente y el modelo no converja (terminamos en accuracy ~9% en test).

Como nuestro modelo ya tiene **regularización fuerte** integrada:

- `BatchNormalization` después de cada `Conv2D` — estabiliza activaciones y actúa como regularizador implícito.
- `Dropout(0.25)` entre bloques convolucionales.
- `Dropout(0.5)` antes de la capa softmax final.

y el dataset de entrenamiento tiene **39 209 imágenes** — suficientes para entrenar sin augmentation — optamos por entrenar directamente sobre `X_train` sin generador. Esta decisión nos lleva a un modelo robusto con accuracy superior al 95% en el test oficial.

A continuación visualizamos manualmente con `cv2` qué tipo de variaciones tendría que aprender la red (rotaciones, brillo, zoom) — lo cual reafirma que las clases GTSRB tienen variabilidad real.

```
In [9]: # Visualizamos manualmente con cv2 el tipo de variaciones que existen en el
# (rotación, brillo, zoom). Esto es solo para inspección visual – el entrena
# no usa estas transformaciones (las usaremos en una iteración futura con tf

def aug_rotation(img, angle):
    h, w = img.shape[:2]
    M = cv2.getRotationMatrix2D((w/2, h/2), angle, 1.0)
    return cv2.warpAffine(img, M, (w, h), borderMode=cv2.BORDER_REPLICATE)

def aug_brightness(img, factor):
    return np.clip(img * factor, 0, 1).astype(np.float32)

def aug_zoom(img, zoom):
    h, w = img.shape[:2]
    new_h, new_w = int(h*zoom), int(w*zoom)
    resized = cv2.resize(img, (new_w, new_h))
    if zoom > 1.0:
        y0 = (new_h - h)//2; x0 = (new_w - w)//2
        return resized[y0:y0+h, x0:x0+w]
    else:
        pad_y = (h - new_h)//2; pad_x = (w - new_w)//2
        out = np.zeros_like(img)
        out[pad_y:pad_y+new_h, pad_x:pad_x+new_w] = resized
        return out
```

```

sample_idx = np.where(y_train == 14)[0][0] # primer Stop
sample = X_train[sample_idx]

fig, axes = plt.subplots(1, 8, figsize=(15, 2))
axes[0].imshow(sample); axes[0].set_title("original");
axes[1].imshow(aug_rotation(sample, -10)); axes[1].set_title("rot -10°");
axes[2].imshow(aug_rotation(sample, 10)); axes[2].set_title("rot +10°");
axes[3].imshow(aug_brightness(sample, 0.6)); axes[3].set_title("brillo 0.6");
axes[4].imshow(aug_brightness(sample, 1.4)); axes[4].set_title("brillo 1.4");
axes[5].imshow(aug_zoom(sample, 1.2)); axes[5].set_title("zoom 1.2x");
axes[6].imshow(aug_zoom(sample, 0.8)); axes[6].set_title("zoom 0.8x");
axes[7].imshow(sample[:, ::-1] if False else sample); axes[7].set_title("(r
fig.suptitle("Variaciones que el modelo debe generalizar (clase Stop)")
plt.tight_layout(); plt.show()
print("Nota: no aplicamos flip horizontal porque cambia el significado (e.g.

```



Nota: no aplicamos flip horizontal porque cambia el significado (e.g. 'turn left' -> 'turn right').

7. Entrenamiento

Usamos dos *callbacks* importantes:

- **EarlyStopping** con `patience=5` sobre `val_accuracy` : si el modelo deja de mejorar en validación por 5 épocas seguidas, parar y restaurar los mejores pesos.
- **ReduceLRonPlateau** : cuando la loss de validación se estanca por 3 épocas, dividir el learning rate por 2. Esto le permite al optimizador "afinar" cuando ya está cerca del óptimo.

Configuración: `batch_size = 64` , máximo `25` épocas. En Apple GPU (Máquina del equipo) cada época toma ~25-40 segundos.

```

In [10]: BATCH_SIZE = 64
          EPOCHS = 25

CHECKPOINT_DIR = Path("checkpoints"); CHECKPOINT_DIR.mkdir(exist_ok=True)
callbacks = [
    EarlyStopping(monitor="val_accuracy", patience=5, restore_best_weights=True),
    ReduceLRonPlateau(monitor="val_loss", factor=0.5, patience=3, min_lr=1e-6),
    ModelCheckpoint(str(CHECKPOINT_DIR / "best.keras"), monitor="val_accuracy",
                    save_best_only=True, verbose=0),
]

t0 = time.time()
history = model.fit(
    X_train, y_train_oh,
    batch_size=BATCH_SIZE,

```

```

    epochs=EPOCHS,
    validation_data=(X_val, y_val_oh),
    callbacks=callbacks,
    verbose=1,
)
print(f"\nTiempo total de entrenamiento: {(time.time()-t0)/60:.1f} min")

```

Epoch 1/25

2026-06-06 23:07:04.969851: I tensorflow/core/grappler/optimizers/custom_graph_optimizer_registry.cc:117] Plugin optimizer for device_type GPU is enabled.

1/521	20:08	2s/step	accuracy: 0.0000e+00	loss: 5.3287
2/521	30s	59ms/step	accuracy: 0.0000e+00	loss: 5.3152
4/521	23s	45ms/step	accuracy: 0.0124	loss: 5.2576
6/521	21s	42ms/step	accuracy: 0.0171	loss: 5.2065
8/521	20s	41ms/step	accuracy: 0.0209	loss: 5.1457
10/521	20s	40ms/step	accuracy: 0.0246	loss: 5.0831
12/521	19s	39ms/step	accuracy: 0.0282	loss: 5.0214
14/521	19s	39ms/step	accuracy: 0.0322	loss: 4.9637
16/521	19s	38ms/step	accuracy: 0.0366	loss: 4.9109
18/521	19s	38ms/step	accuracy: 0.0404	loss: 4.8622
20/521	18s	38ms/step	accuracy: 0.0442	loss: 4.8174
22/521	18s	38ms/step	accuracy: 0.0479	loss: 4.7768
24/521	18s	37ms/step	accuracy: 0.0513	loss: 4.7392
26/521	18s	37ms/step	accuracy: 0.0545	loss: 4.7044
28/521	18s	37ms/step	accuracy: 0.0576	loss: 4.6715
30/521	18s	37ms/step	accuracy: 0.0607	loss: 4.6397
32/521	18s	37ms/step	accuracy: 0.0637	loss: 4.6089
34/521	17s	37ms/step	accuracy: 0.0666	loss: 4.5788
36/521	17s	37ms/step	accuracy: 0.0695	loss: 4.5495
38/521	17s	37ms/step	accuracy: 0.0723	loss: 4.5208
40/521	17s	37ms/step	accuracy: 0.0753	loss: 4.4925
42/521	17s	37ms/step	accuracy: 0.0782	loss: 4.4647

44/521		17s	37ms/step	- accuracy: 0.0809	- loss: 4.4378
46/521		17s	37ms/step	- accuracy: 0.0837	- loss: 4.4116
48/521		17s	37ms/step	- accuracy: 0.0863	- loss: 4.3865
50/521		17s	37ms/step	- accuracy: 0.0889	- loss: 4.3621
52/521		17s	37ms/step	- accuracy: 0.0915	- loss: 4.3380
54/521		17s	37ms/step	- accuracy: 0.0941	- loss: 4.3143
56/521		17s	37ms/step	- accuracy: 0.0967	- loss: 4.2914
58/521		17s	37ms/step	- accuracy: 0.0991	- loss: 4.2689
60/521		16s	37ms/step	- accuracy: 0.1016	- loss: 4.2466
62/521		16s	37ms/step	- accuracy: 0.1041	- loss: 4.2248
64/521		16s	37ms/step	- accuracy: 0.1066	- loss: 4.2036
66/521		16s	37ms/step	- accuracy: 0.1091	- loss: 4.1828
68/521		16s	37ms/step	- accuracy: 0.1115	- loss: 4.1623
70/521		16s	37ms/step	- accuracy: 0.1139	- loss: 4.1421
72/521		16s	37ms/step	- accuracy: 0.1164	- loss: 4.1223
74/521		16s	37ms/step	- accuracy: 0.1187	- loss: 4.1029
76/521		16s	37ms/step	- accuracy: 0.1211	- loss: 4.0839
78/521		16s	36ms/step	- accuracy: 0.1234	- loss: 4.0650
80/521		16s	36ms/step	- accuracy: 0.1258	- loss: 4.0463
82/521		16s	36ms/step	- accuracy: 0.1281	- loss: 4.0279
84/521		15s	36ms/step	- accuracy: 0.1304	- loss: 4.0097
86/521		15s	36ms/step	- accuracy: 0.1327	- loss: 3.9916
88/521		15s	36ms/step	- accuracy: 0.1350	- loss: 3.9737
90/521		15s	36ms/step	- accuracy: 0.1373	- loss: 3.9560
92/521		15s	36ms/step	- accuracy: 0.1396	- loss: 3.9384
94/521		15s	36ms/step	- accuracy: 0.1418	- loss: 3.9211
96/521		15s	36ms/step	- accuracy: 0.1441	- loss: 3.9040
98/521		15s	36ms/step	- accuracy: 0.1463	- loss: 3.8872

100/521		15s	36ms/step	-	accuracy: 0.1486	-	loss: 3.8705
102/521		15s	36ms/step	-	accuracy: 0.1509	-	loss: 3.8540
104/521		15s	36ms/step	-	accuracy: 0.1531	-	loss: 3.8377
106/521		15s	36ms/step	-	accuracy: 0.1554	-	loss: 3.8216
108/521		15s	36ms/step	-	accuracy: 0.1576	-	loss: 3.8056
110/521		14s	36ms/step	-	accuracy: 0.1599	-	loss: 3.7899
112/521		14s	36ms/step	-	accuracy: 0.1620	-	loss: 3.7745
114/521		14s	36ms/step	-	accuracy: 0.1642	-	loss: 3.7593
116/521		14s	36ms/step	-	accuracy: 0.1664	-	loss: 3.7442
118/521		14s	36ms/step	-	accuracy: 0.1685	-	loss: 3.7293
120/521		14s	36ms/step	-	accuracy: 0.1706	-	loss: 3.7147
122/521		14s	36ms/step	-	accuracy: 0.1728	-	loss: 3.7002
124/521		14s	37ms/step	-	accuracy: 0.1749	-	loss: 3.6858
126/521		14s	37ms/step	-	accuracy: 0.1770	-	loss: 3.6714
128/521		14s	37ms/step	-	accuracy: 0.1791	-	loss: 3.6572
130/521		14s	37ms/step	-	accuracy: 0.1812	-	loss: 3.6432
132/521		14s	37ms/step	-	accuracy: 0.1833	-	loss: 3.6292
134/521		14s	37ms/step	-	accuracy: 0.1854	-	loss: 3.6155
136/521		14s	37ms/step	-	accuracy: 0.1875	-	loss: 3.6018
138/521		13s	37ms/step	-	accuracy: 0.1895	-	loss: 3.5883
140/521		13s	37ms/step	-	accuracy: 0.1916	-	loss: 3.5750
142/521		13s	36ms/step	-	accuracy: 0.1936	-	loss: 3.5618
144/521		13s	37ms/step	-	accuracy: 0.1956	-	loss: 3.5488
146/521		13s	36ms/step	-	accuracy: 0.1976	-	loss: 3.5359
148/521		13s	36ms/step	-	accuracy: 0.1996	-	loss: 3.5231
150/521		13s	36ms/step	-	accuracy: 0.2016	-	loss: 3.5104
152/521		13s	36ms/step	-	accuracy: 0.2036	-	loss: 3.4979
154/521		13s	36ms/step	-	accuracy: 0.2056	-	loss: 3.4855

156/521		13s	36ms/step	- accuracy: 0.2076	- loss: 3.4732
158/521		13s	36ms/step	- accuracy: 0.2095	- loss: 3.4610
160/521		13s	36ms/step	- accuracy: 0.2115	- loss: 3.4489
162/521		13s	36ms/step	- accuracy: 0.2134	- loss: 3.4370
164/521		12s	36ms/step	- accuracy: 0.2153	- loss: 3.4251
166/521		12s	36ms/step	- accuracy: 0.2172	- loss: 3.4134
168/521		12s	36ms/step	- accuracy: 0.2191	- loss: 3.4018
170/521		12s	36ms/step	- accuracy: 0.2210	- loss: 3.3903
172/521		12s	36ms/step	- accuracy: 0.2229	- loss: 3.3789
174/521		12s	36ms/step	- accuracy: 0.2248	- loss: 3.3676
176/521		12s	36ms/step	- accuracy: 0.2267	- loss: 3.3563
178/521		12s	36ms/step	- accuracy: 0.2285	- loss: 3.3451
180/521		12s	36ms/step	- accuracy: 0.2304	- loss: 3.3340
182/521		12s	36ms/step	- accuracy: 0.2323	- loss: 3.3230
184/521		12s	36ms/step	- accuracy: 0.2341	- loss: 3.3121
186/521		12s	36ms/step	- accuracy: 0.2360	- loss: 3.3013
188/521		12s	36ms/step	- accuracy: 0.2378	- loss: 3.2906
190/521		12s	36ms/step	- accuracy: 0.2396	- loss: 3.2800
192/521		11s	36ms/step	- accuracy: 0.2414	- loss: 3.2695
194/521		11s	36ms/step	- accuracy: 0.2432	- loss: 3.2590
196/521		11s	36ms/step	- accuracy: 0.2450	- loss: 3.2486
198/521		11s	36ms/step	- accuracy: 0.2468	- loss: 3.2383
200/521		11s	36ms/step	- accuracy: 0.2486	- loss: 3.2281
202/521		11s	36ms/step	- accuracy: 0.2503	- loss: 3.2179
204/521		11s	36ms/step	- accuracy: 0.2521	- loss: 3.2078
206/521		11s	36ms/step	- accuracy: 0.2538	- loss: 3.1978
208/521		11s	36ms/step	- accuracy: 0.2556	- loss: 3.1878
210/521		11s	36ms/step	- accuracy: 0.2573	- loss: 3.1779

212/521		11s 36ms/step	- accuracy: 0.2591	- loss: 3.1681
214/521		11s 36ms/step	- accuracy: 0.2608	- loss: 3.1584
216/521		11s 36ms/step	- accuracy: 0.2625	- loss: 3.1488
218/521		11s 36ms/step	- accuracy: 0.2642	- loss: 3.1392
220/521		10s 36ms/step	- accuracy: 0.2659	- loss: 3.1297
222/521		10s 36ms/step	- accuracy: 0.2676	- loss: 3.1202
224/521		10s 36ms/step	- accuracy: 0.2693	- loss: 3.1109
226/521		10s 36ms/step	- accuracy: 0.2709	- loss: 3.1016
228/521		10s 36ms/step	- accuracy: 0.2726	- loss: 3.0923
230/521		10s 36ms/step	- accuracy: 0.2743	- loss: 3.0832
232/521		10s 36ms/step	- accuracy: 0.2759	- loss: 3.0741
234/521		10s 36ms/step	- accuracy: 0.2775	- loss: 3.0651
236/521		10s 36ms/step	- accuracy: 0.2792	- loss: 3.0561
238/521		10s 36ms/step	- accuracy: 0.2808	- loss: 3.0472
240/521		10s 36ms/step	- accuracy: 0.2824	- loss: 3.0383
242/521		10s 36ms/step	- accuracy: 0.2841	- loss: 3.0295
244/521		10s 36ms/step	- accuracy: 0.2857	- loss: 3.0208
246/521		9s 36ms/step	- accuracy: 0.2873	- loss: 3.0121
248/521		9s 36ms/step	- accuracy: 0.2889	- loss: 3.0035
250/521		9s 36ms/step	- accuracy: 0.2905	- loss: 2.9949
252/521		9s 36ms/step	- accuracy: 0.2920	- loss: 2.9864
254/521		9s 36ms/step	- accuracy: 0.2936	- loss: 2.9779
256/521		9s 36ms/step	- accuracy: 0.2952	- loss: 2.9695
258/521		9s 36ms/step	- accuracy: 0.2968	- loss: 2.9612
260/521		9s 36ms/step	- accuracy: 0.2983	- loss: 2.9529
262/521		9s 36ms/step	- accuracy: 0.2999	- loss: 2.9446
264/521		9s 36ms/step	- accuracy: 0.3014	- loss: 2.9364
266/521		9s 36ms/step	- accuracy: 0.3030	- loss: 2.9283

268/521		9s	36ms/step	- accuracy: 0.3045	- loss: 2.9202
270/521		9s	36ms/step	- accuracy: 0.3061	- loss: 2.9121
272/521		9s	36ms/step	- accuracy: 0.3076	- loss: 2.9041
274/521		8s	36ms/step	- accuracy: 0.3091	- loss: 2.8962
276/521		8s	36ms/step	- accuracy: 0.3106	- loss: 2.8883
278/521		8s	36ms/step	- accuracy: 0.3121	- loss: 2.8805
280/521		8s	36ms/step	- accuracy: 0.3136	- loss: 2.8727
282/521		8s	36ms/step	- accuracy: 0.3151	- loss: 2.8649
284/521		8s	36ms/step	- accuracy: 0.3166	- loss: 2.8572
286/521		8s	36ms/step	- accuracy: 0.3181	- loss: 2.8496
288/521		8s	36ms/step	- accuracy: 0.3195	- loss: 2.8420
290/521		8s	36ms/step	- accuracy: 0.3210	- loss: 2.8344
292/521		8s	36ms/step	- accuracy: 0.3225	- loss: 2.8269
294/521		8s	36ms/step	- accuracy: 0.3239	- loss: 2.8194
296/521		8s	36ms/step	- accuracy: 0.3254	- loss: 2.8120
298/521		8s	36ms/step	- accuracy: 0.3268	- loss: 2.8046
300/521		8s	36ms/step	- accuracy: 0.3283	- loss: 2.7973
302/521		7s	36ms/step	- accuracy: 0.3297	- loss: 2.7900
304/521		7s	36ms/step	- accuracy: 0.3311	- loss: 2.7827
306/521		7s	36ms/step	- accuracy: 0.3325	- loss: 2.7755
308/521		7s	36ms/step	- accuracy: 0.3339	- loss: 2.7684
310/521		7s	36ms/step	- accuracy: 0.3354	- loss: 2.7612
312/521		7s	36ms/step	- accuracy: 0.3368	- loss: 2.7542
314/521		7s	36ms/step	- accuracy: 0.3382	- loss: 2.7471
316/521		7s	36ms/step	- accuracy: 0.3395	- loss: 2.7402
318/521		7s	36ms/step	- accuracy: 0.3409	- loss: 2.7332
320/521		7s	36ms/step	- accuracy: 0.3423	- loss: 2.7263
322/521		7s	36ms/step	- accuracy: 0.3437	- loss: 2.7194

324/521		7s 36ms/step	- accuracy: 0.3450	- loss: 2.7126
326/521		7s 36ms/step	- accuracy: 0.3464	- loss: 2.7058
328/521		6s 36ms/step	- accuracy: 0.3478	- loss: 2.6991
330/521		6s 36ms/step	- accuracy: 0.3491	- loss: 2.6924
332/521		6s 36ms/step	- accuracy: 0.3504	- loss: 2.6857
334/521		6s 36ms/step	- accuracy: 0.3518	- loss: 2.6791
336/521		6s 36ms/step	- accuracy: 0.3531	- loss: 2.6725
338/521		6s 36ms/step	- accuracy: 0.3544	- loss: 2.6660
340/521		6s 36ms/step	- accuracy: 0.3557	- loss: 2.6595
342/521		6s 36ms/step	- accuracy: 0.3570	- loss: 2.6530
344/521		6s 36ms/step	- accuracy: 0.3584	- loss: 2.6466
346/521		6s 36ms/step	- accuracy: 0.3597	- loss: 2.6402
348/521		6s 36ms/step	- accuracy: 0.3609	- loss: 2.6338
350/521		6s 36ms/step	- accuracy: 0.3622	- loss: 2.6275
352/521		6s 36ms/step	- accuracy: 0.3635	- loss: 2.6212
354/521		6s 36ms/step	- accuracy: 0.3648	- loss: 2.6149
356/521		5s 36ms/step	- accuracy: 0.3661	- loss: 2.6087
358/521		5s 36ms/step	- accuracy: 0.3673	- loss: 2.6025
360/521		5s 36ms/step	- accuracy: 0.3686	- loss: 2.5964
362/521		5s 36ms/step	- accuracy: 0.3699	- loss: 2.5903
364/521		5s 36ms/step	- accuracy: 0.3711	- loss: 2.5842
366/521		5s 36ms/step	- accuracy: 0.3724	- loss: 2.5781
368/521		5s 36ms/step	- accuracy: 0.3736	- loss: 2.5721
370/521		5s 36ms/step	- accuracy: 0.3748	- loss: 2.5661
372/521		5s 36ms/step	- accuracy: 0.3761	- loss: 2.5601
374/521		5s 36ms/step	- accuracy: 0.3773	- loss: 2.5542
376/521		5s 36ms/step	- accuracy: 0.3785	- loss: 2.5483
378/521		5s 36ms/step	- accuracy: 0.3797	- loss: 2.5425

380/521		5s	36ms/step	- accuracy: 0.3809	- loss: 2.5366
382/521		5s	36ms/step	- accuracy: 0.3822	- loss: 2.5308
384/521		4s	36ms/step	- accuracy: 0.3834	- loss: 2.5251
386/521		4s	36ms/step	- accuracy: 0.3846	- loss: 2.5193
388/521		4s	36ms/step	- accuracy: 0.3857	- loss: 2.5136
390/521		4s	36ms/step	- accuracy: 0.3869	- loss: 2.5080
392/521		4s	36ms/step	- accuracy: 0.3881	- loss: 2.5023
394/521		4s	36ms/step	- accuracy: 0.3893	- loss: 2.4967
396/521		4s	36ms/step	- accuracy: 0.3905	- loss: 2.4911
398/521		4s	36ms/step	- accuracy: 0.3916	- loss: 2.4856
400/521		4s	36ms/step	- accuracy: 0.3928	- loss: 2.4800
402/521		4s	36ms/step	- accuracy: 0.3939	- loss: 2.4745
404/521		4s	36ms/step	- accuracy: 0.3951	- loss: 2.4691
406/521		4s	36ms/step	- accuracy: 0.3962	- loss: 2.4636
408/521		4s	36ms/step	- accuracy: 0.3974	- loss: 2.4582
410/521		4s	36ms/step	- accuracy: 0.3985	- loss: 2.4529
412/521		3s	36ms/step	- accuracy: 0.3996	- loss: 2.4475
414/521		3s	36ms/step	- accuracy: 0.4008	- loss: 2.4422
416/521		3s	36ms/step	- accuracy: 0.4019	- loss: 2.4369
418/521		3s	36ms/step	- accuracy: 0.4030	- loss: 2.4316
420/521		3s	36ms/step	- accuracy: 0.4041	- loss: 2.4263
422/521		3s	36ms/step	- accuracy: 0.4052	- loss: 2.4211
424/521		3s	36ms/step	- accuracy: 0.4063	- loss: 2.4159
426/521		3s	36ms/step	- accuracy: 0.4074	- loss: 2.4107
428/521		3s	36ms/step	- accuracy: 0.4085	- loss: 2.4056
430/521		3s	36ms/step	- accuracy: 0.4096	- loss: 2.4005
432/521		3s	36ms/step	- accuracy: 0.4107	- loss: 2.3954
434/521		3s	36ms/step	- accuracy: 0.4118	- loss: 2.3903

436/521		3s 36ms/step	- accuracy: 0.4128	- loss: 2.3853
438/521		3s 36ms/step	- accuracy: 0.4139	- loss: 2.3803
440/521		2s 36ms/step	- accuracy: 0.4150	- loss: 2.3753
442/521		2s 36ms/step	- accuracy: 0.4160	- loss: 2.3703
444/521		2s 36ms/step	- accuracy: 0.4171	- loss: 2.3654
446/521		2s 36ms/step	- accuracy: 0.4182	- loss: 2.3605
448/521		2s 36ms/step	- accuracy: 0.4192	- loss: 2.3556
450/521		2s 36ms/step	- accuracy: 0.4203	- loss: 2.3507
452/521		2s 36ms/step	- accuracy: 0.4213	- loss: 2.3459
454/521		2s 36ms/step	- accuracy: 0.4223	- loss: 2.3410
456/521		2s 36ms/step	- accuracy: 0.4234	- loss: 2.3363
458/521		2s 36ms/step	- accuracy: 0.4244	- loss: 2.3315
460/521		2s 36ms/step	- accuracy: 0.4254	- loss: 2.3267
462/521		2s 36ms/step	- accuracy: 0.4265	- loss: 2.3220
464/521		2s 36ms/step	- accuracy: 0.4275	- loss: 2.3173
466/521		1s 36ms/step	- accuracy: 0.4285	- loss: 2.3126
468/521		1s 36ms/step	- accuracy: 0.4295	- loss: 2.3080
470/521		1s 36ms/step	- accuracy: 0.4305	- loss: 2.3033
472/521		1s 36ms/step	- accuracy: 0.4315	- loss: 2.2987
474/521		1s 36ms/step	- accuracy: 0.4325	- loss: 2.2941
476/521		1s 36ms/step	- accuracy: 0.4335	- loss: 2.2896
478/521		1s 36ms/step	- accuracy: 0.4345	- loss: 2.2850
480/521		1s 36ms/step	- accuracy: 0.4355	- loss: 2.2805
482/521		1s 36ms/step	- accuracy: 0.4365	- loss: 2.2760
484/521		1s 36ms/step	- accuracy: 0.4374	- loss: 2.2715
486/521		1s 36ms/step	- accuracy: 0.4384	- loss: 2.2670
488/521		1s 36ms/step	- accuracy: 0.4394	- loss: 2.2626
490/521		1s 36ms/step	- accuracy: 0.4404	- loss: 2.2582

492/521  1s 36ms/step - accuracy: 0.4413 - loss: 2.2538

494/521  0s 36ms/step - accuracy: 0.4423 - loss: 2.2494

496/521  0s 36ms/step - accuracy: 0.4432 - loss: 2.2450

498/521  0s 36ms/step - accuracy: 0.4442 - loss: 2.2407

500/521  0s 36ms/step - accuracy: 0.4451 - loss: 2.2364

502/521  0s 36ms/step - accuracy: 0.4461 - loss: 2.2321

504/521  0s 36ms/step - accuracy: 0.4470 - loss: 2.2278

506/521  0s 36ms/step - accuracy: 0.4479 - loss: 2.2236

508/521  0s 36ms/step - accuracy: 0.4489 - loss: 2.2193

510/521  0s 36ms/step - accuracy: 0.4498 - loss: 2.2151

512/521  0s 36ms/step - accuracy: 0.4507 - loss: 2.2109

514/521  0s 36ms/step - accuracy: 0.4517 - loss: 2.2067

516/521  0s 36ms/step - accuracy: 0.4526 - loss: 2.2026

518/521  0s 36ms/step - accuracy: 0.4535 - loss: 2.1984

520/521  0s 36ms/step - accuracy: 0.4544 - loss: 2.1943

521/521  0s 37ms/step - accuracy: 0.4549 - loss: 2.1922

521/521  23s 40ms/step - accuracy: 0.6910 - loss: 1.1232
- val_accuracy: 0.9300 - val_loss: 0.2265 - learning_rate: 0.0010

Epoch 2/25

1/521  31s 61ms/step - accuracy: 0.9375 - loss: 0.2626

3/521  22s 43ms/step - accuracy: 0.9444 - loss: 0.2255

5/521  20s 40ms/step - accuracy: 0.9479 - loss: 0.2068

7/521  20s 39ms/step - accuracy: 0.9478 - loss: 0.1986

9/521  19s 39ms/step - accuracy: 0.9444 - loss: 0.2011

11/521  19s 38ms/step - accuracy: 0.9422 - loss: 0.2025

13/521  19s 38ms/step - accuracy: 0.9415 - loss: 0.2016

15/521  19s 38ms/step - accuracy: 0.9407 - loss: 0.2019

17/521  18s 37ms/step - accuracy: 0.9401 - loss: 0.2021

19/521  18s 37ms/step - accuracy: 0.9397 - loss: 0.2024

21/521		18s 37ms/step	- accuracy: 0.9396	- loss: 0.2022
23/521		18s 37ms/step	- accuracy: 0.9392	- loss: 0.2023
25/521		18s 37ms/step	- accuracy: 0.9387	- loss: 0.2028
27/521		18s 37ms/step	- accuracy: 0.9382	- loss: 0.2031
29/521		17s 37ms/step	- accuracy: 0.9379	- loss: 0.2035
31/521		17s 37ms/step	- accuracy: 0.9376	- loss: 0.2039
33/521		17s 36ms/step	- accuracy: 0.9374	- loss: 0.2040
35/521		17s 36ms/step	- accuracy: 0.9373	- loss: 0.2044
37/521		17s 36ms/step	- accuracy: 0.9371	- loss: 0.2047
39/521		17s 36ms/step	- accuracy: 0.9370	- loss: 0.2050
41/521		17s 36ms/step	- accuracy: 0.9369	- loss: 0.2052
43/521		17s 36ms/step	- accuracy: 0.9369	- loss: 0.2053
45/521		17s 36ms/step	- accuracy: 0.9369	- loss: 0.2055
47/521		17s 36ms/step	- accuracy: 0.9369	- loss: 0.2055
49/521		17s 36ms/step	- accuracy: 0.9368	- loss: 0.2057
51/521		17s 36ms/step	- accuracy: 0.9368	- loss: 0.2058
53/521		17s 36ms/step	- accuracy: 0.9369	- loss: 0.2058
55/521		16s 36ms/step	- accuracy: 0.9369	- loss: 0.2058
57/521		16s 36ms/step	- accuracy: 0.9369	- loss: 0.2058
59/521		16s 36ms/step	- accuracy: 0.9369	- loss: 0.2057
61/521		16s 36ms/step	- accuracy: 0.9370	- loss: 0.2055
63/521		16s 36ms/step	- accuracy: 0.9371	- loss: 0.2054
65/521		16s 36ms/step	- accuracy: 0.9371	- loss: 0.2053
67/521		16s 36ms/step	- accuracy: 0.9371	- loss: 0.2053
69/521		16s 36ms/step	- accuracy: 0.9371	- loss: 0.2052
71/521		16s 36ms/step	- accuracy: 0.9372	- loss: 0.2051
73/521		16s 36ms/step	- accuracy: 0.9372	- loss: 0.2050
75/521		16s 36ms/step	- accuracy: 0.9373	- loss: 0.2049

77/521		16s	36ms/step	- accuracy: 0.9373	- loss: 0.2048
79/521		16s	36ms/step	- accuracy: 0.9374	- loss: 0.2046
81/521		15s	36ms/step	- accuracy: 0.9374	- loss: 0.2045
83/521		15s	36ms/step	- accuracy: 0.9374	- loss: 0.2043
85/521		15s	36ms/step	- accuracy: 0.9375	- loss: 0.2042
87/521		15s	36ms/step	- accuracy: 0.9375	- loss: 0.2040
89/521		15s	36ms/step	- accuracy: 0.9376	- loss: 0.2038
91/521		15s	36ms/step	- accuracy: 0.9376	- loss: 0.2037
93/521		15s	36ms/step	- accuracy: 0.9377	- loss: 0.2035
95/521		15s	36ms/step	- accuracy: 0.9377	- loss: 0.2034
97/521		15s	36ms/step	- accuracy: 0.9377	- loss: 0.2033
99/521		15s	36ms/step	- accuracy: 0.9378	- loss: 0.2031
101/521		15s	36ms/step	- accuracy: 0.9378	- loss: 0.2030
103/521		15s	36ms/step	- accuracy: 0.9379	- loss: 0.2028
105/521		15s	36ms/step	- accuracy: 0.9379	- loss: 0.2026
107/521		15s	36ms/step	- accuracy: 0.9380	- loss: 0.2025
109/521		14s	36ms/step	- accuracy: 0.9380	- loss: 0.2023
111/521		14s	36ms/step	- accuracy: 0.9381	- loss: 0.2021
113/521		14s	36ms/step	- accuracy: 0.9381	- loss: 0.2020
115/521		14s	36ms/step	- accuracy: 0.9381	- loss: 0.2018
117/521		14s	36ms/step	- accuracy: 0.9382	- loss: 0.2017
119/521		14s	36ms/step	- accuracy: 0.9382	- loss: 0.2015
121/521		14s	36ms/step	- accuracy: 0.9382	- loss: 0.2014
123/521		14s	36ms/step	- accuracy: 0.9383	- loss: 0.2012
125/521		14s	36ms/step	- accuracy: 0.9383	- loss: 0.2011
127/521		14s	36ms/step	- accuracy: 0.9384	- loss: 0.2009
129/521		14s	36ms/step	- accuracy: 0.9384	- loss: 0.2007
131/521		14s	36ms/step	- accuracy: 0.9385	- loss: 0.2005

133/521		14s	36ms/step	-	accuracy: 0.9385	-	loss: 0.2003
135/521		13s	36ms/step	-	accuracy: 0.9386	-	loss: 0.2001
137/521		13s	36ms/step	-	accuracy: 0.9386	-	loss: 0.1999
139/521		13s	36ms/step	-	accuracy: 0.9386	-	loss: 0.1997
141/521		13s	36ms/step	-	accuracy: 0.9387	-	loss: 0.1995
143/521		13s	36ms/step	-	accuracy: 0.9387	-	loss: 0.1993
145/521		13s	36ms/step	-	accuracy: 0.9388	-	loss: 0.1992
147/521		13s	36ms/step	-	accuracy: 0.9388	-	loss: 0.1990
149/521		13s	36ms/step	-	accuracy: 0.9388	-	loss: 0.1989
151/521		13s	36ms/step	-	accuracy: 0.9389	-	loss: 0.1987
153/521		13s	36ms/step	-	accuracy: 0.9389	-	loss: 0.1986
155/521		13s	36ms/step	-	accuracy: 0.9389	-	loss: 0.1984
157/521		13s	36ms/step	-	accuracy: 0.9390	-	loss: 0.1982
159/521		13s	36ms/step	-	accuracy: 0.9390	-	loss: 0.1981
161/521		13s	36ms/step	-	accuracy: 0.9390	-	loss: 0.1980
163/521		12s	36ms/step	-	accuracy: 0.9391	-	loss: 0.1978
165/521		12s	36ms/step	-	accuracy: 0.9391	-	loss: 0.1976
167/521		12s	36ms/step	-	accuracy: 0.9391	-	loss: 0.1975
169/521		12s	36ms/step	-	accuracy: 0.9392	-	loss: 0.1973
171/521		12s	36ms/step	-	accuracy: 0.9392	-	loss: 0.1972
173/521		12s	36ms/step	-	accuracy: 0.9393	-	loss: 0.1970
175/521		12s	36ms/step	-	accuracy: 0.9393	-	loss: 0.1968
177/521		12s	36ms/step	-	accuracy: 0.9393	-	loss: 0.1966
179/521		12s	36ms/step	-	accuracy: 0.9394	-	loss: 0.1965
181/521		12s	36ms/step	-	accuracy: 0.9394	-	loss: 0.1963
183/521		12s	36ms/step	-	accuracy: 0.9395	-	loss: 0.1961
185/521		12s	36ms/step	-	accuracy: 0.9395	-	loss: 0.1959
187/521		12s	36ms/step	-	accuracy: 0.9396	-	loss: 0.1957

189/521		12s	36ms/step	- accuracy: 0.9396	- loss: 0.1955
191/521		11s	36ms/step	- accuracy: 0.9397	- loss: 0.1953
193/521		11s	36ms/step	- accuracy: 0.9397	- loss: 0.1952
195/521		11s	36ms/step	- accuracy: 0.9398	- loss: 0.1950
197/521		11s	36ms/step	- accuracy: 0.9398	- loss: 0.1948
199/521		11s	36ms/step	- accuracy: 0.9399	- loss: 0.1946
201/521		11s	36ms/step	- accuracy: 0.9399	- loss: 0.1944
203/521		11s	36ms/step	- accuracy: 0.9400	- loss: 0.1942
205/521		11s	36ms/step	- accuracy: 0.9400	- loss: 0.1940
207/521		11s	36ms/step	- accuracy: 0.9401	- loss: 0.1938
209/521		11s	36ms/step	- accuracy: 0.9401	- loss: 0.1936
211/521		11s	36ms/step	- accuracy: 0.9402	- loss: 0.1934
213/521		11s	36ms/step	- accuracy: 0.9402	- loss: 0.1933
215/521		11s	36ms/step	- accuracy: 0.9403	- loss: 0.1931
217/521		11s	36ms/step	- accuracy: 0.9403	- loss: 0.1929
219/521		10s	36ms/step	- accuracy: 0.9404	- loss: 0.1927
221/521		10s	36ms/step	- accuracy: 0.9404	- loss: 0.1925
223/521		10s	36ms/step	- accuracy: 0.9405	- loss: 0.1923
225/521		10s	36ms/step	- accuracy: 0.9406	- loss: 0.1921
227/521		10s	36ms/step	- accuracy: 0.9406	- loss: 0.1919
229/521		10s	36ms/step	- accuracy: 0.9407	- loss: 0.1917
231/521		10s	36ms/step	- accuracy: 0.9407	- loss: 0.1915
233/521		10s	36ms/step	- accuracy: 0.9408	- loss: 0.1913
235/521		10s	36ms/step	- accuracy: 0.9408	- loss: 0.1911
237/521		10s	36ms/step	- accuracy: 0.9409	- loss: 0.1909
239/521		10s	36ms/step	- accuracy: 0.9409	- loss: 0.1907
241/521		10s	36ms/step	- accuracy: 0.9410	- loss: 0.1905
243/521		10s	36ms/step	- accuracy: 0.9410	- loss: 0.1903

245/521		10s	36ms/step	-	accuracy: 0.9411	-	loss: 0.1901
247/521		9s	36ms/step	-	accuracy: 0.9411	-	loss: 0.1899
249/521		9s	36ms/step	-	accuracy: 0.9412	-	loss: 0.1897
251/521		9s	36ms/step	-	accuracy: 0.9412	-	loss: 0.1895
253/521		9s	36ms/step	-	accuracy: 0.9413	-	loss: 0.1892
255/521		9s	36ms/step	-	accuracy: 0.9413	-	loss: 0.1890
257/521		9s	36ms/step	-	accuracy: 0.9414	-	loss: 0.1888
259/521		9s	36ms/step	-	accuracy: 0.9415	-	loss: 0.1886
261/521		9s	36ms/step	-	accuracy: 0.9415	-	loss: 0.1884
263/521		9s	36ms/step	-	accuracy: 0.9416	-	loss: 0.1882
265/521		9s	36ms/step	-	accuracy: 0.9416	-	loss: 0.1880
267/521		9s	36ms/step	-	accuracy: 0.9417	-	loss: 0.1878
269/521		9s	36ms/step	-	accuracy: 0.9417	-	loss: 0.1876
271/521		9s	36ms/step	-	accuracy: 0.9418	-	loss: 0.1874
273/521		8s	36ms/step	-	accuracy: 0.9419	-	loss: 0.1872
275/521		8s	36ms/step	-	accuracy: 0.9419	-	loss: 0.1870
277/521		8s	36ms/step	-	accuracy: 0.9420	-	loss: 0.1868
279/521		8s	36ms/step	-	accuracy: 0.9420	-	loss: 0.1866
281/521		8s	36ms/step	-	accuracy: 0.9421	-	loss: 0.1864
283/521		8s	36ms/step	-	accuracy: 0.9421	-	loss: 0.1862
285/521		8s	36ms/step	-	accuracy: 0.9422	-	loss: 0.1860
287/521		8s	36ms/step	-	accuracy: 0.9423	-	loss: 0.1858
289/521		8s	36ms/step	-	accuracy: 0.9423	-	loss: 0.1856
291/521		8s	36ms/step	-	accuracy: 0.9424	-	loss: 0.1854
293/521		8s	36ms/step	-	accuracy: 0.9424	-	loss: 0.1851
295/521		8s	36ms/step	-	accuracy: 0.9425	-	loss: 0.1849
297/521		8s	36ms/step	-	accuracy: 0.9425	-	loss: 0.1847
299/521		8s	36ms/step	-	accuracy: 0.9426	-	loss: 0.1845

301/521		7s 36ms/step	- accuracy: 0.9427	- loss: 0.1843
303/521		7s 36ms/step	- accuracy: 0.9427	- loss: 0.1841
305/521		7s 36ms/step	- accuracy: 0.9428	- loss: 0.1839
307/521		7s 36ms/step	- accuracy: 0.9428	- loss: 0.1837
309/521		7s 36ms/step	- accuracy: 0.9429	- loss: 0.1835
311/521		7s 36ms/step	- accuracy: 0.9429	- loss: 0.1834
313/521		7s 36ms/step	- accuracy: 0.9430	- loss: 0.1832
315/521		7s 36ms/step	- accuracy: 0.9430	- loss: 0.1830
317/521		7s 36ms/step	- accuracy: 0.9431	- loss: 0.1828
319/521		7s 36ms/step	- accuracy: 0.9431	- loss: 0.1826
321/521		7s 36ms/step	- accuracy: 0.9432	- loss: 0.1824
323/521		7s 36ms/step	- accuracy: 0.9432	- loss: 0.1822
325/521		7s 36ms/step	- accuracy: 0.9433	- loss: 0.1820
327/521		7s 36ms/step	- accuracy: 0.9434	- loss: 0.1819
329/521		6s 36ms/step	- accuracy: 0.9434	- loss: 0.1817
331/521		6s 36ms/step	- accuracy: 0.9435	- loss: 0.1815
333/521		6s 36ms/step	- accuracy: 0.9435	- loss: 0.1813
335/521		6s 36ms/step	- accuracy: 0.9436	- loss: 0.1811
337/521		6s 36ms/step	- accuracy: 0.9436	- loss: 0.1809
339/521		6s 36ms/step	- accuracy: 0.9437	- loss: 0.1807
341/521		6s 36ms/step	- accuracy: 0.9437	- loss: 0.1805
343/521		6s 36ms/step	- accuracy: 0.9438	- loss: 0.1803
345/521		6s 36ms/step	- accuracy: 0.9438	- loss: 0.1802
347/521		6s 36ms/step	- accuracy: 0.9439	- loss: 0.1800
349/521		6s 36ms/step	- accuracy: 0.9439	- loss: 0.1798
351/521		6s 36ms/step	- accuracy: 0.9440	- loss: 0.1796
353/521		6s 36ms/step	- accuracy: 0.9440	- loss: 0.1794
355/521		6s 36ms/step	- accuracy: 0.9441	- loss: 0.1792

357/521		5s 36ms/step	- accuracy: 0.9441	- loss: 0.1791
359/521		5s 36ms/step	- accuracy: 0.9442	- loss: 0.1789
361/521		5s 36ms/step	- accuracy: 0.9442	- loss: 0.1787
363/521		5s 36ms/step	- accuracy: 0.9443	- loss: 0.1785
365/521		5s 36ms/step	- accuracy: 0.9443	- loss: 0.1783
367/521		5s 36ms/step	- accuracy: 0.9444	- loss: 0.1782
369/521		5s 36ms/step	- accuracy: 0.9444	- loss: 0.1780
371/521		5s 36ms/step	- accuracy: 0.9445	- loss: 0.1778
373/521		5s 36ms/step	- accuracy: 0.9445	- loss: 0.1776
375/521		5s 36ms/step	- accuracy: 0.9446	- loss: 0.1774
377/521		5s 36ms/step	- accuracy: 0.9446	- loss: 0.1773
379/521		5s 36ms/step	- accuracy: 0.9447	- loss: 0.1771
381/521		5s 36ms/step	- accuracy: 0.9447	- loss: 0.1769
383/521		5s 36ms/step	- accuracy: 0.9448	- loss: 0.1767
385/521		4s 36ms/step	- accuracy: 0.9448	- loss: 0.1766
387/521		4s 36ms/step	- accuracy: 0.9449	- loss: 0.1764
389/521		4s 36ms/step	- accuracy: 0.9449	- loss: 0.1762
391/521		4s 36ms/step	- accuracy: 0.9450	- loss: 0.1760
393/521		4s 36ms/step	- accuracy: 0.9450	- loss: 0.1759
395/521		4s 36ms/step	- accuracy: 0.9451	- loss: 0.1757
397/521		4s 36ms/step	- accuracy: 0.9451	- loss: 0.1755
399/521		4s 36ms/step	- accuracy: 0.9452	- loss: 0.1753
401/521		4s 36ms/step	- accuracy: 0.9452	- loss: 0.1752
403/521		4s 36ms/step	- accuracy: 0.9453	- loss: 0.1750
405/521		4s 36ms/step	- accuracy: 0.9453	- loss: 0.1748
407/521		4s 36ms/step	- accuracy: 0.9454	- loss: 0.1747
409/521		4s 36ms/step	- accuracy: 0.9455	- loss: 0.1745
411/521		3s 36ms/step	- accuracy: 0.9455	- loss: 0.1743

413/521		3s	36ms/step	- accuracy: 0.9456	- loss: 0.1741
415/521		3s	36ms/step	- accuracy: 0.9456	- loss: 0.1740
417/521		3s	36ms/step	- accuracy: 0.9457	- loss: 0.1738
419/521		3s	36ms/step	- accuracy: 0.9457	- loss: 0.1736
421/521		3s	36ms/step	- accuracy: 0.9458	- loss: 0.1735
423/521		3s	36ms/step	- accuracy: 0.9458	- loss: 0.1733
425/521		3s	36ms/step	- accuracy: 0.9459	- loss: 0.1731
427/521		3s	36ms/step	- accuracy: 0.9459	- loss: 0.1730
429/521		3s	36ms/step	- accuracy: 0.9460	- loss: 0.1728
431/521		3s	36ms/step	- accuracy: 0.9460	- loss: 0.1726
433/521		3s	36ms/step	- accuracy: 0.9460	- loss: 0.1725
435/521		3s	36ms/step	- accuracy: 0.9461	- loss: 0.1723
437/521		3s	36ms/step	- accuracy: 0.9461	- loss: 0.1722
439/521		2s	36ms/step	- accuracy: 0.9462	- loss: 0.1720
441/521		2s	36ms/step	- accuracy: 0.9462	- loss: 0.1718
443/521		2s	36ms/step	- accuracy: 0.9463	- loss: 0.1717
445/521		2s	36ms/step	- accuracy: 0.9463	- loss: 0.1715
447/521		2s	36ms/step	- accuracy: 0.9464	- loss: 0.1713
449/521		2s	36ms/step	- accuracy: 0.9464	- loss: 0.1712
451/521		2s	36ms/step	- accuracy: 0.9465	- loss: 0.1710
453/521		2s	36ms/step	- accuracy: 0.9465	- loss: 0.1709
455/521		2s	36ms/step	- accuracy: 0.9466	- loss: 0.1707
457/521		2s	36ms/step	- accuracy: 0.9466	- loss: 0.1705
459/521		2s	36ms/step	- accuracy: 0.9467	- loss: 0.1704
461/521		2s	36ms/step	- accuracy: 0.9467	- loss: 0.1702
463/521		2s	36ms/step	- accuracy: 0.9468	- loss: 0.1701
465/521		2s	36ms/step	- accuracy: 0.9468	- loss: 0.1699
467/521		1s	36ms/step	- accuracy: 0.9469	- loss: 0.1697

469/521		1s 36ms/step	- accuracy: 0.9469	- loss: 0.1696
471/521		1s 36ms/step	- accuracy: 0.9470	- loss: 0.1694
473/521		1s 36ms/step	- accuracy: 0.9470	- loss: 0.1693
474/521		1s 36ms/step	- accuracy: 0.9470	- loss: 0.1692
476/521		1s 36ms/step	- accuracy: 0.9471	- loss: 0.1690
478/521		1s 36ms/step	- accuracy: 0.9471	- loss: 0.1689
480/521		1s 36ms/step	- accuracy: 0.9472	- loss: 0.1687
482/521		1s 36ms/step	- accuracy: 0.9472	- loss: 0.1686
484/521		1s 36ms/step	- accuracy: 0.9473	- loss: 0.1684
486/521		1s 36ms/step	- accuracy: 0.9473	- loss: 0.1683
488/521		1s 36ms/step	- accuracy: 0.9474	- loss: 0.1681
490/521		1s 36ms/step	- accuracy: 0.9474	- loss: 0.1680
492/521		1s 36ms/step	- accuracy: 0.9474	- loss: 0.1678
494/521		0s 36ms/step	- accuracy: 0.9475	- loss: 0.1677
496/521		0s 36ms/step	- accuracy: 0.9475	- loss: 0.1675
498/521		0s 36ms/step	- accuracy: 0.9476	- loss: 0.1674
500/521		0s 36ms/step	- accuracy: 0.9476	- loss: 0.1672
502/521		0s 36ms/step	- accuracy: 0.9477	- loss: 0.1671
504/521		0s 36ms/step	- accuracy: 0.9477	- loss: 0.1669
506/521		0s 36ms/step	- accuracy: 0.9478	- loss: 0.1668
508/521		0s 36ms/step	- accuracy: 0.9478	- loss: 0.1666
510/521		0s 36ms/step	- accuracy: 0.9479	- loss: 0.1665
512/521		0s 36ms/step	- accuracy: 0.9479	- loss: 0.1663
514/521		0s 36ms/step	- accuracy: 0.9479	- loss: 0.1662
516/521		0s 36ms/step	- accuracy: 0.9480	- loss: 0.1661
518/521		0s 36ms/step	- accuracy: 0.9480	- loss: 0.1659
520/521		0s 36ms/step	- accuracy: 0.9481	- loss: 0.1658
521/521		0s 36ms/step	- accuracy: 0.9481	- loss: 0.1657

521/521  20s 38ms/step - accuracy: 0.9594 - loss: 0.1289
- val_accuracy: 0.9908 - val_loss: 0.0324 - learning_rate: 0.0010
Epoch 3/25

1/521		27s 52ms/step - accuracy: 0.9531 - loss: 0.2011
3/521		19s 37ms/step - accuracy: 0.9670 - loss: 0.1429
5/521		18s 37ms/step - accuracy: 0.9733 - loss: 0.1154
7/521		18s 36ms/step - accuracy: 0.9758 - loss: 0.1028
9/521		18s 36ms/step - accuracy: 0.9759 - loss: 0.0999
11/521		18s 36ms/step - accuracy: 0.9757 - loss: 0.0976
13/521		18s 36ms/step - accuracy: 0.9759 - loss: 0.0952
15/521		18s 36ms/step - accuracy: 0.9762 - loss: 0.0933
17/521		18s 36ms/step - accuracy: 0.9762 - loss: 0.0925
19/521		18s 36ms/step - accuracy: 0.9762 - loss: 0.0920
21/521		18s 36ms/step - accuracy: 0.9762 - loss: 0.0915
23/521		17s 36ms/step - accuracy: 0.9761 - loss: 0.0912
25/521		17s 36ms/step - accuracy: 0.9759 - loss: 0.0913
27/521		17s 36ms/step - accuracy: 0.9758 - loss: 0.0911
29/521		17s 36ms/step - accuracy: 0.9757 - loss: 0.0913
31/521		17s 36ms/step - accuracy: 0.9756 - loss: 0.0913
33/521		17s 36ms/step - accuracy: 0.9754 - loss: 0.0913
35/521		17s 36ms/step - accuracy: 0.9753 - loss: 0.0915
37/521		17s 36ms/step - accuracy: 0.9752 - loss: 0.0915
39/521		17s 36ms/step - accuracy: 0.9751 - loss: 0.0914
41/521		17s 36ms/step - accuracy: 0.9750 - loss: 0.0915
43/521		17s 36ms/step - accuracy: 0.9750 - loss: 0.0914
45/521		17s 36ms/step - accuracy: 0.9751 - loss: 0.0912
47/521		17s 36ms/step - accuracy: 0.9751 - loss: 0.0910
49/521		17s 36ms/step - accuracy: 0.9751 - loss: 0.0909
51/521		17s 36ms/step - accuracy: 0.9751 - loss: 0.0909

53/521		17s	36ms/step	-	accuracy: 0.9751	-	loss: 0.0908
55/521		16s	36ms/step	-	accuracy: 0.9751	-	loss: 0.0908
57/521		16s	36ms/step	-	accuracy: 0.9752	-	loss: 0.0907
59/521		16s	36ms/step	-	accuracy: 0.9752	-	loss: 0.0907
61/521		16s	37ms/step	-	accuracy: 0.9752	-	loss: 0.0906
63/521		16s	37ms/step	-	accuracy: 0.9752	-	loss: 0.0905
65/521		16s	37ms/step	-	accuracy: 0.9752	-	loss: 0.0904
67/521		16s	37ms/step	-	accuracy: 0.9752	-	loss: 0.0903
69/521		16s	37ms/step	-	accuracy: 0.9752	-	loss: 0.0902
71/521		16s	37ms/step	-	accuracy: 0.9753	-	loss: 0.0901
73/521		16s	37ms/step	-	accuracy: 0.9753	-	loss: 0.0900
75/521		16s	37ms/step	-	accuracy: 0.9753	-	loss: 0.0900
77/521		16s	37ms/step	-	accuracy: 0.9753	-	loss: 0.0899
79/521		16s	37ms/step	-	accuracy: 0.9753	-	loss: 0.0898
81/521		16s	37ms/step	-	accuracy: 0.9754	-	loss: 0.0897
83/521		16s	37ms/step	-	accuracy: 0.9754	-	loss: 0.0896
85/521		15s	37ms/step	-	accuracy: 0.9754	-	loss: 0.0895
87/521		15s	37ms/step	-	accuracy: 0.9754	-	loss: 0.0894
89/521		15s	37ms/step	-	accuracy: 0.9754	-	loss: 0.0892
91/521		15s	37ms/step	-	accuracy: 0.9755	-	loss: 0.0891
93/521		15s	37ms/step	-	accuracy: 0.9755	-	loss: 0.0890
95/521		15s	37ms/step	-	accuracy: 0.9755	-	loss: 0.0889
97/521		15s	37ms/step	-	accuracy: 0.9755	-	loss: 0.0888
99/521		15s	37ms/step	-	accuracy: 0.9755	-	loss: 0.0888
101/521		15s	37ms/step	-	accuracy: 0.9755	-	loss: 0.0887
103/521		15s	37ms/step	-	accuracy: 0.9755	-	loss: 0.0885
105/521		15s	37ms/step	-	accuracy: 0.9756	-	loss: 0.0884
107/521		15s	37ms/step	-	accuracy: 0.9756	-	loss: 0.0883

109/521		15s 37ms/step	- accuracy: 0.9756	- loss: 0.0882
111/521		15s 37ms/step	- accuracy: 0.9756	- loss: 0.0881
113/521		14s 37ms/step	- accuracy: 0.9756	- loss: 0.0880
115/521		14s 37ms/step	- accuracy: 0.9756	- loss: 0.0879
117/521		14s 37ms/step	- accuracy: 0.9757	- loss: 0.0878
119/521		14s 37ms/step	- accuracy: 0.9757	- loss: 0.0877
121/521		14s 37ms/step	- accuracy: 0.9757	- loss: 0.0876
123/521		14s 37ms/step	- accuracy: 0.9757	- loss: 0.0875
125/521		14s 37ms/step	- accuracy: 0.9757	- loss: 0.0874
127/521		14s 37ms/step	- accuracy: 0.9758	- loss: 0.0873
129/521		14s 37ms/step	- accuracy: 0.9758	- loss: 0.0872
131/521		14s 37ms/step	- accuracy: 0.9758	- loss: 0.0871
133/521		14s 37ms/step	- accuracy: 0.9758	- loss: 0.0869
135/521		14s 37ms/step	- accuracy: 0.9759	- loss: 0.0868
137/521		14s 37ms/step	- accuracy: 0.9759	- loss: 0.0867
139/521		14s 37ms/step	- accuracy: 0.9759	- loss: 0.0866
141/521		13s 37ms/step	- accuracy: 0.9759	- loss: 0.0865
143/521		13s 37ms/step	- accuracy: 0.9759	- loss: 0.0864
145/521		13s 37ms/step	- accuracy: 0.9759	- loss: 0.0863
147/521		13s 37ms/step	- accuracy: 0.9759	- loss: 0.0862
149/521		13s 37ms/step	- accuracy: 0.9760	- loss: 0.0861
151/521		13s 37ms/step	- accuracy: 0.9760	- loss: 0.0860
153/521		13s 37ms/step	- accuracy: 0.9760	- loss: 0.0859
155/521		13s 37ms/step	- accuracy: 0.9760	- loss: 0.0858
157/521		13s 37ms/step	- accuracy: 0.9760	- loss: 0.0857
159/521		13s 37ms/step	- accuracy: 0.9760	- loss: 0.0856
161/521		13s 37ms/step	- accuracy: 0.9761	- loss: 0.0855
163/521		13s 37ms/step	- accuracy: 0.9761	- loss: 0.0854

165/521		13s	37ms/step	-	accuracy: 0.9761	-	loss: 0.0853
167/521		13s	37ms/step	-	accuracy: 0.9761	-	loss: 0.0852
169/521		12s	37ms/step	-	accuracy: 0.9761	-	loss: 0.0851
171/521		12s	37ms/step	-	accuracy: 0.9761	-	loss: 0.0850
173/521		12s	37ms/step	-	accuracy: 0.9762	-	loss: 0.0849
175/521		12s	37ms/step	-	accuracy: 0.9762	-	loss: 0.0848
177/521		12s	37ms/step	-	accuracy: 0.9762	-	loss: 0.0847
179/521		12s	37ms/step	-	accuracy: 0.9762	-	loss: 0.0846
181/521		12s	37ms/step	-	accuracy: 0.9762	-	loss: 0.0845
183/521		12s	37ms/step	-	accuracy: 0.9763	-	loss: 0.0844
185/521		12s	37ms/step	-	accuracy: 0.9763	-	loss: 0.0843
187/521		12s	37ms/step	-	accuracy: 0.9763	-	loss: 0.0842
189/521		12s	37ms/step	-	accuracy: 0.9763	-	loss: 0.0841
191/521		12s	37ms/step	-	accuracy: 0.9764	-	loss: 0.0840
193/521		12s	37ms/step	-	accuracy: 0.9764	-	loss: 0.0839
195/521		11s	37ms/step	-	accuracy: 0.9764	-	loss: 0.0838
197/521		11s	37ms/step	-	accuracy: 0.9764	-	loss: 0.0837
199/521		11s	37ms/step	-	accuracy: 0.9764	-	loss: 0.0836
201/521		11s	37ms/step	-	accuracy: 0.9765	-	loss: 0.0835
203/521		11s	37ms/step	-	accuracy: 0.9765	-	loss: 0.0834
205/521		11s	37ms/step	-	accuracy: 0.9765	-	loss: 0.0833
207/521		11s	37ms/step	-	accuracy: 0.9765	-	loss: 0.0832
209/521		11s	37ms/step	-	accuracy: 0.9765	-	loss: 0.0831
211/521		11s	37ms/step	-	accuracy: 0.9766	-	loss: 0.0831
213/521		11s	37ms/step	-	accuracy: 0.9766	-	loss: 0.0830
215/521		11s	37ms/step	-	accuracy: 0.9766	-	loss: 0.0829
217/521		11s	37ms/step	-	accuracy: 0.9766	-	loss: 0.0828
219/521		11s	37ms/step	-	accuracy: 0.9766	-	loss: 0.0827

221/521		11s 37ms/step	- accuracy: 0.9766	- loss: 0.0826
223/521		10s 37ms/step	- accuracy: 0.9766	- loss: 0.0825
225/521		10s 37ms/step	- accuracy: 0.9767	- loss: 0.0825
227/521		10s 37ms/step	- accuracy: 0.9767	- loss: 0.0824
229/521		10s 37ms/step	- accuracy: 0.9767	- loss: 0.0823
231/521		10s 37ms/step	- accuracy: 0.9767	- loss: 0.0822
233/521		10s 37ms/step	- accuracy: 0.9767	- loss: 0.0821
235/521		10s 37ms/step	- accuracy: 0.9767	- loss: 0.0821
237/521		10s 37ms/step	- accuracy: 0.9768	- loss: 0.0820
239/521		10s 37ms/step	- accuracy: 0.9768	- loss: 0.0819
241/521		10s 37ms/step	- accuracy: 0.9768	- loss: 0.0818
243/521		10s 37ms/step	- accuracy: 0.9768	- loss: 0.0817
245/521		10s 37ms/step	- accuracy: 0.9768	- loss: 0.0817
247/521		10s 37ms/step	- accuracy: 0.9768	- loss: 0.0816
249/521		9s 37ms/step	- accuracy: 0.9769	- loss: 0.0815
251/521		9s 37ms/step	- accuracy: 0.9769	- loss: 0.0815
253/521		9s 37ms/step	- accuracy: 0.9769	- loss: 0.0814
255/521		9s 37ms/step	- accuracy: 0.9769	- loss: 0.0813
257/521		9s 37ms/step	- accuracy: 0.9769	- loss: 0.0813
259/521		9s 37ms/step	- accuracy: 0.9769	- loss: 0.0812
261/521		9s 37ms/step	- accuracy: 0.9769	- loss: 0.0811
263/521		9s 37ms/step	- accuracy: 0.9770	- loss: 0.0811
265/521		9s 37ms/step	- accuracy: 0.9770	- loss: 0.0810
267/521		9s 37ms/step	- accuracy: 0.9770	- loss: 0.0809
269/521		9s 37ms/step	- accuracy: 0.9770	- loss: 0.0809
271/521		9s 37ms/step	- accuracy: 0.9770	- loss: 0.0808
273/521		9s 37ms/step	- accuracy: 0.9770	- loss: 0.0807
275/521		9s 37ms/step	- accuracy: 0.9770	- loss: 0.0807

277/521		8s 37ms/step	- accuracy: 0.9770	- loss: 0.0806
279/521		8s 37ms/step	- accuracy: 0.9771	- loss: 0.0806
281/521		8s 37ms/step	- accuracy: 0.9771	- loss: 0.0805
283/521		8s 37ms/step	- accuracy: 0.9771	- loss: 0.0804
285/521		8s 37ms/step	- accuracy: 0.9771	- loss: 0.0804
287/521		8s 37ms/step	- accuracy: 0.9771	- loss: 0.0803
289/521		8s 37ms/step	- accuracy: 0.9771	- loss: 0.0802
291/521		8s 37ms/step	- accuracy: 0.9771	- loss: 0.0802
293/521		8s 37ms/step	- accuracy: 0.9772	- loss: 0.0801
295/521		8s 37ms/step	- accuracy: 0.9772	- loss: 0.0800
297/521		8s 37ms/step	- accuracy: 0.9772	- loss: 0.0800
299/521		8s 37ms/step	- accuracy: 0.9772	- loss: 0.0799
301/521		8s 37ms/step	- accuracy: 0.9772	- loss: 0.0799
303/521		8s 37ms/step	- accuracy: 0.9772	- loss: 0.0798
305/521		7s 37ms/step	- accuracy: 0.9772	- loss: 0.0797
307/521		7s 37ms/step	- accuracy: 0.9772	- loss: 0.0797
309/521		7s 37ms/step	- accuracy: 0.9773	- loss: 0.0796
311/521		7s 37ms/step	- accuracy: 0.9773	- loss: 0.0795
313/521		7s 37ms/step	- accuracy: 0.9773	- loss: 0.0795
315/521		7s 37ms/step	- accuracy: 0.9773	- loss: 0.0794
317/521		7s 37ms/step	- accuracy: 0.9773	- loss: 0.0794
319/521		7s 37ms/step	- accuracy: 0.9773	- loss: 0.0793
321/521		7s 37ms/step	- accuracy: 0.9773	- loss: 0.0792
323/521		7s 37ms/step	- accuracy: 0.9774	- loss: 0.0792
325/521		7s 37ms/step	- accuracy: 0.9774	- loss: 0.0791
327/521		7s 37ms/step	- accuracy: 0.9774	- loss: 0.0791
329/521		7s 37ms/step	- accuracy: 0.9774	- loss: 0.0790
331/521		6s 37ms/step	- accuracy: 0.9774	- loss: 0.0789

333/521		6s 37ms/step - accuracy: 0.9774 - loss: 0.0789
335/521		6s 37ms/step - accuracy: 0.9774 - loss: 0.0788
337/521		6s 37ms/step - accuracy: 0.9775 - loss: 0.0787
339/521		6s 37ms/step - accuracy: 0.9775 - loss: 0.0787
341/521		6s 37ms/step - accuracy: 0.9775 - loss: 0.0786
343/521		6s 37ms/step - accuracy: 0.9775 - loss: 0.0786
345/521		6s 37ms/step - accuracy: 0.9775 - loss: 0.0785
347/521		6s 37ms/step - accuracy: 0.9775 - loss: 0.0784
349/521		6s 37ms/step - accuracy: 0.9775 - loss: 0.0784
351/521		6s 37ms/step - accuracy: 0.9775 - loss: 0.0783
353/521		6s 37ms/step - accuracy: 0.9776 - loss: 0.0783
355/521		6s 37ms/step - accuracy: 0.9776 - loss: 0.0782
357/521		6s 37ms/step - accuracy: 0.9776 - loss: 0.0782
359/521		5s 37ms/step - accuracy: 0.9776 - loss: 0.0781
361/521		5s 37ms/step - accuracy: 0.9776 - loss: 0.0780
363/521		5s 37ms/step - accuracy: 0.9776 - loss: 0.0780
365/521		5s 37ms/step - accuracy: 0.9776 - loss: 0.0779
367/521		5s 37ms/step - accuracy: 0.9776 - loss: 0.0779
369/521		5s 37ms/step - accuracy: 0.9777 - loss: 0.0778
371/521		5s 37ms/step - accuracy: 0.9777 - loss: 0.0778
373/521		5s 37ms/step - accuracy: 0.9777 - loss: 0.0777
375/521		5s 37ms/step - accuracy: 0.9777 - loss: 0.0777
377/521		5s 37ms/step - accuracy: 0.9777 - loss: 0.0776
379/521		5s 37ms/step - accuracy: 0.9777 - loss: 0.0776
381/521		5s 37ms/step - accuracy: 0.9777 - loss: 0.0775
383/521		5s 37ms/step - accuracy: 0.9778 - loss: 0.0774
385/521		4s 37ms/step - accuracy: 0.9778 - loss: 0.0774
387/521		4s 37ms/step - accuracy: 0.9778 - loss: 0.0773

389/521		4s 37ms/step	- accuracy: 0.9778	- loss: 0.0773
391/521		4s 37ms/step	- accuracy: 0.9778	- loss: 0.0772
393/521		4s 37ms/step	- accuracy: 0.9778	- loss: 0.0772
395/521		4s 37ms/step	- accuracy: 0.9778	- loss: 0.0771
397/521		4s 37ms/step	- accuracy: 0.9778	- loss: 0.0771
399/521		4s 37ms/step	- accuracy: 0.9779	- loss: 0.0770
401/521		4s 37ms/step	- accuracy: 0.9779	- loss: 0.0770
403/521		4s 37ms/step	- accuracy: 0.9779	- loss: 0.0769
405/521		4s 37ms/step	- accuracy: 0.9779	- loss: 0.0768
407/521		4s 37ms/step	- accuracy: 0.9779	- loss: 0.0768
409/521		4s 37ms/step	- accuracy: 0.9779	- loss: 0.0767
411/521		4s 37ms/step	- accuracy: 0.9779	- loss: 0.0767
413/521		3s 37ms/step	- accuracy: 0.9779	- loss: 0.0766
415/521		3s 37ms/step	- accuracy: 0.9780	- loss: 0.0766
417/521		3s 37ms/step	- accuracy: 0.9780	- loss: 0.0765
419/521		3s 37ms/step	- accuracy: 0.9780	- loss: 0.0765
421/521		3s 37ms/step	- accuracy: 0.9780	- loss: 0.0764
423/521		3s 37ms/step	- accuracy: 0.9780	- loss: 0.0764
425/521		3s 37ms/step	- accuracy: 0.9780	- loss: 0.0763
427/521		3s 37ms/step	- accuracy: 0.9780	- loss: 0.0763
429/521		3s 37ms/step	- accuracy: 0.9780	- loss: 0.0762
431/521		3s 37ms/step	- accuracy: 0.9781	- loss: 0.0762
433/521		3s 37ms/step	- accuracy: 0.9781	- loss: 0.0761
435/521		3s 37ms/step	- accuracy: 0.9781	- loss: 0.0761
437/521		3s 37ms/step	- accuracy: 0.9781	- loss: 0.0760
439/521		3s 37ms/step	- accuracy: 0.9781	- loss: 0.0760
441/521		2s 37ms/step	- accuracy: 0.9781	- loss: 0.0759
443/521		2s 37ms/step	- accuracy: 0.9781	- loss: 0.0759

445/521		2s	37ms/step	-	accuracy: 0.9781	-	loss: 0.0758
447/521		2s	37ms/step	-	accuracy: 0.9782	-	loss: 0.0758
448/521		2s	37ms/step	-	accuracy: 0.9782	-	loss: 0.0757
449/521		2s	37ms/step	-	accuracy: 0.9782	-	loss: 0.0757
451/521		2s	37ms/step	-	accuracy: 0.9782	-	loss: 0.0757
453/521		2s	37ms/step	-	accuracy: 0.9782	-	loss: 0.0756
455/521		2s	37ms/step	-	accuracy: 0.9782	-	loss: 0.0756
457/521		2s	37ms/step	-	accuracy: 0.9782	-	loss: 0.0755
459/521		2s	37ms/step	-	accuracy: 0.9782	-	loss: 0.0755
461/521		2s	37ms/step	-	accuracy: 0.9782	-	loss: 0.0754
463/521		2s	37ms/step	-	accuracy: 0.9782	-	loss: 0.0754
465/521		2s	37ms/step	-	accuracy: 0.9783	-	loss: 0.0753
467/521		1s	37ms/step	-	accuracy: 0.9783	-	loss: 0.0753
469/521		1s	37ms/step	-	accuracy: 0.9783	-	loss: 0.0753
471/521		1s	37ms/step	-	accuracy: 0.9783	-	loss: 0.0752
473/521		1s	37ms/step	-	accuracy: 0.9783	-	loss: 0.0752
475/521		1s	37ms/step	-	accuracy: 0.9783	-	loss: 0.0751
477/521		1s	37ms/step	-	accuracy: 0.9783	-	loss: 0.0751
479/521		1s	37ms/step	-	accuracy: 0.9783	-	loss: 0.0750
481/521		1s	37ms/step	-	accuracy: 0.9783	-	loss: 0.0750
483/521		1s	37ms/step	-	accuracy: 0.9783	-	loss: 0.0749
485/521		1s	37ms/step	-	accuracy: 0.9784	-	loss: 0.0749
487/521		1s	37ms/step	-	accuracy: 0.9784	-	loss: 0.0749
489/521		1s	37ms/step	-	accuracy: 0.9784	-	loss: 0.0748
491/521		1s	37ms/step	-	accuracy: 0.9784	-	loss: 0.0748
493/521		1s	37ms/step	-	accuracy: 0.9784	-	loss: 0.0747
495/521		0s	37ms/step	-	accuracy: 0.9784	-	loss: 0.0747
497/521		0s	37ms/step	-	accuracy: 0.9784	-	loss: 0.0746

499/521		0s 37ms/step	- accuracy: 0.9784	- loss: 0.0746
501/521		0s 37ms/step	- accuracy: 0.9784	- loss: 0.0746
503/521		0s 37ms/step	- accuracy: 0.9785	- loss: 0.0745
505/521		0s 37ms/step	- accuracy: 0.9785	- loss: 0.0745
507/521		0s 37ms/step	- accuracy: 0.9785	- loss: 0.0744
509/521		0s 37ms/step	- accuracy: 0.9785	- loss: 0.0744
511/521		0s 37ms/step	- accuracy: 0.9785	- loss: 0.0744
513/521		0s 37ms/step	- accuracy: 0.9785	- loss: 0.0743
514/521		0s 37ms/step	- accuracy: 0.9785	- loss: 0.0743
516/521		0s 37ms/step	- accuracy: 0.9785	- loss: 0.0743
518/521		0s 37ms/step	- accuracy: 0.9785	- loss: 0.0742
520/521		0s 37ms/step	- accuracy: 0.9785	- loss: 0.0742
521/521		20s 39ms/step	- accuracy: 0.9810	- loss: 0.0642
- val_accuracy: 0.9932 - val_loss: 0.0251 - learning_rate: 0.0010				
Epoch 4/25				
1/521		28s 55ms/step	- accuracy: 0.9688	- loss: 0.0834
3/521		18s 36ms/step	- accuracy: 0.9679	- loss: 0.0860
5/521		18s 36ms/step	- accuracy: 0.9737	- loss: 0.0733
7/521		18s 36ms/step	- accuracy: 0.9764	- loss: 0.0668
9/521		18s 36ms/step	- accuracy: 0.9776	- loss: 0.0643
11/521		18s 36ms/step	- accuracy: 0.9787	- loss: 0.0620
13/521		18s 36ms/step	- accuracy: 0.9793	- loss: 0.0601
15/521		18s 37ms/step	- accuracy: 0.9797	- loss: 0.0586
17/521		18s 37ms/step	- accuracy: 0.9800	- loss: 0.0572
19/521		18s 37ms/step	- accuracy: 0.9805	- loss: 0.0560
21/521		18s 37ms/step	- accuracy: 0.9807	- loss: 0.0551
23/521		18s 37ms/step	- accuracy: 0.9810	- loss: 0.0544
25/521		18s 37ms/step	- accuracy: 0.9812	- loss: 0.0538
27/521		18s 37ms/step	- accuracy: 0.9815	- loss: 0.0531

29/521		17s	37ms/step	-	accuracy: 0.9817	-	loss: 0.0525
31/521		17s	37ms/step	-	accuracy: 0.9819	-	loss: 0.0519
33/521		17s	36ms/step	-	accuracy: 0.9820	-	loss: 0.0514
35/521		17s	36ms/step	-	accuracy: 0.9821	-	loss: 0.0511
37/521		17s	37ms/step	-	accuracy: 0.9821	-	loss: 0.0509
39/521		17s	37ms/step	-	accuracy: 0.9822	-	loss: 0.0506
41/521		17s	37ms/step	-	accuracy: 0.9822	-	loss: 0.0505
43/521		17s	37ms/step	-	accuracy: 0.9823	-	loss: 0.0504
45/521		17s	37ms/step	-	accuracy: 0.9824	-	loss: 0.0502
47/521		17s	37ms/step	-	accuracy: 0.9824	-	loss: 0.0501
49/521		17s	37ms/step	-	accuracy: 0.9825	-	loss: 0.0500
51/521		17s	37ms/step	-	accuracy: 0.9826	-	loss: 0.0499
53/521		17s	37ms/step	-	accuracy: 0.9826	-	loss: 0.0498
55/521		17s	37ms/step	-	accuracy: 0.9827	-	loss: 0.0497
57/521		16s	37ms/step	-	accuracy: 0.9828	-	loss: 0.0496
59/521		16s	36ms/step	-	accuracy: 0.9829	-	loss: 0.0496
61/521		16s	36ms/step	-	accuracy: 0.9830	-	loss: 0.0494
63/521		16s	36ms/step	-	accuracy: 0.9830	-	loss: 0.0493
65/521		16s	36ms/step	-	accuracy: 0.9831	-	loss: 0.0492
67/521		16s	36ms/step	-	accuracy: 0.9832	-	loss: 0.0491
69/521		16s	36ms/step	-	accuracy: 0.9832	-	loss: 0.0490
71/521		16s	36ms/step	-	accuracy: 0.9833	-	loss: 0.0489
73/521		16s	36ms/step	-	accuracy: 0.9834	-	loss: 0.0488
75/521		16s	36ms/step	-	accuracy: 0.9835	-	loss: 0.0487
77/521		16s	36ms/step	-	accuracy: 0.9835	-	loss: 0.0486
79/521		16s	36ms/step	-	accuracy: 0.9836	-	loss: 0.0485
81/521		16s	36ms/step	-	accuracy: 0.9837	-	loss: 0.0484
83/521		15s	37ms/step	-	accuracy: 0.9837	-	loss: 0.0483

85/521		15s	37ms/step	- accuracy: 0.9838	- loss: 0.0481
87/521		15s	36ms/step	- accuracy: 0.9839	- loss: 0.0480
89/521		15s	36ms/step	- accuracy: 0.9839	- loss: 0.0480
91/521		15s	36ms/step	- accuracy: 0.9839	- loss: 0.0479
93/521		15s	37ms/step	- accuracy: 0.9840	- loss: 0.0478
95/521		15s	37ms/step	- accuracy: 0.9840	- loss: 0.0477
97/521		15s	37ms/step	- accuracy: 0.9841	- loss: 0.0476
99/521		15s	37ms/step	- accuracy: 0.9841	- loss: 0.0475
101/521		15s	37ms/step	- accuracy: 0.9841	- loss: 0.0475
103/521		15s	37ms/step	- accuracy: 0.9842	- loss: 0.0474
105/521		15s	37ms/step	- accuracy: 0.9842	- loss: 0.0473
107/521		15s	37ms/step	- accuracy: 0.9843	- loss: 0.0472
109/521		15s	37ms/step	- accuracy: 0.9843	- loss: 0.0471
111/521		14s	37ms/step	- accuracy: 0.9843	- loss: 0.0471
113/521		14s	37ms/step	- accuracy: 0.9844	- loss: 0.0470
115/521		14s	37ms/step	- accuracy: 0.9844	- loss: 0.0470
117/521		14s	37ms/step	- accuracy: 0.9844	- loss: 0.0469
119/521		14s	37ms/step	- accuracy: 0.9844	- loss: 0.0469
121/521		14s	37ms/step	- accuracy: 0.9845	- loss: 0.0469
123/521		14s	37ms/step	- accuracy: 0.9845	- loss: 0.0468
125/521		14s	37ms/step	- accuracy: 0.9845	- loss: 0.0468
127/521		14s	37ms/step	- accuracy: 0.9845	- loss: 0.0467
129/521		14s	37ms/step	- accuracy: 0.9845	- loss: 0.0467
131/521		14s	37ms/step	- accuracy: 0.9845	- loss: 0.0467
133/521		14s	37ms/step	- accuracy: 0.9846	- loss: 0.0466
135/521		14s	37ms/step	- accuracy: 0.9846	- loss: 0.0466
137/521		14s	37ms/step	- accuracy: 0.9846	- loss: 0.0466
139/521		13s	37ms/step	- accuracy: 0.9846	- loss: 0.0465

141/521		13s 37ms/step	- accuracy: 0.9846	- loss: 0.0465
143/521		13s 37ms/step	- accuracy: 0.9846	- loss: 0.0465
145/521		13s 37ms/step	- accuracy: 0.9847	- loss: 0.0464
147/521		13s 37ms/step	- accuracy: 0.9847	- loss: 0.0464
149/521		13s 37ms/step	- accuracy: 0.9847	- loss: 0.0464
151/521		13s 37ms/step	- accuracy: 0.9847	- loss: 0.0464
153/521		13s 37ms/step	- accuracy: 0.9847	- loss: 0.0464
155/521		13s 37ms/step	- accuracy: 0.9847	- loss: 0.0464
157/521		13s 37ms/step	- accuracy: 0.9847	- loss: 0.0464
159/521		13s 37ms/step	- accuracy: 0.9847	- loss: 0.0464
161/521		13s 37ms/step	- accuracy: 0.9848	- loss: 0.0463
163/521		13s 37ms/step	- accuracy: 0.9848	- loss: 0.0463
165/521		13s 37ms/step	- accuracy: 0.9848	- loss: 0.0463
167/521		12s 37ms/step	- accuracy: 0.9848	- loss: 0.0463
169/521		12s 37ms/step	- accuracy: 0.9848	- loss: 0.0463
171/521		12s 37ms/step	- accuracy: 0.9848	- loss: 0.0463
173/521		12s 36ms/step	- accuracy: 0.9848	- loss: 0.0463
175/521		12s 36ms/step	- accuracy: 0.9848	- loss: 0.0462
177/521		12s 36ms/step	- accuracy: 0.9848	- loss: 0.0462
179/521		12s 36ms/step	- accuracy: 0.9848	- loss: 0.0462
181/521		12s 36ms/step	- accuracy: 0.9848	- loss: 0.0462
183/521		12s 36ms/step	- accuracy: 0.9849	- loss: 0.0462
185/521		12s 36ms/step	- accuracy: 0.9849	- loss: 0.0461
187/521		12s 36ms/step	- accuracy: 0.9849	- loss: 0.0461
189/521		12s 36ms/step	- accuracy: 0.9849	- loss: 0.0461
191/521		12s 36ms/step	- accuracy: 0.9849	- loss: 0.0461
193/521		11s 36ms/step	- accuracy: 0.9849	- loss: 0.0461
195/521		11s 36ms/step	- accuracy: 0.9849	- loss: 0.0461

197/521		11s 36ms/step	- accuracy: 0.9849	- loss: 0.0460
199/521		11s 37ms/step	- accuracy: 0.9849	- loss: 0.0460
201/521		11s 37ms/step	- accuracy: 0.9849	- loss: 0.0460
203/521		11s 37ms/step	- accuracy: 0.9849	- loss: 0.0460
205/521		11s 37ms/step	- accuracy: 0.9849	- loss: 0.0460
207/521		11s 37ms/step	- accuracy: 0.9849	- loss: 0.0460
209/521		11s 37ms/step	- accuracy: 0.9849	- loss: 0.0459
211/521		11s 37ms/step	- accuracy: 0.9849	- loss: 0.0459
213/521		11s 37ms/step	- accuracy: 0.9849	- loss: 0.0459
215/521		11s 37ms/step	- accuracy: 0.9850	- loss: 0.0459
217/521		11s 37ms/step	- accuracy: 0.9850	- loss: 0.0459
219/521		11s 37ms/step	- accuracy: 0.9850	- loss: 0.0459
221/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0458
223/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0458
225/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0458
227/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0458
229/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0458
231/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0458
233/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0457
235/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0457
237/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0457
239/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0457
241/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0457
243/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0456
245/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0456
247/521		10s 37ms/step	- accuracy: 0.9850	- loss: 0.0456
249/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0456
251/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0456

253/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0456
255/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0456
257/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0455
259/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0455
261/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0455
263/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0455
265/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0455
267/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0455
269/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0455
271/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0454
273/521		9s 37ms/step	- accuracy: 0.9850	- loss: 0.0454
275/521		9s 37ms/step	- accuracy: 0.9851	- loss: 0.0454
277/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0454
279/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0454
281/521		8s 37ms/step	- accuracy: 0.9850	- loss: 0.0454
283/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0454
285/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0454
287/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0454
289/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0454
291/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0454
293/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0453
295/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0453
297/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0453
299/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0453
301/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0453
303/521		8s 37ms/step	- accuracy: 0.9851	- loss: 0.0453
305/521		7s 37ms/step	- accuracy: 0.9851	- loss: 0.0453
307/521		7s 37ms/step	- accuracy: 0.9851	- loss: 0.0453

309/521		7s	37ms/step	- accuracy: 0.9851	- loss: 0.0452
311/521		7s	37ms/step	- accuracy: 0.9851	- loss: 0.0452
313/521		7s	37ms/step	- accuracy: 0.9851	- loss: 0.0452
315/521		7s	37ms/step	- accuracy: 0.9851	- loss: 0.0452
317/521		7s	37ms/step	- accuracy: 0.9851	- loss: 0.0452
319/521		7s	37ms/step	- accuracy: 0.9851	- loss: 0.0452
321/521		7s	37ms/step	- accuracy: 0.9851	- loss: 0.0452
323/521		7s	37ms/step	- accuracy: 0.9851	- loss: 0.0452
325/521		7s	37ms/step	- accuracy: 0.9851	- loss: 0.0452
327/521		7s	37ms/step	- accuracy: 0.9851	- loss: 0.0451
329/521		7s	37ms/step	- accuracy: 0.9851	- loss: 0.0451
331/521		6s	37ms/step	- accuracy: 0.9851	- loss: 0.0451
333/521		6s	37ms/step	- accuracy: 0.9851	- loss: 0.0451
335/521		6s	37ms/step	- accuracy: 0.9851	- loss: 0.0451
337/521		6s	37ms/step	- accuracy: 0.9851	- loss: 0.0451
339/521		6s	37ms/step	- accuracy: 0.9851	- loss: 0.0451
341/521		6s	37ms/step	- accuracy: 0.9852	- loss: 0.0451
343/521		6s	37ms/step	- accuracy: 0.9852	- loss: 0.0450
345/521		6s	37ms/step	- accuracy: 0.9852	- loss: 0.0450
347/521		6s	37ms/step	- accuracy: 0.9852	- loss: 0.0450
349/521		6s	37ms/step	- accuracy: 0.9852	- loss: 0.0450
351/521		6s	37ms/step	- accuracy: 0.9852	- loss: 0.0450
353/521		6s	37ms/step	- accuracy: 0.9852	- loss: 0.0450
355/521		6s	37ms/step	- accuracy: 0.9852	- loss: 0.0450
357/521		6s	37ms/step	- accuracy: 0.9852	- loss: 0.0450
359/521		5s	37ms/step	- accuracy: 0.9852	- loss: 0.0449
361/521		5s	37ms/step	- accuracy: 0.9852	- loss: 0.0449
363/521		5s	37ms/step	- accuracy: 0.9852	- loss: 0.0449

365/521		5s 37ms/step	- accuracy: 0.9852	- loss: 0.0449
367/521		5s 37ms/step	- accuracy: 0.9852	- loss: 0.0449
369/521		5s 37ms/step	- accuracy: 0.9852	- loss: 0.0449
371/521		5s 37ms/step	- accuracy: 0.9852	- loss: 0.0449
373/521		5s 37ms/step	- accuracy: 0.9852	- loss: 0.0449
375/521		5s 37ms/step	- accuracy: 0.9852	- loss: 0.0449
377/521		5s 37ms/step	- accuracy: 0.9852	- loss: 0.0448
379/521		5s 37ms/step	- accuracy: 0.9852	- loss: 0.0448
381/521		5s 37ms/step	- accuracy: 0.9852	- loss: 0.0448
383/521		5s 37ms/step	- accuracy: 0.9853	- loss: 0.0448
385/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0448
387/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0448
389/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0448
391/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0448
393/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0448
395/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0448
397/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0448
399/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0448
401/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0448
403/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0448
405/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0447
407/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0447
409/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0447
411/521		4s 37ms/step	- accuracy: 0.9853	- loss: 0.0447
413/521		3s 37ms/step	- accuracy: 0.9853	- loss: 0.0447
415/521		3s 37ms/step	- accuracy: 0.9853	- loss: 0.0447
417/521		3s 37ms/step	- accuracy: 0.9853	- loss: 0.0447
419/521		3s 37ms/step	- accuracy: 0.9853	- loss: 0.0447

421/521  3s 37ms/step - accuracy: 0.9853 - loss: 0.0447

423/521  3s 37ms/step - accuracy: 0.9853 - loss: 0.0447

425/521  3s 37ms/step - accuracy: 0.9853 - loss: 0.0447

427/521  3s 37ms/step - accuracy: 0.9854 - loss: 0.0447

429/521  3s 37ms/step - accuracy: 0.9854 - loss: 0.0447

431/521  3s 37ms/step - accuracy: 0.9854 - loss: 0.0447

433/521  3s 37ms/step - accuracy: 0.9854 - loss: 0.0446

435/521  3s 37ms/step - accuracy: 0.9854 - loss: 0.0446

437/521  3s 37ms/step - accuracy: 0.9854 - loss: 0.0446

439/521  3s 37ms/step - accuracy: 0.9854 - loss: 0.0446

441/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

443/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

445/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

447/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

449/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

451/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

453/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

455/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

457/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

459/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

461/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

463/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

465/521  2s 37ms/step - accuracy: 0.9854 - loss: 0.0446

467/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0446

469/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0446

471/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0446

473/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0446

475/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0446

477/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0446

479/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0446

481/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0446

483/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0445

485/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0445

487/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0445

489/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0445

491/521  1s 37ms/step - accuracy: 0.9854 - loss: 0.0445

493/521  1s 37ms/step - accuracy: 0.9855 - loss: 0.0445

495/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

497/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

499/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

501/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

503/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

505/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

507/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

509/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

511/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

513/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

515/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

517/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

519/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

521/521  0s 37ms/step - accuracy: 0.9855 - loss: 0.0445

521/521  20s 39ms/step - accuracy: 0.9862 - loss: 0.0434
- val_accuracy: 0.9968 - val_loss: 0.0127 - learning_rate: 0.0010

Epoch 5/25

1/521  30s 58ms/step - accuracy: 0.9844 - loss: 0.0737

3/521  19s 37ms/step - accuracy: 0.9887 - loss: 0.0500

5/521  19s 37ms/step - accuracy: 0.9885 - loss: 0.0447

7/521		18s	37ms/step	-	accuracy: 0.9880	-	loss: 0.0427
9/521		18s	37ms/step	-	accuracy: 0.9876	-	loss: 0.0418
11/521		18s	36ms/step	-	accuracy: 0.9872	-	loss: 0.0412
13/521		18s	36ms/step	-	accuracy: 0.9869	-	loss: 0.0408
15/521		18s	36ms/step	-	accuracy: 0.9865	-	loss: 0.0407
17/521		18s	36ms/step	-	accuracy: 0.9862	-	loss: 0.0410
19/521		18s	37ms/step	-	accuracy: 0.9859	-	loss: 0.0412
21/521		18s	37ms/step	-	accuracy: 0.9858	-	loss: 0.0411
23/521		18s	37ms/step	-	accuracy: 0.9858	-	loss: 0.0408
25/521		18s	37ms/step	-	accuracy: 0.9859	-	loss: 0.0405
27/521		18s	37ms/step	-	accuracy: 0.9860	-	loss: 0.0401
29/521		18s	37ms/step	-	accuracy: 0.9861	-	loss: 0.0400
31/521		17s	37ms/step	-	accuracy: 0.9862	-	loss: 0.0399
33/521		17s	37ms/step	-	accuracy: 0.9862	-	loss: 0.0399
35/521		17s	37ms/step	-	accuracy: 0.9863	-	loss: 0.0399
37/521		17s	37ms/step	-	accuracy: 0.9863	-	loss: 0.0399
39/521		17s	37ms/step	-	accuracy: 0.9864	-	loss: 0.0398
41/521		17s	37ms/step	-	accuracy: 0.9864	-	loss: 0.0399
43/521		17s	37ms/step	-	accuracy: 0.9865	-	loss: 0.0399
45/521		17s	37ms/step	-	accuracy: 0.9865	-	loss: 0.0399
47/521		17s	37ms/step	-	accuracy: 0.9866	-	loss: 0.0398
49/521		17s	36ms/step	-	accuracy: 0.9866	-	loss: 0.0399
51/521		17s	36ms/step	-	accuracy: 0.9866	-	loss: 0.0400
53/521		17s	36ms/step	-	accuracy: 0.9866	-	loss: 0.0400
55/521		16s	36ms/step	-	accuracy: 0.9866	-	loss: 0.0402
57/521		16s	36ms/step	-	accuracy: 0.9866	-	loss: 0.0403
59/521		16s	36ms/step	-	accuracy: 0.9866	-	loss: 0.0403
61/521		16s	36ms/step	-	accuracy: 0.9865	-	loss: 0.0404

63/521		16s	36ms/step	-	accuracy: 0.9865	-	loss: 0.0405
65/521		16s	36ms/step	-	accuracy: 0.9865	-	loss: 0.0407
67/521		16s	36ms/step	-	accuracy: 0.9864	-	loss: 0.0408
69/521		16s	36ms/step	-	accuracy: 0.9864	-	loss: 0.0408
71/521		16s	36ms/step	-	accuracy: 0.9864	-	loss: 0.0409
73/521		16s	36ms/step	-	accuracy: 0.9864	-	loss: 0.0410
75/521		16s	36ms/step	-	accuracy: 0.9864	-	loss: 0.0410
77/521		16s	36ms/step	-	accuracy: 0.9863	-	loss: 0.0410
79/521		16s	36ms/step	-	accuracy: 0.9863	-	loss: 0.0410
81/521		16s	36ms/step	-	accuracy: 0.9863	-	loss: 0.0411
83/521		15s	36ms/step	-	accuracy: 0.9863	-	loss: 0.0411
85/521		15s	36ms/step	-	accuracy: 0.9863	-	loss: 0.0411
87/521		15s	36ms/step	-	accuracy: 0.9863	-	loss: 0.0411
89/521		15s	36ms/step	-	accuracy: 0.9864	-	loss: 0.0410
91/521		15s	36ms/step	-	accuracy: 0.9864	-	loss: 0.0410
93/521		15s	36ms/step	-	accuracy: 0.9864	-	loss: 0.0410
95/521		15s	36ms/step	-	accuracy: 0.9864	-	loss: 0.0409
97/521		15s	36ms/step	-	accuracy: 0.9864	-	loss: 0.0409
99/521		15s	36ms/step	-	accuracy: 0.9865	-	loss: 0.0408
101/521		15s	36ms/step	-	accuracy: 0.9865	-	loss: 0.0408
103/521		15s	36ms/step	-	accuracy: 0.9865	-	loss: 0.0407
105/521		15s	36ms/step	-	accuracy: 0.9865	-	loss: 0.0406
107/521		15s	36ms/step	-	accuracy: 0.9866	-	loss: 0.0406
109/521		15s	36ms/step	-	accuracy: 0.9866	-	loss: 0.0405
111/521		14s	36ms/step	-	accuracy: 0.9866	-	loss: 0.0405
113/521		14s	36ms/step	-	accuracy: 0.9866	-	loss: 0.0404
115/521		14s	36ms/step	-	accuracy: 0.9867	-	loss: 0.0403
117/521		14s	36ms/step	-	accuracy: 0.9867	-	loss: 0.0403

119/521		14s 36ms/step	- accuracy: 0.9867	- loss: 0.0402
121/521		14s 36ms/step	- accuracy: 0.9867	- loss: 0.0402
123/521		14s 36ms/step	- accuracy: 0.9867	- loss: 0.0401
125/521		14s 36ms/step	- accuracy: 0.9868	- loss: 0.0401
127/521		14s 36ms/step	- accuracy: 0.9868	- loss: 0.0400
129/521		14s 36ms/step	- accuracy: 0.9868	- loss: 0.0400
131/521		14s 36ms/step	- accuracy: 0.9868	- loss: 0.0400
133/521		14s 36ms/step	- accuracy: 0.9868	- loss: 0.0399
135/521		14s 36ms/step	- accuracy: 0.9869	- loss: 0.0399
137/521		13s 36ms/step	- accuracy: 0.9869	- loss: 0.0398
139/521		13s 36ms/step	- accuracy: 0.9869	- loss: 0.0398
141/521		13s 36ms/step	- accuracy: 0.9869	- loss: 0.0398
143/521		13s 36ms/step	- accuracy: 0.9869	- loss: 0.0397
145/521		13s 36ms/step	- accuracy: 0.9869	- loss: 0.0397
147/521		13s 36ms/step	- accuracy: 0.9869	- loss: 0.0397
149/521		13s 36ms/step	- accuracy: 0.9869	- loss: 0.0397
151/521		13s 36ms/step	- accuracy: 0.9870	- loss: 0.0397
153/521		13s 36ms/step	- accuracy: 0.9870	- loss: 0.0396
155/521		13s 36ms/step	- accuracy: 0.9870	- loss: 0.0396
157/521		13s 36ms/step	- accuracy: 0.9870	- loss: 0.0396
159/521		13s 36ms/step	- accuracy: 0.9870	- loss: 0.0396
161/521		13s 36ms/step	- accuracy: 0.9870	- loss: 0.0396
163/521		13s 36ms/step	- accuracy: 0.9870	- loss: 0.0395
165/521		12s 36ms/step	- accuracy: 0.9871	- loss: 0.0395
167/521		12s 36ms/step	- accuracy: 0.9871	- loss: 0.0395
169/521		12s 36ms/step	- accuracy: 0.9871	- loss: 0.0395
171/521		12s 36ms/step	- accuracy: 0.9871	- loss: 0.0394
173/521		12s 36ms/step	- accuracy: 0.9871	- loss: 0.0394

175/521		12s	36ms/step	-	accuracy: 0.9871	-	loss: 0.0394
177/521		12s	36ms/step	-	accuracy: 0.9871	-	loss: 0.0394
179/521		12s	36ms/step	-	accuracy: 0.9872	-	loss: 0.0393
181/521		12s	36ms/step	-	accuracy: 0.9872	-	loss: 0.0393
183/521		12s	36ms/step	-	accuracy: 0.9872	-	loss: 0.0393
185/521		12s	36ms/step	-	accuracy: 0.9872	-	loss: 0.0392
187/521		12s	36ms/step	-	accuracy: 0.9872	-	loss: 0.0392
189/521		12s	36ms/step	-	accuracy: 0.9872	-	loss: 0.0391
191/521		12s	36ms/step	-	accuracy: 0.9873	-	loss: 0.0391
193/521		11s	36ms/step	-	accuracy: 0.9873	-	loss: 0.0391
195/521		11s	36ms/step	-	accuracy: 0.9873	-	loss: 0.0390
197/521		11s	36ms/step	-	accuracy: 0.9873	-	loss: 0.0390
199/521		11s	36ms/step	-	accuracy: 0.9873	-	loss: 0.0390
201/521		11s	36ms/step	-	accuracy: 0.9874	-	loss: 0.0389
203/521		11s	36ms/step	-	accuracy: 0.9874	-	loss: 0.0389
205/521		11s	36ms/step	-	accuracy: 0.9874	-	loss: 0.0389
207/521		11s	36ms/step	-	accuracy: 0.9874	-	loss: 0.0389
209/521		11s	36ms/step	-	accuracy: 0.9874	-	loss: 0.0388
211/521		11s	36ms/step	-	accuracy: 0.9874	-	loss: 0.0388
213/521		11s	36ms/step	-	accuracy: 0.9874	-	loss: 0.0388
215/521		11s	36ms/step	-	accuracy: 0.9875	-	loss: 0.0387
217/521		11s	36ms/step	-	accuracy: 0.9875	-	loss: 0.0387
219/521		10s	36ms/step	-	accuracy: 0.9875	-	loss: 0.0387
221/521		10s	36ms/step	-	accuracy: 0.9875	-	loss: 0.0387
223/521		10s	36ms/step	-	accuracy: 0.9875	-	loss: 0.0386
225/521		10s	36ms/step	-	accuracy: 0.9875	-	loss: 0.0386
227/521		10s	36ms/step	-	accuracy: 0.9875	-	loss: 0.0386
229/521		10s	36ms/step	-	accuracy: 0.9876	-	loss: 0.0386

231/521		10s 36ms/step	- accuracy: 0.9876	- loss: 0.0385
233/521		10s 36ms/step	- accuracy: 0.9876	- loss: 0.0385
235/521		10s 36ms/step	- accuracy: 0.9876	- loss: 0.0385
237/521		10s 36ms/step	- accuracy: 0.9876	- loss: 0.0384
239/521		10s 36ms/step	- accuracy: 0.9876	- loss: 0.0384
241/521		10s 36ms/step	- accuracy: 0.9876	- loss: 0.0384
243/521		10s 36ms/step	- accuracy: 0.9876	- loss: 0.0384
245/521		10s 36ms/step	- accuracy: 0.9877	- loss: 0.0383
247/521		9s 36ms/step	- accuracy: 0.9877	- loss: 0.0383
249/521		9s 36ms/step	- accuracy: 0.9877	- loss: 0.0383
251/521		9s 36ms/step	- accuracy: 0.9877	- loss: 0.0382
253/521		9s 36ms/step	- accuracy: 0.9877	- loss: 0.0382
255/521		9s 36ms/step	- accuracy: 0.9877	- loss: 0.0382
257/521		9s 36ms/step	- accuracy: 0.9877	- loss: 0.0382
259/521		9s 36ms/step	- accuracy: 0.9877	- loss: 0.0381
261/521		9s 36ms/step	- accuracy: 0.9878	- loss: 0.0381
263/521		9s 36ms/step	- accuracy: 0.9878	- loss: 0.0381
265/521		9s 36ms/step	- accuracy: 0.9878	- loss: 0.0380
267/521		9s 36ms/step	- accuracy: 0.9878	- loss: 0.0380
269/521		9s 36ms/step	- accuracy: 0.9878	- loss: 0.0380
271/521		9s 36ms/step	- accuracy: 0.9878	- loss: 0.0380
273/521		9s 36ms/step	- accuracy: 0.9878	- loss: 0.0380
275/521		8s 36ms/step	- accuracy: 0.9878	- loss: 0.0379
277/521		8s 36ms/step	- accuracy: 0.9878	- loss: 0.0379
279/521		8s 36ms/step	- accuracy: 0.9878	- loss: 0.0379
281/521		8s 36ms/step	- accuracy: 0.9878	- loss: 0.0379
283/521		8s 36ms/step	- accuracy: 0.9879	- loss: 0.0378
285/521		8s 36ms/step	- accuracy: 0.9879	- loss: 0.0378

287/521		8s 36ms/step	- accuracy: 0.9879	- loss: 0.0378
289/521		8s 36ms/step	- accuracy: 0.9879	- loss: 0.0378
291/521		8s 36ms/step	- accuracy: 0.9879	- loss: 0.0378
293/521		8s 36ms/step	- accuracy: 0.9879	- loss: 0.0378
295/521		8s 36ms/step	- accuracy: 0.9879	- loss: 0.0377
297/521		8s 36ms/step	- accuracy: 0.9879	- loss: 0.0377
299/521		8s 36ms/step	- accuracy: 0.9879	- loss: 0.0377
301/521		8s 36ms/step	- accuracy: 0.9879	- loss: 0.0377
303/521		7s 36ms/step	- accuracy: 0.9879	- loss: 0.0377
305/521		7s 36ms/step	- accuracy: 0.9879	- loss: 0.0376
307/521		7s 36ms/step	- accuracy: 0.9879	- loss: 0.0376
309/521		7s 36ms/step	- accuracy: 0.9879	- loss: 0.0376
311/521		7s 36ms/step	- accuracy: 0.9879	- loss: 0.0376
313/521		7s 36ms/step	- accuracy: 0.9879	- loss: 0.0376
315/521		7s 36ms/step	- accuracy: 0.9880	- loss: 0.0376
317/521		7s 36ms/step	- accuracy: 0.9880	- loss: 0.0375
319/521		7s 36ms/step	- accuracy: 0.9880	- loss: 0.0375
321/521		7s 36ms/step	- accuracy: 0.9880	- loss: 0.0375
323/521		7s 36ms/step	- accuracy: 0.9880	- loss: 0.0375
325/521		7s 36ms/step	- accuracy: 0.9880	- loss: 0.0375
327/521		7s 36ms/step	- accuracy: 0.9880	- loss: 0.0375
329/521		6s 36ms/step	- accuracy: 0.9880	- loss: 0.0375
331/521		6s 36ms/step	- accuracy: 0.9880	- loss: 0.0374
333/521		6s 36ms/step	- accuracy: 0.9880	- loss: 0.0374
335/521		6s 36ms/step	- accuracy: 0.9880	- loss: 0.0374
337/521		6s 36ms/step	- accuracy: 0.9880	- loss: 0.0374
339/521		6s 36ms/step	- accuracy: 0.9880	- loss: 0.0374
341/521		6s 36ms/step	- accuracy: 0.9880	- loss: 0.0374

343/521		6s 36ms/step - accuracy: 0.9880 - loss: 0.0374
345/521		6s 36ms/step - accuracy: 0.9880 - loss: 0.0374
347/521		6s 36ms/step - accuracy: 0.9880 - loss: 0.0373
349/521		6s 36ms/step - accuracy: 0.9880 - loss: 0.0373
351/521		6s 36ms/step - accuracy: 0.9880 - loss: 0.0373
353/521		6s 36ms/step - accuracy: 0.9880 - loss: 0.0373
355/521		6s 36ms/step - accuracy: 0.9880 - loss: 0.0373
357/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0373
359/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0373
361/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0373
363/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0372
365/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0372
367/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0372
369/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0372
371/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0372
373/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0372
375/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0372
377/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0372
379/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0371
381/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0371
383/521		5s 36ms/step - accuracy: 0.9881 - loss: 0.0371
385/521		4s 36ms/step - accuracy: 0.9881 - loss: 0.0371
387/521		4s 36ms/step - accuracy: 0.9881 - loss: 0.0371
389/521		4s 36ms/step - accuracy: 0.9881 - loss: 0.0371
391/521		4s 36ms/step - accuracy: 0.9881 - loss: 0.0371
393/521		4s 36ms/step - accuracy: 0.9881 - loss: 0.0370
395/521		4s 36ms/step - accuracy: 0.9881 - loss: 0.0370
397/521		4s 36ms/step - accuracy: 0.9881 - loss: 0.0370

399/521		4s 36ms/step	- accuracy: 0.9882	- loss: 0.0370
401/521		4s 36ms/step	- accuracy: 0.9882	- loss: 0.0370
403/521		4s 36ms/step	- accuracy: 0.9882	- loss: 0.0370
405/521		4s 36ms/step	- accuracy: 0.9882	- loss: 0.0370
407/521		4s 36ms/step	- accuracy: 0.9882	- loss: 0.0369
409/521		4s 36ms/step	- accuracy: 0.9882	- loss: 0.0369
411/521		4s 36ms/step	- accuracy: 0.9882	- loss: 0.0369
413/521		3s 36ms/step	- accuracy: 0.9882	- loss: 0.0369
415/521		3s 36ms/step	- accuracy: 0.9882	- loss: 0.0369
417/521		3s 36ms/step	- accuracy: 0.9882	- loss: 0.0369
419/521		3s 36ms/step	- accuracy: 0.9882	- loss: 0.0368
421/521		3s 37ms/step	- accuracy: 0.9882	- loss: 0.0368
423/521		3s 37ms/step	- accuracy: 0.9882	- loss: 0.0368
425/521		3s 37ms/step	- accuracy: 0.9882	- loss: 0.0368
427/521		3s 37ms/step	- accuracy: 0.9882	- loss: 0.0368
429/521		3s 37ms/step	- accuracy: 0.9882	- loss: 0.0368
431/521		3s 37ms/step	- accuracy: 0.9882	- loss: 0.0367
433/521		3s 37ms/step	- accuracy: 0.9882	- loss: 0.0367
435/521		3s 37ms/step	- accuracy: 0.9882	- loss: 0.0367
437/521		3s 37ms/step	- accuracy: 0.9883	- loss: 0.0367
439/521		2s 37ms/step	- accuracy: 0.9883	- loss: 0.0367
441/521		2s 37ms/step	- accuracy: 0.9883	- loss: 0.0367
443/521		2s 37ms/step	- accuracy: 0.9883	- loss: 0.0367
445/521		2s 37ms/step	- accuracy: 0.9883	- loss: 0.0366
447/521		2s 37ms/step	- accuracy: 0.9883	- loss: 0.0366
449/521		2s 37ms/step	- accuracy: 0.9883	- loss: 0.0366
451/521		2s 37ms/step	- accuracy: 0.9883	- loss: 0.0366
453/521		2s 37ms/step	- accuracy: 0.9883	- loss: 0.0366

455/521		2s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0366
457/521		2s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0366
459/521		2s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0365
461/521		2s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0365
463/521		2s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0365
465/521		2s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0365
467/521		1s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0365
469/521		1s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0365
471/521		1s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0365
473/521		1s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0365
475/521		1s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0365
477/521		1s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0364
479/521		1s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0364
481/521		1s	37ms/step	-	accuracy: 0.9883	-	loss: 0.0364
483/521		1s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0364
485/521		1s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0364
487/521		1s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0364
489/521		1s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0364
491/521		1s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0364
493/521		1s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0363
495/521		0s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0363
497/521		0s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0363
499/521		0s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0363
501/521		0s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0363
503/521		0s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0363
505/521		0s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0363
507/521		0s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0363
509/521		0s	37ms/step	-	accuracy: 0.9884	-	loss: 0.0363

511/521  0s 37ms/step - accuracy: 0.9884 - loss: 0.0362

513/521  0s 37ms/step - accuracy: 0.9884 - loss: 0.0362

515/521  0s 37ms/step - accuracy: 0.9884 - loss: 0.0362

517/521  0s 37ms/step - accuracy: 0.9884 - loss: 0.0362

519/521  0s 37ms/step - accuracy: 0.9884 - loss: 0.0362

521/521  0s 37ms/step - accuracy: 0.9884 - loss: 0.0362

521/521  20s 38ms/step - accuracy: 0.9894 - loss: 0.0333
- val_accuracy: 0.9966 - val_loss: 0.0112 - learning_rate: 0.0010

Epoch 6/25

1/521  31s 61ms/step - accuracy: 0.9844 - loss: 0.1069

3/521  21s 42ms/step - accuracy: 0.9844 - loss: 0.0730

5/521  20s 40ms/step - accuracy: 0.9850 - loss: 0.0598

7/521  19s 39ms/step - accuracy: 0.9855 - loss: 0.0521

9/521  19s 38ms/step - accuracy: 0.9859 - loss: 0.0478

11/521  19s 38ms/step - accuracy: 0.9863 - loss: 0.0446

13/521  18s 37ms/step - accuracy: 0.9868 - loss: 0.0420

15/521  18s 37ms/step - accuracy: 0.9874 - loss: 0.0397

17/521  18s 37ms/step - accuracy: 0.9879 - loss: 0.0379

19/521  18s 37ms/step - accuracy: 0.9880 - loss: 0.0366

21/521  18s 37ms/step - accuracy: 0.9882 - loss: 0.0357

23/521  18s 36ms/step - accuracy: 0.9884 - loss: 0.0349

25/521  18s 36ms/step - accuracy: 0.9886 - loss: 0.0343

27/521  17s 36ms/step - accuracy: 0.9887 - loss: 0.0337

29/521  17s 36ms/step - accuracy: 0.9888 - loss: 0.0332

31/521  17s 36ms/step - accuracy: 0.9890 - loss: 0.0327

33/521  17s 36ms/step - accuracy: 0.9891 - loss: 0.0322

35/521  17s 36ms/step - accuracy: 0.9892 - loss: 0.0318

37/521  17s 36ms/step - accuracy: 0.9893 - loss: 0.0314

39/521  17s 36ms/step - accuracy: 0.9894 - loss: 0.0310

41/521		17s	36ms/step	- accuracy: 0.9895	- loss: 0.0306
43/521		17s	36ms/step	- accuracy: 0.9896	- loss: 0.0303
45/521		17s	37ms/step	- accuracy: 0.9897	- loss: 0.0300
47/521		17s	37ms/step	- accuracy: 0.9897	- loss: 0.0297
49/521		17s	37ms/step	- accuracy: 0.9898	- loss: 0.0295
51/521		17s	37ms/step	- accuracy: 0.9898	- loss: 0.0294
53/521		17s	37ms/step	- accuracy: 0.9898	- loss: 0.0293
55/521		17s	37ms/step	- accuracy: 0.9899	- loss: 0.0291
57/521		17s	37ms/step	- accuracy: 0.9899	- loss: 0.0290
59/521		16s	37ms/step	- accuracy: 0.9899	- loss: 0.0289
61/521		16s	37ms/step	- accuracy: 0.9900	- loss: 0.0287
63/521		16s	37ms/step	- accuracy: 0.9900	- loss: 0.0286
65/521		16s	37ms/step	- accuracy: 0.9901	- loss: 0.0284
67/521		16s	37ms/step	- accuracy: 0.9901	- loss: 0.0283
69/521		16s	37ms/step	- accuracy: 0.9902	- loss: 0.0281
71/521		16s	37ms/step	- accuracy: 0.9902	- loss: 0.0280
73/521		16s	37ms/step	- accuracy: 0.9903	- loss: 0.0279
75/521		16s	37ms/step	- accuracy: 0.9903	- loss: 0.0278
77/521		16s	37ms/step	- accuracy: 0.9903	- loss: 0.0277
79/521		16s	37ms/step	- accuracy: 0.9904	- loss: 0.0276
81/521		16s	37ms/step	- accuracy: 0.9904	- loss: 0.0274
83/521		16s	37ms/step	- accuracy: 0.9905	- loss: 0.0273
85/521		15s	37ms/step	- accuracy: 0.9906	- loss: 0.0272
87/521		15s	37ms/step	- accuracy: 0.9906	- loss: 0.0270
89/521		15s	37ms/step	- accuracy: 0.9907	- loss: 0.0269
91/521		15s	37ms/step	- accuracy: 0.9907	- loss: 0.0268
93/521		15s	37ms/step	- accuracy: 0.9908	- loss: 0.0267
95/521		15s	37ms/step	- accuracy: 0.9908	- loss: 0.0266

97/521		15s	37ms/step	-	accuracy: 0.9909	-	loss: 0.0266
99/521		15s	37ms/step	-	accuracy: 0.9909	-	loss: 0.0265
101/521		15s	37ms/step	-	accuracy: 0.9909	-	loss: 0.0264
103/521		15s	37ms/step	-	accuracy: 0.9909	-	loss: 0.0264
105/521		15s	37ms/step	-	accuracy: 0.9909	-	loss: 0.0264
107/521		15s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0264
109/521		15s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0264
111/521		15s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0264
113/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0264
115/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0265
117/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0265
119/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0265
121/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0266
123/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0266
125/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0266
127/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0267
129/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0267
131/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0267
133/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0267
135/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0268
137/521		14s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0268
139/521		13s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0268
141/521		13s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0269
143/521		13s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0269
145/521		13s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0269
147/521		13s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0270
149/521		13s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0270
151/521		13s	37ms/step	-	accuracy: 0.9910	-	loss: 0.0270

153/521		13s 37ms/step	- accuracy: 0.9910	- loss: 0.0271
155/521		13s 37ms/step	- accuracy: 0.9910	- loss: 0.0271
157/521		13s 37ms/step	- accuracy: 0.9910	- loss: 0.0272
159/521		13s 37ms/step	- accuracy: 0.9910	- loss: 0.0272
161/521		13s 37ms/step	- accuracy: 0.9910	- loss: 0.0272
163/521		13s 37ms/step	- accuracy: 0.9909	- loss: 0.0272
165/521		13s 37ms/step	- accuracy: 0.9909	- loss: 0.0273
167/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0273
169/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0273
171/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0273
173/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0274
175/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0274
177/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0274
179/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0274
181/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0274
183/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0275
185/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0275
187/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0275
189/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0275
191/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0275
193/521		12s 37ms/step	- accuracy: 0.9909	- loss: 0.0275
195/521		11s 37ms/step	- accuracy: 0.9909	- loss: 0.0275
197/521		11s 37ms/step	- accuracy: 0.9909	- loss: 0.0275
199/521		11s 37ms/step	- accuracy: 0.9909	- loss: 0.0275
201/521		11s 37ms/step	- accuracy: 0.9910	- loss: 0.0275
203/521		11s 37ms/step	- accuracy: 0.9910	- loss: 0.0275
205/521		11s 37ms/step	- accuracy: 0.9910	- loss: 0.0275
207/521		11s 37ms/step	- accuracy: 0.9910	- loss: 0.0276

209/521		11s 37ms/step	- accuracy: 0.9910	- loss: 0.0276
211/521		11s 37ms/step	- accuracy: 0.9910	- loss: 0.0276
213/521		11s 37ms/step	- accuracy: 0.9909	- loss: 0.0276
215/521		11s 37ms/step	- accuracy: 0.9909	- loss: 0.0276
217/521		11s 37ms/step	- accuracy: 0.9909	- loss: 0.0276
219/521		11s 37ms/step	- accuracy: 0.9909	- loss: 0.0276
221/521		11s 37ms/step	- accuracy: 0.9909	- loss: 0.0277
223/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0277
225/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0277
227/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0277
229/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0277
231/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0278
233/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0278
235/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0278
237/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0278
239/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0278
241/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0279
243/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0279
245/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0279
247/521		10s 37ms/step	- accuracy: 0.9909	- loss: 0.0279
249/521		9s 37ms/step	- accuracy: 0.9909	- loss: 0.0279
251/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0280
253/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0280
255/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0280
257/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0280
259/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0280
261/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0281
263/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0281

265/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0281
267/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0281
269/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0281
271/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0281
273/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0282
275/521		9s 37ms/step	- accuracy: 0.9908	- loss: 0.0282
277/521		8s 37ms/step	- accuracy: 0.9908	- loss: 0.0282
279/521		8s 37ms/step	- accuracy: 0.9908	- loss: 0.0282
281/521		8s 37ms/step	- accuracy: 0.9908	- loss: 0.0282
283/521		8s 37ms/step	- accuracy: 0.9908	- loss: 0.0282
285/521		8s 37ms/step	- accuracy: 0.9908	- loss: 0.0282
287/521		8s 37ms/step	- accuracy: 0.9908	- loss: 0.0282
289/521		8s 37ms/step	- accuracy: 0.9908	- loss: 0.0283
291/521		8s 37ms/step	- accuracy: 0.9908	- loss: 0.0283
293/521		8s 37ms/step	- accuracy: 0.9908	- loss: 0.0283
295/521		8s 37ms/step	- accuracy: 0.9907	- loss: 0.0283
297/521		8s 37ms/step	- accuracy: 0.9907	- loss: 0.0283
299/521		8s 37ms/step	- accuracy: 0.9907	- loss: 0.0283
301/521		8s 37ms/step	- accuracy: 0.9907	- loss: 0.0283
303/521		8s 37ms/step	- accuracy: 0.9907	- loss: 0.0283
305/521		7s 37ms/step	- accuracy: 0.9907	- loss: 0.0284
307/521		7s 37ms/step	- accuracy: 0.9907	- loss: 0.0284
309/521		7s 37ms/step	- accuracy: 0.9907	- loss: 0.0284
311/521		7s 37ms/step	- accuracy: 0.9907	- loss: 0.0284
313/521		7s 37ms/step	- accuracy: 0.9907	- loss: 0.0284
315/521		7s 37ms/step	- accuracy: 0.9907	- loss: 0.0284
317/521		7s 37ms/step	- accuracy: 0.9907	- loss: 0.0284
319/521		7s 37ms/step	- accuracy: 0.9907	- loss: 0.0284

321/521		7s	37ms/step	- accuracy: 0.9907	- loss: 0.0285
323/521		7s	37ms/step	- accuracy: 0.9907	- loss: 0.0285
325/521		7s	37ms/step	- accuracy: 0.9907	- loss: 0.0285
327/521		7s	37ms/step	- accuracy: 0.9907	- loss: 0.0285
329/521		7s	37ms/step	- accuracy: 0.9907	- loss: 0.0285
331/521		6s	37ms/step	- accuracy: 0.9907	- loss: 0.0285
333/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0285
335/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0285
337/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0286
339/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0286
341/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0286
343/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0286
345/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0286
347/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0286
349/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0286
351/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0286
353/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0287
355/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0287
357/521		6s	37ms/step	- accuracy: 0.9906	- loss: 0.0287
359/521		5s	37ms/step	- accuracy: 0.9906	- loss: 0.0287
361/521		5s	37ms/step	- accuracy: 0.9906	- loss: 0.0287
363/521		5s	37ms/step	- accuracy: 0.9906	- loss: 0.0287
365/521		5s	37ms/step	- accuracy: 0.9906	- loss: 0.0287
367/521		5s	37ms/step	- accuracy: 0.9906	- loss: 0.0287
369/521		5s	37ms/step	- accuracy: 0.9906	- loss: 0.0287
371/521		5s	37ms/step	- accuracy: 0.9906	- loss: 0.0288
373/521		5s	37ms/step	- accuracy: 0.9905	- loss: 0.0288
375/521		5s	37ms/step	- accuracy: 0.9905	- loss: 0.0288

377/521		5s 37ms/step - accuracy: 0.9905 - loss: 0.0288
379/521		5s 37ms/step - accuracy: 0.9905 - loss: 0.0288
381/521		5s 37ms/step - accuracy: 0.9905 - loss: 0.0288
383/521		5s 37ms/step - accuracy: 0.9905 - loss: 0.0288
385/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0288
387/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0288
389/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0288
391/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0288
393/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0288
395/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0289
397/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0289
399/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0289
401/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0289
403/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0289
405/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0289
407/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0289
409/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0289
411/521		4s 37ms/step - accuracy: 0.9905 - loss: 0.0289
413/521		3s 37ms/step - accuracy: 0.9905 - loss: 0.0289
415/521		3s 37ms/step - accuracy: 0.9905 - loss: 0.0289
417/521		3s 37ms/step - accuracy: 0.9905 - loss: 0.0289
419/521		3s 37ms/step - accuracy: 0.9905 - loss: 0.0289
421/521		3s 37ms/step - accuracy: 0.9905 - loss: 0.0289
423/521		3s 37ms/step - accuracy: 0.9905 - loss: 0.0289
425/521		3s 37ms/step - accuracy: 0.9905 - loss: 0.0289
427/521		3s 37ms/step - accuracy: 0.9905 - loss: 0.0289
429/521		3s 37ms/step - accuracy: 0.9905 - loss: 0.0289
431/521		3s 37ms/step - accuracy: 0.9905 - loss: 0.0289

433/521  3s 37ms/step - accuracy: 0.9905 - loss: 0.0289

435/521  3s 37ms/step - accuracy: 0.9905 - loss: 0.0289

437/521  3s 37ms/step - accuracy: 0.9905 - loss: 0.0289

439/521  3s 37ms/step - accuracy: 0.9905 - loss: 0.0289

441/521  2s 37ms/step - accuracy: 0.9905 - loss: 0.0289

443/521  2s 37ms/step - accuracy: 0.9905 - loss: 0.0289

445/521  2s 37ms/step - accuracy: 0.9905 - loss: 0.0289

447/521  2s 37ms/step - accuracy: 0.9905 - loss: 0.0289

449/521  2s 37ms/step - accuracy: 0.9905 - loss: 0.0289

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463/521  2s 37ms/step - accuracy: 0.9905 - loss: 0.0289

465/521  2s 37ms/step - accuracy: 0.9905 - loss: 0.0289

467/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

469/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

471/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

473/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

475/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

477/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

479/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

481/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

483/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

485/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

487/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

489/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

491/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

493/521  1s 37ms/step - accuracy: 0.9905 - loss: 0.0289

495/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

497/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

499/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

501/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

503/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

505/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

507/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

509/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

511/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

513/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

515/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

517/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

519/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

521/521  0s 37ms/step - accuracy: 0.9905 - loss: 0.0289

521/521  20s 39ms/step - accuracy: 0.9907 - loss: 0.0283
- val_accuracy: 0.9963 - val_loss: 0.0130 - learning_rate: 0.0010

Epoch 7/25

1/521  28s 56ms/step - accuracy: 0.9531 - loss: 0.0596

3/521  19s 37ms/step - accuracy: 0.9696 - loss: 0.0448

5/521  19s 37ms/step - accuracy: 0.9747 - loss: 0.0397

7/521  19s 37ms/step - accuracy: 0.9785 - loss: 0.0354

9/521  19s 37ms/step - accuracy: 0.9800 - loss: 0.0341

11/521  18s 37ms/step - accuracy: 0.9815 - loss: 0.0327

13/521  18s 37ms/step - accuracy: 0.9827 - loss: 0.0315

15/521  18s 37ms/step - accuracy: 0.9837 - loss: 0.0305

17/521  18s 37ms/step - accuracy: 0.9844 - loss: 0.0299

19/521		18s 37ms/step	- accuracy: 0.9848	- loss: 0.0296
21/521		18s 37ms/step	- accuracy: 0.9852	- loss: 0.0296
23/521		18s 37ms/step	- accuracy: 0.9853	- loss: 0.0297
25/521		18s 37ms/step	- accuracy: 0.9855	- loss: 0.0297
27/521		18s 37ms/step	- accuracy: 0.9857	- loss: 0.0296
29/521		18s 37ms/step	- accuracy: 0.9859	- loss: 0.0294
31/521		17s 37ms/step	- accuracy: 0.9861	- loss: 0.0293
33/521		17s 37ms/step	- accuracy: 0.9864	- loss: 0.0291
35/521		17s 37ms/step	- accuracy: 0.9866	- loss: 0.0288
37/521		17s 37ms/step	- accuracy: 0.9868	- loss: 0.0286
39/521		17s 36ms/step	- accuracy: 0.9870	- loss: 0.0284
41/521		17s 36ms/step	- accuracy: 0.9872	- loss: 0.0283
43/521		17s 36ms/step	- accuracy: 0.9874	- loss: 0.0281
45/521		17s 36ms/step	- accuracy: 0.9876	- loss: 0.0279
47/521		17s 36ms/step	- accuracy: 0.9877	- loss: 0.0277
49/521		17s 36ms/step	- accuracy: 0.9879	- loss: 0.0275
51/521		17s 36ms/step	- accuracy: 0.9880	- loss: 0.0275
53/521		16s 36ms/step	- accuracy: 0.9882	- loss: 0.0274
55/521		16s 36ms/step	- accuracy: 0.9883	- loss: 0.0273
57/521		16s 36ms/step	- accuracy: 0.9884	- loss: 0.0273
59/521		16s 36ms/step	- accuracy: 0.9885	- loss: 0.0273
61/521		16s 36ms/step	- accuracy: 0.9885	- loss: 0.0273
63/521		16s 36ms/step	- accuracy: 0.9886	- loss: 0.0273
65/521		16s 36ms/step	- accuracy: 0.9887	- loss: 0.0273
67/521		16s 36ms/step	- accuracy: 0.9887	- loss: 0.0273
69/521		16s 36ms/step	- accuracy: 0.9888	- loss: 0.0273
71/521		16s 36ms/step	- accuracy: 0.9889	- loss: 0.0272
73/521		16s 36ms/step	- accuracy: 0.9889	- loss: 0.0272

75/521		16s	36ms/step	- accuracy: 0.9890	- loss: 0.0272
77/521		16s	36ms/step	- accuracy: 0.9891	- loss: 0.0271
79/521		15s	36ms/step	- accuracy: 0.9891	- loss: 0.0271
81/521		15s	36ms/step	- accuracy: 0.9892	- loss: 0.0270
83/521		15s	36ms/step	- accuracy: 0.9893	- loss: 0.0270
85/521		15s	36ms/step	- accuracy: 0.9893	- loss: 0.0269
87/521		15s	36ms/step	- accuracy: 0.9894	- loss: 0.0268
89/521		15s	36ms/step	- accuracy: 0.9895	- loss: 0.0268
91/521		15s	36ms/step	- accuracy: 0.9895	- loss: 0.0267
93/521		15s	36ms/step	- accuracy: 0.9896	- loss: 0.0266
95/521		15s	36ms/step	- accuracy: 0.9897	- loss: 0.0266
97/521		15s	36ms/step	- accuracy: 0.9897	- loss: 0.0265
99/521		15s	36ms/step	- accuracy: 0.9898	- loss: 0.0264
101/521		15s	36ms/step	- accuracy: 0.9898	- loss: 0.0263
103/521		15s	36ms/step	- accuracy: 0.9899	- loss: 0.0263
105/521		15s	36ms/step	- accuracy: 0.9900	- loss: 0.0262
107/521		14s	36ms/step	- accuracy: 0.9900	- loss: 0.0261
109/521		14s	36ms/step	- accuracy: 0.9901	- loss: 0.0260
111/521		14s	36ms/step	- accuracy: 0.9901	- loss: 0.0260
113/521		14s	36ms/step	- accuracy: 0.9902	- loss: 0.0259
115/521		14s	36ms/step	- accuracy: 0.9902	- loss: 0.0258
117/521		14s	36ms/step	- accuracy: 0.9903	- loss: 0.0258
119/521		14s	36ms/step	- accuracy: 0.9903	- loss: 0.0257
121/521		14s	36ms/step	- accuracy: 0.9904	- loss: 0.0257
123/521		14s	36ms/step	- accuracy: 0.9904	- loss: 0.0256
125/521		14s	36ms/step	- accuracy: 0.9904	- loss: 0.0255
127/521		14s	36ms/step	- accuracy: 0.9905	- loss: 0.0255
129/521		14s	36ms/step	- accuracy: 0.9905	- loss: 0.0254

131/521		14s	36ms/step	- accuracy: 0.9906	- loss: 0.0253
133/521		13s	36ms/step	- accuracy: 0.9906	- loss: 0.0253
135/521		13s	36ms/step	- accuracy: 0.9906	- loss: 0.0252
137/521		13s	36ms/step	- accuracy: 0.9907	- loss: 0.0252
139/521		13s	36ms/step	- accuracy: 0.9907	- loss: 0.0251
141/521		13s	36ms/step	- accuracy: 0.9907	- loss: 0.0251
143/521		13s	36ms/step	- accuracy: 0.9908	- loss: 0.0251
145/521		13s	36ms/step	- accuracy: 0.9908	- loss: 0.0250
147/521		13s	36ms/step	- accuracy: 0.9908	- loss: 0.0250
149/521		13s	36ms/step	- accuracy: 0.9909	- loss: 0.0250
151/521		13s	36ms/step	- accuracy: 0.9909	- loss: 0.0249
153/521		13s	36ms/step	- accuracy: 0.9909	- loss: 0.0249
155/521		13s	36ms/step	- accuracy: 0.9910	- loss: 0.0249
157/521		13s	36ms/step	- accuracy: 0.9910	- loss: 0.0248
159/521		13s	36ms/step	- accuracy: 0.9910	- loss: 0.0248
161/521		12s	36ms/step	- accuracy: 0.9910	- loss: 0.0248
163/521		12s	36ms/step	- accuracy: 0.9910	- loss: 0.0248
165/521		12s	36ms/step	- accuracy: 0.9911	- loss: 0.0248
167/521		12s	36ms/step	- accuracy: 0.9911	- loss: 0.0247
169/521		12s	36ms/step	- accuracy: 0.9911	- loss: 0.0247
171/521		12s	36ms/step	- accuracy: 0.9911	- loss: 0.0247
173/521		12s	36ms/step	- accuracy: 0.9911	- loss: 0.0247
175/521		12s	36ms/step	- accuracy: 0.9911	- loss: 0.0247
177/521		12s	36ms/step	- accuracy: 0.9912	- loss: 0.0247
179/521		12s	36ms/step	- accuracy: 0.9912	- loss: 0.0247
181/521		12s	36ms/step	- accuracy: 0.9912	- loss: 0.0247
183/521		12s	36ms/step	- accuracy: 0.9912	- loss: 0.0247
185/521		12s	36ms/step	- accuracy: 0.9912	- loss: 0.0247

187/521		12s	36ms/step	-	accuracy: 0.9912	-	loss: 0.0247
189/521		11s	36ms/step	-	accuracy: 0.9912	-	loss: 0.0246
191/521		11s	36ms/step	-	accuracy: 0.9913	-	loss: 0.0246
193/521		11s	36ms/step	-	accuracy: 0.9913	-	loss: 0.0246
195/521		11s	36ms/step	-	accuracy: 0.9913	-	loss: 0.0246
197/521		11s	36ms/step	-	accuracy: 0.9913	-	loss: 0.0246
199/521		11s	36ms/step	-	accuracy: 0.9913	-	loss: 0.0246
201/521		11s	36ms/step	-	accuracy: 0.9913	-	loss: 0.0246
203/521		11s	36ms/step	-	accuracy: 0.9913	-	loss: 0.0246
205/521		11s	36ms/step	-	accuracy: 0.9913	-	loss: 0.0246
207/521		11s	36ms/step	-	accuracy: 0.9913	-	loss: 0.0246
209/521		11s	36ms/step	-	accuracy: 0.9914	-	loss: 0.0246
211/521		11s	36ms/step	-	accuracy: 0.9914	-	loss: 0.0246
213/521		11s	36ms/step	-	accuracy: 0.9914	-	loss: 0.0246
215/521		11s	36ms/step	-	accuracy: 0.9914	-	loss: 0.0246
217/521		10s	36ms/step	-	accuracy: 0.9914	-	loss: 0.0246
219/521		10s	36ms/step	-	accuracy: 0.9914	-	loss: 0.0246
221/521		10s	36ms/step	-	accuracy: 0.9914	-	loss: 0.0246
223/521		10s	36ms/step	-	accuracy: 0.9914	-	loss: 0.0246
225/521		10s	36ms/step	-	accuracy: 0.9914	-	loss: 0.0246
227/521		10s	36ms/step	-	accuracy: 0.9914	-	loss: 0.0246
229/521		10s	36ms/step	-	accuracy: 0.9915	-	loss: 0.0246
231/521		10s	36ms/step	-	accuracy: 0.9915	-	loss: 0.0246
233/521		10s	36ms/step	-	accuracy: 0.9915	-	loss: 0.0246
235/521		10s	36ms/step	-	accuracy: 0.9915	-	loss: 0.0246
237/521		10s	36ms/step	-	accuracy: 0.9915	-	loss: 0.0246
239/521		10s	36ms/step	-	accuracy: 0.9915	-	loss: 0.0246
241/521		10s	36ms/step	-	accuracy: 0.9915	-	loss: 0.0245

243/521		10s 36ms/step	- accuracy: 0.9915	- loss: 0.0245
245/521		9s 36ms/step	- accuracy: 0.9915	- loss: 0.0245
247/521		9s 36ms/step	- accuracy: 0.9915	- loss: 0.0245
249/521		9s 36ms/step	- accuracy: 0.9915	- loss: 0.0245
251/521		9s 36ms/step	- accuracy: 0.9915	- loss: 0.0245
253/521		9s 36ms/step	- accuracy: 0.9915	- loss: 0.0245
255/521		9s 36ms/step	- accuracy: 0.9915	- loss: 0.0245
257/521		9s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
259/521		9s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
261/521		9s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
263/521		9s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
265/521		9s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
267/521		9s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
269/521		9s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
271/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
273/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
275/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
277/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
279/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
281/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
283/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
285/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
287/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
289/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
291/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
293/521		8s 36ms/step	- accuracy: 0.9916	- loss: 0.0245
295/521		8s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
297/521		8s 36ms/step	- accuracy: 0.9917	- loss: 0.0245

299/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
301/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
303/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
305/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
307/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
309/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
311/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
313/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
315/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
317/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
319/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
321/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
323/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
325/521		7s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
327/521		6s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
329/521		6s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
331/521		6s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
333/521		6s 36ms/step	- accuracy: 0.9917	- loss: 0.0245
335/521		6s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
337/521		6s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
339/521		6s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
341/521		6s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
343/521		6s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
345/521		6s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
347/521		6s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
349/521		6s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
351/521		6s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
353/521		6s 36ms/step	- accuracy: 0.9918	- loss: 0.0244

355/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
357/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
359/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
361/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
363/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
365/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
367/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
369/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
371/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
373/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
375/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
377/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
379/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
381/521		5s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
383/521		4s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
385/521		4s 36ms/step	- accuracy: 0.9918	- loss: 0.0244
387/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
389/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
391/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
393/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
395/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
397/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
399/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
401/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
403/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
405/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
407/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
409/521		4s 36ms/step	- accuracy: 0.9919	- loss: 0.0243

411/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
413/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
415/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
417/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
419/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
421/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
423/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
425/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
427/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
429/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
431/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
433/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
435/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
437/521		3s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
439/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
441/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
443/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
445/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
447/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
449/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
451/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
453/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
455/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
457/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
459/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
461/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
463/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243
465/521		2s 36ms/step	- accuracy: 0.9919	- loss: 0.0243

467/521		1s	36ms/step	- accuracy: 0.9919	- loss: 0.0243
469/521		1s	36ms/step	- accuracy: 0.9919	- loss: 0.0243
471/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0243
473/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0243
475/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0243
477/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0243
479/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0243
481/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0243
483/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0243
485/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0243
487/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
489/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
491/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
493/521		1s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
495/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
497/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
499/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
501/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
503/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
505/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
507/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
509/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
511/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
513/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
515/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
517/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
519/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244
521/521		0s	36ms/step	- accuracy: 0.9920	- loss: 0.0244

521/521  20s 38ms/step - accuracy: 0.9921 - loss: 0.0252
- val_accuracy: 0.9949 - val_loss: 0.0152 - learning_rate: 0.0010
Epoch 8/25

1/521		31s 60ms/step - accuracy: 0.9688 - loss: 0.0534
3/521		21s 42ms/step - accuracy: 0.9766 - loss: 0.0410
5/521		20s 40ms/step - accuracy: 0.9811 - loss: 0.0344
7/521		20s 39ms/step - accuracy: 0.9830 - loss: 0.0310
9/521		19s 39ms/step - accuracy: 0.9839 - loss: 0.0303
11/521		19s 38ms/step - accuracy: 0.9848 - loss: 0.0292
13/521		19s 38ms/step - accuracy: 0.9856 - loss: 0.0280
15/521		18s 38ms/step - accuracy: 0.9863 - loss: 0.0273
17/521		18s 37ms/step - accuracy: 0.9869 - loss: 0.0271
19/521		18s 37ms/step - accuracy: 0.9874 - loss: 0.0268
21/521		18s 37ms/step - accuracy: 0.9879 - loss: 0.0264
23/521		18s 37ms/step - accuracy: 0.9881 - loss: 0.0262
25/521		18s 37ms/step - accuracy: 0.9883 - loss: 0.0262
27/521		18s 37ms/step - accuracy: 0.9883 - loss: 0.0263
29/521		18s 37ms/step - accuracy: 0.9884 - loss: 0.0263
31/521		18s 37ms/step - accuracy: 0.9884 - loss: 0.0263
33/521		17s 37ms/step - accuracy: 0.9884 - loss: 0.0263
35/521		17s 37ms/step - accuracy: 0.9885 - loss: 0.0263
37/521		17s 37ms/step - accuracy: 0.9885 - loss: 0.0263
39/521		17s 37ms/step - accuracy: 0.9886 - loss: 0.0262
41/521		17s 37ms/step - accuracy: 0.9886 - loss: 0.0262
43/521		17s 37ms/step - accuracy: 0.9887 - loss: 0.0261
45/521		17s 37ms/step - accuracy: 0.9887 - loss: 0.0261
47/521		17s 37ms/step - accuracy: 0.9887 - loss: 0.0261
49/521		17s 37ms/step - accuracy: 0.9887 - loss: 0.0262
51/521		17s 37ms/step - accuracy: 0.9887 - loss: 0.0263

53/521		17s	37ms/step	- accuracy: 0.9886	- loss: 0.0264
55/521		17s	37ms/step	- accuracy: 0.9886	- loss: 0.0264
57/521		16s	37ms/step	- accuracy: 0.9886	- loss: 0.0265
59/521		16s	37ms/step	- accuracy: 0.9886	- loss: 0.0265
61/521		16s	37ms/step	- accuracy: 0.9886	- loss: 0.0266
63/521		16s	37ms/step	- accuracy: 0.9886	- loss: 0.0266
65/521		16s	37ms/step	- accuracy: 0.9886	- loss: 0.0266
67/521		16s	37ms/step	- accuracy: 0.9886	- loss: 0.0267
69/521		16s	37ms/step	- accuracy: 0.9886	- loss: 0.0267
71/521		16s	37ms/step	- accuracy: 0.9886	- loss: 0.0267
73/521		16s	37ms/step	- accuracy: 0.9886	- loss: 0.0268
75/521		16s	37ms/step	- accuracy: 0.9887	- loss: 0.0268
77/521		16s	37ms/step	- accuracy: 0.9887	- loss: 0.0267
79/521		16s	37ms/step	- accuracy: 0.9887	- loss: 0.0267
81/521		16s	37ms/step	- accuracy: 0.9887	- loss: 0.0267
83/521		16s	37ms/step	- accuracy: 0.9887	- loss: 0.0267
85/521		15s	37ms/step	- accuracy: 0.9887	- loss: 0.0268
87/521		15s	37ms/step	- accuracy: 0.9887	- loss: 0.0268
89/521		15s	37ms/step	- accuracy: 0.9887	- loss: 0.0269
91/521		15s	37ms/step	- accuracy: 0.9887	- loss: 0.0269
93/521		15s	37ms/step	- accuracy: 0.9887	- loss: 0.0269
95/521		15s	37ms/step	- accuracy: 0.9888	- loss: 0.0269
97/521		15s	37ms/step	- accuracy: 0.9888	- loss: 0.0270
99/521		15s	37ms/step	- accuracy: 0.9888	- loss: 0.0270
101/521		15s	37ms/step	- accuracy: 0.9888	- loss: 0.0270
103/521		15s	37ms/step	- accuracy: 0.9888	- loss: 0.0270
105/521		15s	37ms/step	- accuracy: 0.9888	- loss: 0.0270
107/521		15s	37ms/step	- accuracy: 0.9888	- loss: 0.0270

109/521		15s 37ms/step	- accuracy: 0.9889	- loss: 0.0270
111/521		15s 37ms/step	- accuracy: 0.9889	- loss: 0.0271
113/521		14s 37ms/step	- accuracy: 0.9889	- loss: 0.0271
115/521		14s 37ms/step	- accuracy: 0.9889	- loss: 0.0271
117/521		14s 37ms/step	- accuracy: 0.9889	- loss: 0.0271
119/521		14s 37ms/step	- accuracy: 0.9890	- loss: 0.0271
121/521		14s 37ms/step	- accuracy: 0.9890	- loss: 0.0271
123/521		14s 37ms/step	- accuracy: 0.9890	- loss: 0.0271
125/521		14s 37ms/step	- accuracy: 0.9890	- loss: 0.0271
127/521		14s 37ms/step	- accuracy: 0.9890	- loss: 0.0272
129/521		14s 37ms/step	- accuracy: 0.9890	- loss: 0.0272
131/521		14s 37ms/step	- accuracy: 0.9891	- loss: 0.0272
133/521		14s 37ms/step	- accuracy: 0.9891	- loss: 0.0272
135/521		14s 37ms/step	- accuracy: 0.9891	- loss: 0.0272
137/521		14s 37ms/step	- accuracy: 0.9891	- loss: 0.0271
139/521		14s 37ms/step	- accuracy: 0.9891	- loss: 0.0271
141/521		13s 37ms/step	- accuracy: 0.9892	- loss: 0.0271
143/521		13s 37ms/step	- accuracy: 0.9892	- loss: 0.0271
145/521		13s 37ms/step	- accuracy: 0.9892	- loss: 0.0271
147/521		13s 37ms/step	- accuracy: 0.9892	- loss: 0.0271
149/521		13s 37ms/step	- accuracy: 0.9893	- loss: 0.0271
151/521		13s 37ms/step	- accuracy: 0.9893	- loss: 0.0271
153/521		13s 37ms/step	- accuracy: 0.9893	- loss: 0.0271
155/521		13s 37ms/step	- accuracy: 0.9893	- loss: 0.0271
157/521		13s 37ms/step	- accuracy: 0.9893	- loss: 0.0271
159/521		13s 37ms/step	- accuracy: 0.9894	- loss: 0.0270
161/521		13s 37ms/step	- accuracy: 0.9894	- loss: 0.0270
163/521		13s 37ms/step	- accuracy: 0.9894	- loss: 0.0270

165/521		13s	37ms/step	- accuracy: 0.9894	- loss: 0.0270
167/521		13s	37ms/step	- accuracy: 0.9894	- loss: 0.0270
169/521		12s	37ms/step	- accuracy: 0.9895	- loss: 0.0270
171/521		12s	37ms/step	- accuracy: 0.9895	- loss: 0.0270
173/521		12s	37ms/step	- accuracy: 0.9895	- loss: 0.0270
175/521		12s	37ms/step	- accuracy: 0.9895	- loss: 0.0270
177/521		12s	37ms/step	- accuracy: 0.9895	- loss: 0.0270
179/521		12s	37ms/step	- accuracy: 0.9895	- loss: 0.0270
181/521		12s	37ms/step	- accuracy: 0.9896	- loss: 0.0270
183/521		12s	37ms/step	- accuracy: 0.9896	- loss: 0.0269
185/521		12s	37ms/step	- accuracy: 0.9896	- loss: 0.0269
187/521		12s	37ms/step	- accuracy: 0.9896	- loss: 0.0269
189/521		12s	37ms/step	- accuracy: 0.9896	- loss: 0.0269
191/521		12s	37ms/step	- accuracy: 0.9896	- loss: 0.0269
193/521		12s	37ms/step	- accuracy: 0.9897	- loss: 0.0269
195/521		12s	37ms/step	- accuracy: 0.9897	- loss: 0.0269
197/521		11s	37ms/step	- accuracy: 0.9897	- loss: 0.0269
199/521		11s	37ms/step	- accuracy: 0.9897	- loss: 0.0269
201/521		11s	37ms/step	- accuracy: 0.9897	- loss: 0.0269
203/521		11s	37ms/step	- accuracy: 0.9897	- loss: 0.0269
205/521		11s	37ms/step	- accuracy: 0.9897	- loss: 0.0269
207/521		11s	37ms/step	- accuracy: 0.9898	- loss: 0.0269
209/521		11s	37ms/step	- accuracy: 0.9898	- loss: 0.0269
211/521		11s	37ms/step	- accuracy: 0.9898	- loss: 0.0269
213/521		11s	37ms/step	- accuracy: 0.9898	- loss: 0.0269
215/521		11s	37ms/step	- accuracy: 0.9898	- loss: 0.0269
217/521		11s	37ms/step	- accuracy: 0.9898	- loss: 0.0269
219/521		11s	37ms/step	- accuracy: 0.9898	- loss: 0.0268

221/521		11s 37ms/step	- accuracy: 0.9898	- loss: 0.0268
223/521		10s 37ms/step	- accuracy: 0.9898	- loss: 0.0268
225/521		10s 37ms/step	- accuracy: 0.9898	- loss: 0.0268
227/521		10s 37ms/step	- accuracy: 0.9898	- loss: 0.0268
229/521		10s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
231/521		10s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
233/521		10s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
235/521		10s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
237/521		10s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
239/521		10s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
241/521		10s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
243/521		10s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
245/521		10s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
247/521		10s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
249/521		10s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
251/521		9s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
253/521		9s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
255/521		9s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
257/521		9s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
259/521		9s 37ms/step	- accuracy: 0.9899	- loss: 0.0268
261/521		9s 37ms/step	- accuracy: 0.9900	- loss: 0.0268
263/521		9s 37ms/step	- accuracy: 0.9900	- loss: 0.0268
265/521		9s 37ms/step	- accuracy: 0.9900	- loss: 0.0268
267/521		9s 37ms/step	- accuracy: 0.9900	- loss: 0.0268
269/521		9s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
271/521		9s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
273/521		9s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
275/521		9s 37ms/step	- accuracy: 0.9900	- loss: 0.0269

277/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
279/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
281/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
283/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
285/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
287/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
289/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
291/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
293/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
295/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
297/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
299/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
301/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
303/521		8s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
305/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
307/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0269
309/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270
311/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270
313/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270
315/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270
317/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270
319/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270
321/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270
323/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270
325/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270
327/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270
329/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270
331/521		7s 37ms/step	- accuracy: 0.9900	- loss: 0.0270

333/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0270
335/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0270
337/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0270
339/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0270
341/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0270
343/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0270
345/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0271
347/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0271
349/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0271
351/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0271
353/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0271
355/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0271
357/521		6s 37ms/step - accuracy: 0.9900 - loss: 0.0271
359/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0271
361/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0271
363/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0271
365/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0271
367/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0271
369/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0271
371/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0271
373/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0271
375/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0271
377/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0271
379/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0272
381/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0272
383/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0272
385/521		5s 37ms/step - accuracy: 0.9900 - loss: 0.0272
387/521		4s 37ms/step - accuracy: 0.9900 - loss: 0.0272

389/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0272
391/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0272
393/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0272
395/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0272
397/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0272
399/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0272
401/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0272
403/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0272
405/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0272
407/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0272
409/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
411/521		4s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
413/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
415/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
417/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
419/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
420/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
422/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
424/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
426/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
428/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
430/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
432/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
434/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
436/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
438/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
440/521		3s 37ms/step	- accuracy: 0.9900	- loss: 0.0273
442/521		2s 37ms/step	- accuracy: 0.9900	- loss: 0.0274

444/521  2s 37ms/step - accuracy: 0.9900 - loss: 0.0274

446/521  2s 37ms/step - accuracy: 0.9900 - loss: 0.0274

448/521  2s 37ms/step - accuracy: 0.9900 - loss: 0.0274

450/521  2s 37ms/step - accuracy: 0.9900 - loss: 0.0274

452/521  2s 37ms/step - accuracy: 0.9900 - loss: 0.0274

454/521  2s 37ms/step - accuracy: 0.9900 - loss: 0.0274

456/521  2s 37ms/step - accuracy: 0.9900 - loss: 0.0274

458/521  2s 37ms/step - accuracy: 0.9900 - loss: 0.0274

460/521  2s 37ms/step - accuracy: 0.9900 - loss: 0.0274

462/521  2s 37ms/step - accuracy: 0.9900 - loss: 0.0274

464/521  2s 37ms/step - accuracy: 0.9900 - loss: 0.0274

466/521  2s 37ms/step - accuracy: 0.9901 - loss: 0.0274

468/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

470/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

472/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

474/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

476/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

478/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

480/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

482/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

484/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

486/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

488/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

490/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

492/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

494/521  1s 37ms/step - accuracy: 0.9901 - loss: 0.0274

496/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0274

498/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0274

500/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0274

502/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0274

504/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0274

506/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0274

508/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0274

510/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0274

512/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0274

514/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0274

516/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0275

518/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0275

520/521  0s 37ms/step - accuracy: 0.9901 - loss: 0.0275
Epoch 8: ReduceLROnPlateau reducing learning rate to 0.0005000000237487257.

521/521  20s 39ms/step - accuracy: 0.9904 - loss: 0.0280
- val_accuracy: 0.9951 - val_loss: 0.0162 - learning_rate: 0.0010
Epoch 9/25

1/521  31s 61ms/step - accuracy: 1.0000 - loss: 0.0192

3/521  22s 44ms/step - accuracy: 0.9957 - loss: 0.0171

5/521  22s 43ms/step - accuracy: 0.9932 - loss: 0.0217

7/521  21s 41ms/step - accuracy: 0.9927 - loss: 0.0232

9/521  20s 40ms/step - accuracy: 0.9925 - loss: 0.0241

11/521  20s 40ms/step - accuracy: 0.9925 - loss: 0.0242

13/521  19s 39ms/step - accuracy: 0.9927 - loss: 0.0239

15/521  19s 39ms/step - accuracy: 0.9930 - loss: 0.0235

17/521  19s 39ms/step - accuracy: 0.9931 - loss: 0.0230

19/521  19s 39ms/step - accuracy: 0.9933 - loss: 0.0225

21/521  19s 38ms/step - accuracy: 0.9934 - loss: 0.0221

23/521  19s 38ms/step - accuracy: 0.9935 - loss: 0.0219

25/521  18s 38ms/step - accuracy: 0.9936 - loss: 0.0218

27/521  18s 38ms/step - accuracy: 0.9936 - loss: 0.0218

29/521  18s 38ms/step - accuracy: 0.9937 - loss: 0.0217

31/521		18s 38ms/step	- accuracy: 0.9938	- loss: 0.0215
33/521		18s 38ms/step	- accuracy: 0.9939	- loss: 0.0213
35/521		18s 38ms/step	- accuracy: 0.9940	- loss: 0.0211
37/521		18s 38ms/step	- accuracy: 0.9941	- loss: 0.0209
39/521		18s 37ms/step	- accuracy: 0.9942	- loss: 0.0206
41/521		17s 37ms/step	- accuracy: 0.9942	- loss: 0.0204
43/521		17s 37ms/step	- accuracy: 0.9943	- loss: 0.0201
45/521		17s 37ms/step	- accuracy: 0.9944	- loss: 0.0199
47/521		17s 37ms/step	- accuracy: 0.9945	- loss: 0.0197
49/521		17s 37ms/step	- accuracy: 0.9946	- loss: 0.0195
51/521		17s 37ms/step	- accuracy: 0.9946	- loss: 0.0194
53/521		17s 37ms/step	- accuracy: 0.9947	- loss: 0.0192
55/521		17s 37ms/step	- accuracy: 0.9947	- loss: 0.0191
57/521		17s 37ms/step	- accuracy: 0.9948	- loss: 0.0189
59/521		17s 37ms/step	- accuracy: 0.9948	- loss: 0.0188
61/521		17s 37ms/step	- accuracy: 0.9948	- loss: 0.0187
63/521		17s 37ms/step	- accuracy: 0.9949	- loss: 0.0186
65/521		16s 37ms/step	- accuracy: 0.9949	- loss: 0.0185
67/521		16s 37ms/step	- accuracy: 0.9949	- loss: 0.0184
69/521		16s 37ms/step	- accuracy: 0.9950	- loss: 0.0184
71/521		16s 37ms/step	- accuracy: 0.9950	- loss: 0.0184
73/521		16s 37ms/step	- accuracy: 0.9950	- loss: 0.0183
75/521		16s 37ms/step	- accuracy: 0.9950	- loss: 0.0183
77/521		16s 37ms/step	- accuracy: 0.9950	- loss: 0.0182
79/521		16s 37ms/step	- accuracy: 0.9951	- loss: 0.0182
81/521		16s 37ms/step	- accuracy: 0.9951	- loss: 0.0181
83/521		16s 37ms/step	- accuracy: 0.9951	- loss: 0.0181
85/521		16s 37ms/step	- accuracy: 0.9951	- loss: 0.0180

87/521		16s	37ms/step	- accuracy: 0.9952	- loss: 0.0180
89/521		15s	37ms/step	- accuracy: 0.9952	- loss: 0.0179
91/521		15s	37ms/step	- accuracy: 0.9952	- loss: 0.0179
93/521		15s	37ms/step	- accuracy: 0.9952	- loss: 0.0178
95/521		15s	37ms/step	- accuracy: 0.9952	- loss: 0.0178
97/521		15s	37ms/step	- accuracy: 0.9952	- loss: 0.0178
99/521		15s	37ms/step	- accuracy: 0.9953	- loss: 0.0177
101/521		15s	37ms/step	- accuracy: 0.9953	- loss: 0.0177
103/521		15s	37ms/step	- accuracy: 0.9953	- loss: 0.0176
105/521		15s	37ms/step	- accuracy: 0.9953	- loss: 0.0176
107/521		15s	37ms/step	- accuracy: 0.9953	- loss: 0.0175
109/521		15s	37ms/step	- accuracy: 0.9953	- loss: 0.0175
111/521		15s	37ms/step	- accuracy: 0.9953	- loss: 0.0175
113/521		15s	37ms/step	- accuracy: 0.9953	- loss: 0.0174
115/521		15s	37ms/step	- accuracy: 0.9953	- loss: 0.0174
117/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0174
119/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0174
121/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0173
123/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0173
125/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0173
127/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0173
129/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0173
131/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0172
133/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0172
135/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0172
137/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0172
139/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0172
141/521		14s	37ms/step	- accuracy: 0.9953	- loss: 0.0172

143/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0172
145/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0172
147/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0172
149/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0172
151/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0172
153/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0172
155/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0172
157/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0172
159/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0172
161/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0171
163/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0171
165/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0171
167/521		13s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0171
169/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0171
171/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0171
173/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0171
175/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0171
177/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0170
179/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0170
181/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0170
183/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0170
185/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0170
187/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0170
189/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0169
191/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0169
193/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0169
195/521		12s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0169
197/521		11s	37ms/step	-	accuracy: 0.9953	-	loss: 0.0169

199/521		11s 37ms/step	- accuracy: 0.9953	- loss: 0.0168
201/521		11s 37ms/step	- accuracy: 0.9953	- loss: 0.0168
203/521		11s 37ms/step	- accuracy: 0.9953	- loss: 0.0168
205/521		11s 37ms/step	- accuracy: 0.9953	- loss: 0.0168
207/521		11s 37ms/step	- accuracy: 0.9953	- loss: 0.0168
209/521		11s 37ms/step	- accuracy: 0.9953	- loss: 0.0168
211/521		11s 37ms/step	- accuracy: 0.9953	- loss: 0.0167
213/521		11s 37ms/step	- accuracy: 0.9953	- loss: 0.0167
215/521		11s 37ms/step	- accuracy: 0.9953	- loss: 0.0167
217/521		11s 37ms/step	- accuracy: 0.9954	- loss: 0.0167
219/521		11s 37ms/step	- accuracy: 0.9954	- loss: 0.0167
221/521		11s 37ms/step	- accuracy: 0.9954	- loss: 0.0166
223/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0166
225/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0166
227/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0166
229/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0166
231/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0165
233/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0165
235/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0165
237/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0165
239/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0164
241/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0164
243/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0164
245/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0164
247/521		10s 37ms/step	- accuracy: 0.9954	- loss: 0.0164
249/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0163
251/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0163
253/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0163

255/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0163
257/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0163
259/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0162
261/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0162
263/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0162
265/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0162
267/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0162
269/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0162
271/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0161
273/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0161
275/521		9s 37ms/step	- accuracy: 0.9954	- loss: 0.0161
277/521		8s 37ms/step	- accuracy: 0.9954	- loss: 0.0161
279/521		8s 37ms/step	- accuracy: 0.9954	- loss: 0.0161
281/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0161
283/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0160
285/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0160
287/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0160
289/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0160
291/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0160
293/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0160
295/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0160
297/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0159
299/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0159
301/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0159
303/521		8s 37ms/step	- accuracy: 0.9955	- loss: 0.0159
305/521		7s 37ms/step	- accuracy: 0.9955	- loss: 0.0159
307/521		7s 37ms/step	- accuracy: 0.9955	- loss: 0.0159
309/521		7s 37ms/step	- accuracy: 0.9955	- loss: 0.0159

311/521		7s	37ms/step	- accuracy: 0.9955	- loss: 0.0158
313/521		7s	37ms/step	- accuracy: 0.9955	- loss: 0.0158
315/521		7s	37ms/step	- accuracy: 0.9955	- loss: 0.0158
317/521		7s	37ms/step	- accuracy: 0.9955	- loss: 0.0158
319/521		7s	37ms/step	- accuracy: 0.9955	- loss: 0.0158
321/521		7s	37ms/step	- accuracy: 0.9955	- loss: 0.0158
323/521		7s	37ms/step	- accuracy: 0.9955	- loss: 0.0157
325/521		7s	37ms/step	- accuracy: 0.9955	- loss: 0.0157
327/521		7s	37ms/step	- accuracy: 0.9955	- loss: 0.0157
329/521		7s	37ms/step	- accuracy: 0.9955	- loss: 0.0157
331/521		7s	37ms/step	- accuracy: 0.9955	- loss: 0.0157
333/521		6s	37ms/step	- accuracy: 0.9955	- loss: 0.0157
335/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0157
337/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0156
339/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0156
341/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0156
343/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0156
345/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0156
347/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0156
349/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0155
351/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0155
353/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0155
355/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0155
357/521		6s	37ms/step	- accuracy: 0.9956	- loss: 0.0155
359/521		5s	37ms/step	- accuracy: 0.9956	- loss: 0.0155
361/521		5s	37ms/step	- accuracy: 0.9956	- loss: 0.0155
363/521		5s	37ms/step	- accuracy: 0.9956	- loss: 0.0154
365/521		5s	37ms/step	- accuracy: 0.9956	- loss: 0.0154

367/521		5s 37ms/step	- accuracy: 0.9956	- loss: 0.0154
369/521		5s 37ms/step	- accuracy: 0.9956	- loss: 0.0154
371/521		5s 37ms/step	- accuracy: 0.9956	- loss: 0.0154
373/521		5s 37ms/step	- accuracy: 0.9956	- loss: 0.0154
375/521		5s 37ms/step	- accuracy: 0.9956	- loss: 0.0154
377/521		5s 37ms/step	- accuracy: 0.9956	- loss: 0.0153
379/521		5s 37ms/step	- accuracy: 0.9956	- loss: 0.0153
381/521		5s 37ms/step	- accuracy: 0.9956	- loss: 0.0153
383/521		5s 37ms/step	- accuracy: 0.9956	- loss: 0.0153
385/521		5s 37ms/step	- accuracy: 0.9956	- loss: 0.0153
387/521		4s 37ms/step	- accuracy: 0.9956	- loss: 0.0153
389/521		4s 37ms/step	- accuracy: 0.9956	- loss: 0.0153
391/521		4s 37ms/step	- accuracy: 0.9956	- loss: 0.0152
393/521		4s 37ms/step	- accuracy: 0.9957	- loss: 0.0152
395/521		4s 37ms/step	- accuracy: 0.9957	- loss: 0.0152
397/521		4s 37ms/step	- accuracy: 0.9957	- loss: 0.0152
399/521		4s 37ms/step	- accuracy: 0.9957	- loss: 0.0152
401/521		4s 37ms/step	- accuracy: 0.9957	- loss: 0.0152
403/521		4s 37ms/step	- accuracy: 0.9957	- loss: 0.0152
405/521		4s 37ms/step	- accuracy: 0.9957	- loss: 0.0151
407/521		4s 37ms/step	- accuracy: 0.9957	- loss: 0.0151
409/521		4s 37ms/step	- accuracy: 0.9957	- loss: 0.0151
411/521		4s 37ms/step	- accuracy: 0.9957	- loss: 0.0151
413/521		3s 37ms/step	- accuracy: 0.9957	- loss: 0.0151
415/521		3s 37ms/step	- accuracy: 0.9957	- loss: 0.0151
417/521		3s 37ms/step	- accuracy: 0.9957	- loss: 0.0151
419/521		3s 37ms/step	- accuracy: 0.9957	- loss: 0.0150
421/521		3s 37ms/step	- accuracy: 0.9957	- loss: 0.0150

423/521		3s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0150
425/521		3s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0150
427/521		3s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0150
429/521		3s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0150
431/521		3s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0150
433/521		3s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0149
435/521		3s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0149
437/521		3s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0149
439/521		3s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0149
441/521		2s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0149
443/521		2s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0149
445/521		2s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0149
447/521		2s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0148
449/521		2s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0148
451/521		2s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0148
453/521		2s	37ms/step	-	accuracy: 0.9957	-	loss: 0.0148
455/521		2s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0148
457/521		2s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0148
459/521		2s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0148
461/521		2s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0148
463/521		2s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0147
465/521		2s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0147
467/521		1s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0147
469/521		1s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0147
471/521		1s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0147
473/521		1s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0147
475/521		1s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0147
477/521		1s	37ms/step	-	accuracy: 0.9958	-	loss: 0.0146

479/521  1s 37ms/step - accuracy: 0.9958 - loss: 0.0146

481/521  1s 37ms/step - accuracy: 0.9958 - loss: 0.0146

483/521  1s 37ms/step - accuracy: 0.9958 - loss: 0.0146

485/521  1s 37ms/step - accuracy: 0.9958 - loss: 0.0146

487/521  1s 37ms/step - accuracy: 0.9958 - loss: 0.0146

489/521  1s 37ms/step - accuracy: 0.9958 - loss: 0.0146

491/521  1s 37ms/step - accuracy: 0.9958 - loss: 0.0146

493/521  1s 37ms/step - accuracy: 0.9958 - loss: 0.0145

495/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0145

497/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0145

499/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0145

501/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0145

503/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0145

505/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0145

507/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0145

509/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0145

511/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0144

513/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0144

515/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0144

517/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0144

519/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0144

521/521  0s 37ms/step - accuracy: 0.9958 - loss: 0.0144

521/521  20s 39ms/step - accuracy: 0.9964 - loss: 0.0117
- val_accuracy: 0.9985 - val_loss: 0.0050 - learning_rate: 5.0000e-04

Epoch 10/25

1/521  29s 57ms/step - accuracy: 1.0000 - loss: 0.0197

3/521  20s 40ms/step - accuracy: 1.0000 - loss: 0.0139

5/521  19s 38ms/step - accuracy: 1.0000 - loss: 0.0112

7/521  19s 37ms/step - accuracy: 0.9997 - loss: 0.0105

9/521		18s	37ms/step	- accuracy: 0.9989	- loss: 0.0105
11/521		18s	37ms/step	- accuracy: 0.9986	- loss: 0.0104
13/521		18s	37ms/step	- accuracy: 0.9984	- loss: 0.0101
15/521		18s	37ms/step	- accuracy: 0.9983	- loss: 0.0098
17/521		18s	37ms/step	- accuracy: 0.9983	- loss: 0.0095
19/521		18s	37ms/step	- accuracy: 0.9983	- loss: 0.0092
21/521		18s	37ms/step	- accuracy: 0.9983	- loss: 0.0088
23/521		18s	37ms/step	- accuracy: 0.9984	- loss: 0.0086
25/521		18s	37ms/step	- accuracy: 0.9984	- loss: 0.0083
27/521		18s	37ms/step	- accuracy: 0.9984	- loss: 0.0081
29/521		18s	37ms/step	- accuracy: 0.9984	- loss: 0.0080
31/521		18s	37ms/step	- accuracy: 0.9984	- loss: 0.0078
33/521		18s	37ms/step	- accuracy: 0.9984	- loss: 0.0077
35/521		18s	37ms/step	- accuracy: 0.9984	- loss: 0.0075
37/521		17s	37ms/step	- accuracy: 0.9984	- loss: 0.0074
39/521		17s	37ms/step	- accuracy: 0.9984	- loss: 0.0073
41/521		17s	37ms/step	- accuracy: 0.9984	- loss: 0.0072
43/521		17s	37ms/step	- accuracy: 0.9984	- loss: 0.0072
45/521		17s	37ms/step	- accuracy: 0.9983	- loss: 0.0073
47/521		17s	37ms/step	- accuracy: 0.9983	- loss: 0.0074
49/521		17s	37ms/step	- accuracy: 0.9982	- loss: 0.0074
51/521		17s	37ms/step	- accuracy: 0.9982	- loss: 0.0075
53/521		17s	37ms/step	- accuracy: 0.9982	- loss: 0.0075
55/521		17s	37ms/step	- accuracy: 0.9981	- loss: 0.0075
57/521		17s	37ms/step	- accuracy: 0.9981	- loss: 0.0075
59/521		17s	37ms/step	- accuracy: 0.9981	- loss: 0.0075
61/521		17s	37ms/step	- accuracy: 0.9981	- loss: 0.0075
63/521		16s	37ms/step	- accuracy: 0.9980	- loss: 0.0075

65/521		16s 37ms/step	- accuracy: 0.9980	- loss: 0.0075
67/521		16s 37ms/step	- accuracy: 0.9980	- loss: 0.0075
69/521		16s 37ms/step	- accuracy: 0.9980	- loss: 0.0075
71/521		16s 37ms/step	- accuracy: 0.9980	- loss: 0.0076
73/521		16s 37ms/step	- accuracy: 0.9979	- loss: 0.0076
75/521		16s 37ms/step	- accuracy: 0.9979	- loss: 0.0077
77/521		16s 37ms/step	- accuracy: 0.9979	- loss: 0.0077
79/521		16s 37ms/step	- accuracy: 0.9979	- loss: 0.0077
81/521		16s 37ms/step	- accuracy: 0.9979	- loss: 0.0077
83/521		16s 37ms/step	- accuracy: 0.9978	- loss: 0.0078
85/521		16s 37ms/step	- accuracy: 0.9978	- loss: 0.0078
87/521		16s 37ms/step	- accuracy: 0.9978	- loss: 0.0078
89/521		16s 37ms/step	- accuracy: 0.9978	- loss: 0.0079
91/521		15s 37ms/step	- accuracy: 0.9978	- loss: 0.0079
93/521		15s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
95/521		15s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
97/521		15s 37ms/step	- accuracy: 0.9977	- loss: 0.0081
99/521		15s 37ms/step	- accuracy: 0.9977	- loss: 0.0081
101/521		15s 37ms/step	- accuracy: 0.9977	- loss: 0.0082
103/521		15s 37ms/step	- accuracy: 0.9977	- loss: 0.0082
105/521		15s 37ms/step	- accuracy: 0.9976	- loss: 0.0082
107/521		15s 37ms/step	- accuracy: 0.9976	- loss: 0.0083
109/521		15s 37ms/step	- accuracy: 0.9976	- loss: 0.0083
111/521		15s 37ms/step	- accuracy: 0.9976	- loss: 0.0083
113/521		15s 37ms/step	- accuracy: 0.9976	- loss: 0.0084
115/521		15s 37ms/step	- accuracy: 0.9976	- loss: 0.0084
117/521		15s 37ms/step	- accuracy: 0.9976	- loss: 0.0085
119/521		14s 37ms/step	- accuracy: 0.9975	- loss: 0.0085

121/521		14s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0085
123/521		14s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0086
125/521		14s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0086
127/521		14s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0086
129/521		14s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0087
131/521		14s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0087
133/521		14s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0087
135/521		14s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0088
137/521		14s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0088
139/521		14s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0088
141/521		14s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0088
143/521		14s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0089
145/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0089
147/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0089
149/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0089
151/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0090
153/521		13s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0090
155/521		13s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0090
157/521		13s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0090
159/521		13s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0090
161/521		13s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0090
163/521		13s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0091
165/521		13s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0091
167/521		13s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0091
169/521		13s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0091
171/521		12s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0091
173/521		12s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0091
175/521		12s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0091

177/521		12s 37ms/step	- accuracy: 0.9973	- loss: 0.0091
179/521		12s 37ms/step	- accuracy: 0.9973	- loss: 0.0092
181/521		12s 37ms/step	- accuracy: 0.9973	- loss: 0.0092
183/521		12s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
185/521		12s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
187/521		12s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
189/521		12s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
191/521		12s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
193/521		12s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
195/521		12s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
197/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
199/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
201/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
203/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
205/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
207/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
209/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
211/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
213/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0092
215/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
217/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
219/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
221/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
223/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
225/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
227/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
229/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
231/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093

233/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
235/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
237/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
239/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
241/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
243/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
245/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
247/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
249/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
251/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0093
253/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
255/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
257/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
259/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
261/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
263/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
265/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
267/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
269/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
271/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
273/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
275/521		9s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
277/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
279/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
281/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
283/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
285/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
287/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0093

289/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
291/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
293/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
295/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0093
297/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
299/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
301/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
303/521		8s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
305/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
307/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
309/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
311/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
313/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
315/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
317/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
319/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
321/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
323/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
325/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
327/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
329/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
331/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
333/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
335/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
337/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
339/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
341/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0092
343/521		6s 37ms/step	- accuracy: 0.9972	- loss: 0.0092

345/521		6s 37ms/step - accuracy: 0.9972 - loss: 0.0092
347/521		6s 37ms/step - accuracy: 0.9972 - loss: 0.0092
349/521		6s 37ms/step - accuracy: 0.9972 - loss: 0.0092
351/521		6s 37ms/step - accuracy: 0.9972 - loss: 0.0092
353/521		6s 37ms/step - accuracy: 0.9972 - loss: 0.0092
355/521		6s 37ms/step - accuracy: 0.9972 - loss: 0.0091
357/521		6s 37ms/step - accuracy: 0.9972 - loss: 0.0091
359/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
361/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
363/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
365/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
367/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
369/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
371/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
373/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
375/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
377/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
379/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
381/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
383/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
385/521		5s 37ms/step - accuracy: 0.9972 - loss: 0.0091
387/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0091
389/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0091
391/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0091
393/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0091
395/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0091
397/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0091
399/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0091

401/521		4s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
403/521		4s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
405/521		4s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
407/521		4s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
409/521		4s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
411/521		4s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
413/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
415/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
417/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
419/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
421/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
423/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
425/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
427/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
429/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
431/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
433/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
435/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
437/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
439/521		3s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
441/521		2s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
443/521		2s 37ms/step	- accuracy: 0.9972	- loss: 0.0090
445/521		2s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
447/521		2s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
449/521		2s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
451/521		2s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
453/521		2s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
455/521		2s 37ms/step	- accuracy: 0.9972	- loss: 0.0089

457/521		2s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
459/521		2s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
461/521		2s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
463/521		2s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
465/521		2s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
467/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
469/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
471/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
473/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
475/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
477/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
479/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
481/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
483/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
485/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0089
487/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
489/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
491/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
493/521		1s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
495/521		0s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
497/521		0s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
499/521		0s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
501/521		0s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
503/521		0s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
505/521		0s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
507/521		0s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
509/521		0s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088
511/521		0s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0088

513/521  0s 37ms/step - accuracy: 0.9973 - loss: 0.0088

515/521  0s 37ms/step - accuracy: 0.9973 - loss: 0.0088

517/521  0s 37ms/step - accuracy: 0.9973 - loss: 0.0088

519/521  0s 37ms/step - accuracy: 0.9973 - loss: 0.0088

521/521  0s 37ms/step - accuracy: 0.9973 - loss: 0.0088

521/521  20s 39ms/step - accuracy: 0.9979 - loss: 0.0075
- val_accuracy: 0.9985 - val_loss: 0.0046 - learning_rate: 5.0000e-04

Epoch 11/25

1/521  31s 61ms/step - accuracy: 1.0000 - loss: 0.0050

3/521  22s 43ms/step - accuracy: 1.0000 - loss: 0.0034

5/521  21s 41ms/step - accuracy: 1.0000 - loss: 0.0027

7/521  20s 39ms/step - accuracy: 0.9993 - loss: 0.0031

9/521  19s 39ms/step - accuracy: 0.9991 - loss: 0.0032

11/521  19s 38ms/step - accuracy: 0.9990 - loss: 0.0033

13/521  19s 38ms/step - accuracy: 0.9989 - loss: 0.0032

15/521  18s 37ms/step - accuracy: 0.9989 - loss: 0.0034

17/521  18s 37ms/step - accuracy: 0.9988 - loss: 0.0037

19/521  18s 37ms/step - accuracy: 0.9987 - loss: 0.0038

21/521  18s 37ms/step - accuracy: 0.9987 - loss: 0.0040

23/521  18s 37ms/step - accuracy: 0.9987 - loss: 0.0040

25/521  18s 37ms/step - accuracy: 0.9987 - loss: 0.0041

27/521  18s 37ms/step - accuracy: 0.9987 - loss: 0.0041

29/521  17s 37ms/step - accuracy: 0.9986 - loss: 0.0043

31/521  17s 37ms/step - accuracy: 0.9986 - loss: 0.0045

33/521  17s 37ms/step - accuracy: 0.9986 - loss: 0.0047

35/521  17s 37ms/step - accuracy: 0.9985 - loss: 0.0048

37/521  17s 36ms/step - accuracy: 0.9985 - loss: 0.0049

39/521  17s 36ms/step - accuracy: 0.9985 - loss: 0.0050

41/521  17s 36ms/step - accuracy: 0.9984 - loss: 0.0052

43/521		17s 37ms/step	- accuracy: 0.9984	- loss: 0.0052
45/521		17s 37ms/step	- accuracy: 0.9984	- loss: 0.0053
47/521		17s 37ms/step	- accuracy: 0.9984	- loss: 0.0054
49/521		17s 37ms/step	- accuracy: 0.9983	- loss: 0.0054
51/521		17s 37ms/step	- accuracy: 0.9983	- loss: 0.0054
53/521		17s 37ms/step	- accuracy: 0.9983	- loss: 0.0055
55/521		17s 36ms/step	- accuracy: 0.9983	- loss: 0.0055
57/521		16s 36ms/step	- accuracy: 0.9983	- loss: 0.0055
59/521		16s 36ms/step	- accuracy: 0.9983	- loss: 0.0055
61/521		16s 36ms/step	- accuracy: 0.9983	- loss: 0.0056
63/521		16s 36ms/step	- accuracy: 0.9983	- loss: 0.0056
65/521		16s 36ms/step	- accuracy: 0.9983	- loss: 0.0056
67/521		16s 36ms/step	- accuracy: 0.9983	- loss: 0.0056
69/521		16s 36ms/step	- accuracy: 0.9983	- loss: 0.0056
71/521		16s 36ms/step	- accuracy: 0.9983	- loss: 0.0056
73/521		16s 36ms/step	- accuracy: 0.9983	- loss: 0.0056
75/521		16s 37ms/step	- accuracy: 0.9983	- loss: 0.0056
77/521		16s 37ms/step	- accuracy: 0.9983	- loss: 0.0056
79/521		16s 37ms/step	- accuracy: 0.9983	- loss: 0.0056
81/521		16s 37ms/step	- accuracy: 0.9983	- loss: 0.0056
83/521		16s 37ms/step	- accuracy: 0.9983	- loss: 0.0056
85/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0056
87/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0056
89/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0056
91/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0056
93/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0056
95/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0056
97/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0056

99/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
101/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
103/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
105/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
107/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
109/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
111/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
113/521		15s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
115/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
117/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
119/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
121/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
123/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
125/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
127/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
129/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
131/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
133/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
135/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
137/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
139/521		14s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
141/521		13s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
143/521		13s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
145/521		13s 37ms/step	- accuracy: 0.9983	- loss: 0.0058
147/521		13s 37ms/step	- accuracy: 0.9983	- loss: 0.0058
149/521		13s 37ms/step	- accuracy: 0.9983	- loss: 0.0058
151/521		13s 37ms/step	- accuracy: 0.9983	- loss: 0.0058
153/521		13s 37ms/step	- accuracy: 0.9983	- loss: 0.0058

155/521		13s 37ms/step	- accuracy: 0.9983	- loss: 0.0058
157/521		13s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
159/521		13s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
161/521		13s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
163/521		13s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
165/521		13s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
167/521		13s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
169/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
171/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
173/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
175/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
177/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
179/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
181/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
183/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
185/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
187/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
189/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
191/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
193/521		12s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
195/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
197/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
199/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
201/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
203/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
205/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
207/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
209/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0058

211/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0058
213/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
215/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
217/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
219/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
221/521		11s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
223/521		10s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
225/521		10s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
227/521		10s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
229/521		10s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
231/521		10s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
233/521		10s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
235/521		10s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
237/521		10s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
239/521		10s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
241/521		10s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
243/521		10s 37ms/step	- accuracy: 0.9982	- loss: 0.0057
245/521		10s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
247/521		10s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
249/521		10s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
251/521		9s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
253/521		9s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
255/521		9s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
257/521		9s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
259/521		9s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
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263/521		9s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
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267/521		9s 37ms/step - accuracy: 0.9983 - loss: 0.0057
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275/521		9s 37ms/step - accuracy: 0.9983 - loss: 0.0057
277/521		8s 37ms/step - accuracy: 0.9983 - loss: 0.0057
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301/521		8s 37ms/step - accuracy: 0.9983 - loss: 0.0057
303/521		8s 37ms/step - accuracy: 0.9983 - loss: 0.0057
305/521		7s 37ms/step - accuracy: 0.9983 - loss: 0.0057
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323/521		7s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
325/521		7s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
327/521		7s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
329/521		7s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
331/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
333/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
335/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
337/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
339/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
341/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
343/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
345/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
347/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
349/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
351/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
353/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
355/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
357/521		6s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
359/521		5s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
361/521		5s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
363/521		5s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
365/521		5s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
367/521		5s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
369/521		5s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
371/521		5s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
373/521		5s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
375/521		5s 37ms/step	- accuracy: 0.9983	- loss: 0.0057
377/521		5s 37ms/step	- accuracy: 0.9983	- loss: 0.0057

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387/521		4s 37ms/step - accuracy: 0.9983 - loss: 0.0057
389/521		4s 37ms/step - accuracy: 0.9983 - loss: 0.0057
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413/521		3s 37ms/step - accuracy: 0.9983 - loss: 0.0057
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441/521  2s 37ms/step - accuracy: 0.9983 - loss: 0.0057

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467/521  1s 37ms/step - accuracy: 0.9983 - loss: 0.0057

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495/521  0s 37ms/step - accuracy: 0.9983 - loss: 0.0057

497/521  0s 37ms/step - accuracy: 0.9983 - loss: 0.0057

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519/521  0s 37ms/step - accuracy: 0.9983 - loss: 0.0057

521/521  0s 37ms/step - accuracy: 0.9983 - loss: 0.0057

521/521  20s 39ms/step - accuracy: 0.9982 - loss: 0.0061
- val_accuracy: 0.9986 - val_loss: 0.0049 - learning_rate: 5.0000e-04

Epoch 12/25

1/521  29s 57ms/step - accuracy: 0.9844 - loss: 0.0392

3/521  19s 37ms/step - accuracy: 0.9905 - loss: 0.0249

5/521  19s 37ms/step - accuracy: 0.9929 - loss: 0.0188

7/521  18s 37ms/step - accuracy: 0.9942 - loss: 0.0155

9/521  18s 37ms/step - accuracy: 0.9951 - loss: 0.0134

11/521  18s 37ms/step - accuracy: 0.9957 - loss: 0.0121

13/521  18s 37ms/step - accuracy: 0.9960 - loss: 0.0112

15/521  18s 37ms/step - accuracy: 0.9962 - loss: 0.0105

17/521  18s 37ms/step - accuracy: 0.9965 - loss: 0.0099

19/521  18s 37ms/step - accuracy: 0.9966 - loss: 0.0094

21/521		18s	37ms/step	-	accuracy: 0.9968	-	loss: 0.0090
23/521		18s	37ms/step	-	accuracy: 0.9970	-	loss: 0.0086
25/521		18s	37ms/step	-	accuracy: 0.9971	-	loss: 0.0085
27/521		18s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0084
29/521		18s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0083
31/521		18s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0082
33/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
35/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
37/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0082
39/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0082
41/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0082
43/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0083
45/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0083
47/521		17s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0082
49/521		17s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0082
51/521		17s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0082
53/521		17s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0082
55/521		17s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0082
57/521		16s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0082
59/521		16s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0081
61/521		16s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0081
63/521		16s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0080
65/521		16s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0080
67/521		16s	37ms/step	-	accuracy: 0.9976	-	loss: 0.0080
69/521		16s	37ms/step	-	accuracy: 0.9976	-	loss: 0.0080
71/521		16s	37ms/step	-	accuracy: 0.9976	-	loss: 0.0080
73/521		16s	37ms/step	-	accuracy: 0.9976	-	loss: 0.0079
75/521		16s	37ms/step	-	accuracy: 0.9976	-	loss: 0.0079

77/521		16s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
79/521		16s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
81/521		16s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
83/521		15s 36ms/step	- accuracy: 0.9976	- loss: 0.0079
85/521		15s 36ms/step	- accuracy: 0.9976	- loss: 0.0078
87/521		15s 36ms/step	- accuracy: 0.9976	- loss: 0.0078
89/521		15s 36ms/step	- accuracy: 0.9977	- loss: 0.0078
91/521		15s 36ms/step	- accuracy: 0.9977	- loss: 0.0078
93/521		15s 36ms/step	- accuracy: 0.9977	- loss: 0.0078
95/521		15s 36ms/step	- accuracy: 0.9977	- loss: 0.0078
97/521		15s 36ms/step	- accuracy: 0.9977	- loss: 0.0077
99/521		15s 36ms/step	- accuracy: 0.9977	- loss: 0.0077
101/521		15s 36ms/step	- accuracy: 0.9977	- loss: 0.0077
103/521		15s 37ms/step	- accuracy: 0.9977	- loss: 0.0077
105/521		15s 37ms/step	- accuracy: 0.9977	- loss: 0.0077
107/521		15s 37ms/step	- accuracy: 0.9977	- loss: 0.0077
109/521		15s 37ms/step	- accuracy: 0.9977	- loss: 0.0078
111/521		14s 37ms/step	- accuracy: 0.9977	- loss: 0.0078
113/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078
115/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078
117/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078
119/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078
121/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078
123/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078
125/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078
127/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078
129/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078
131/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078

133/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078
135/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0078
137/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
139/521		14s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
141/521		13s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
143/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0079
145/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0079
147/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0079
149/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0079
151/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0079
153/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0079
155/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0079
157/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0079
159/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
161/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
163/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
165/521		13s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
167/521		12s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
169/521		12s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
171/521		12s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
173/521		12s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
175/521		12s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
177/521		12s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
179/521		12s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
181/521		12s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
183/521		12s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
185/521		12s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
187/521		12s 37ms/step	- accuracy: 0.9975	- loss: 0.0081

189/521		12s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0081
191/521		12s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0081
193/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
195/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
197/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
199/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
201/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
203/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
205/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
207/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
209/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
211/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
213/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
215/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
217/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
219/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
221/521		10s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
223/521		10s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
225/521		10s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
227/521		10s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
229/521		10s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
231/521		10s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
233/521		10s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
235/521		10s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
237/521		10s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0081
239/521		10s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0081
241/521		10s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0081
243/521		10s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0081

245/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
247/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
249/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
251/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
253/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
255/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
257/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
259/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
261/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
263/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0081
265/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
267/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
269/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
271/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
273/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
275/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
277/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
279/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
281/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
283/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
285/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
287/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
289/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
291/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
293/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
295/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
297/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
299/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080

301/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
303/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
305/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0079
307/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0079
309/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
311/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
313/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
315/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
317/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
319/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
321/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
323/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
325/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
327/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
329/521		7s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
331/521		6s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
333/521		6s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
335/521		6s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
337/521		6s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
339/521		6s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
341/521		6s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
343/521		6s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
345/521		6s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
347/521		6s 37ms/step	- accuracy: 0.9975	- loss: 0.0080
349/521		6s 37ms/step	- accuracy: 0.9976	- loss: 0.0080
351/521		6s 37ms/step	- accuracy: 0.9976	- loss: 0.0080
353/521		6s 37ms/step	- accuracy: 0.9976	- loss: 0.0080
355/521		6s 37ms/step	- accuracy: 0.9976	- loss: 0.0080

357/521		6s 37ms/step - accuracy: 0.9976 - loss: 0.0080
359/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
361/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
363/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
365/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
367/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
369/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
371/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
373/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
375/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
377/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
379/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
381/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
383/521		5s 37ms/step - accuracy: 0.9976 - loss: 0.0080
385/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
387/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
389/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
391/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
393/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
395/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
397/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
399/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
401/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
403/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
405/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
407/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
409/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0080
411/521		4s 37ms/step - accuracy: 0.9976 - loss: 0.0079

413/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

415/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

417/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

419/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

421/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

423/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

425/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

427/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

429/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

431/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

433/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

435/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

437/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

439/521  3s 37ms/step - accuracy: 0.9976 - loss: 0.0079

441/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

443/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

445/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

447/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

449/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

451/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

453/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

455/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

457/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

459/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

461/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

463/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

465/521  2s 37ms/step - accuracy: 0.9976 - loss: 0.0079

467/521  1s 37ms/step - accuracy: 0.9976 - loss: 0.0079

469/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
471/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
473/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
475/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
477/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
479/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
481/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
483/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
485/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
487/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
489/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
491/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
493/521		1s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
495/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
497/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
499/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
501/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
503/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
505/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
507/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
509/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
511/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
513/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
515/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
517/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
519/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079
521/521		0s 37ms/step	- accuracy: 0.9976	- loss: 0.0079

521/521  20s 38ms/step - accuracy: 0.9978 - loss: 0.0076
- val_accuracy: 0.9986 - val_loss: 0.0064 - learning_rate: 5.0000e-04
Epoch 13/25

1/521		31s 61ms/step - accuracy: 0.9844 - loss: 0.0257
3/521		21s 42ms/step - accuracy: 0.9887 - loss: 0.0188
5/521		20s 41ms/step - accuracy: 0.9904 - loss: 0.0162
7/521		20s 40ms/step - accuracy: 0.9918 - loss: 0.0143
9/521		19s 39ms/step - accuracy: 0.9928 - loss: 0.0129
11/521		19s 38ms/step - accuracy: 0.9936 - loss: 0.0118
13/521		19s 38ms/step - accuracy: 0.9942 - loss: 0.0109
15/521		18s 37ms/step - accuracy: 0.9945 - loss: 0.0104
17/521		18s 37ms/step - accuracy: 0.9947 - loss: 0.0101
19/521		18s 37ms/step - accuracy: 0.9949 - loss: 0.0098
21/521		18s 37ms/step - accuracy: 0.9950 - loss: 0.0096
23/521		18s 36ms/step - accuracy: 0.9951 - loss: 0.0094
25/521		18s 36ms/step - accuracy: 0.9953 - loss: 0.0092
27/521		17s 36ms/step - accuracy: 0.9954 - loss: 0.0089
29/521		17s 36ms/step - accuracy: 0.9955 - loss: 0.0087
31/521		17s 36ms/step - accuracy: 0.9956 - loss: 0.0086
33/521		17s 36ms/step - accuracy: 0.9958 - loss: 0.0084
35/521		17s 36ms/step - accuracy: 0.9959 - loss: 0.0083
37/521		17s 36ms/step - accuracy: 0.9959 - loss: 0.0081
39/521		17s 36ms/step - accuracy: 0.9960 - loss: 0.0080
41/521		17s 36ms/step - accuracy: 0.9961 - loss: 0.0080
43/521		17s 36ms/step - accuracy: 0.9961 - loss: 0.0079
45/521		17s 36ms/step - accuracy: 0.9962 - loss: 0.0078
47/521		17s 37ms/step - accuracy: 0.9963 - loss: 0.0077
49/521		17s 37ms/step - accuracy: 0.9963 - loss: 0.0077
51/521		17s 37ms/step - accuracy: 0.9964 - loss: 0.0077

53/521		17s	37ms/step	- accuracy: 0.9964	- loss: 0.0077
55/521		17s	37ms/step	- accuracy: 0.9964	- loss: 0.0076
57/521		17s	37ms/step	- accuracy: 0.9965	- loss: 0.0076
59/521		16s	37ms/step	- accuracy: 0.9965	- loss: 0.0076
61/521		16s	37ms/step	- accuracy: 0.9965	- loss: 0.0076
63/521		16s	37ms/step	- accuracy: 0.9966	- loss: 0.0075
65/521		16s	37ms/step	- accuracy: 0.9966	- loss: 0.0075
67/521		16s	37ms/step	- accuracy: 0.9967	- loss: 0.0075
69/521		16s	37ms/step	- accuracy: 0.9967	- loss: 0.0074
71/521		16s	37ms/step	- accuracy: 0.9967	- loss: 0.0074
73/521		16s	37ms/step	- accuracy: 0.9967	- loss: 0.0074
75/521		16s	37ms/step	- accuracy: 0.9968	- loss: 0.0074
77/521		16s	37ms/step	- accuracy: 0.9968	- loss: 0.0073
79/521		16s	37ms/step	- accuracy: 0.9968	- loss: 0.0073
81/521		16s	37ms/step	- accuracy: 0.9969	- loss: 0.0073
83/521		16s	37ms/step	- accuracy: 0.9969	- loss: 0.0073
85/521		16s	37ms/step	- accuracy: 0.9969	- loss: 0.0073
87/521		15s	37ms/step	- accuracy: 0.9969	- loss: 0.0073
89/521		15s	37ms/step	- accuracy: 0.9970	- loss: 0.0073
91/521		15s	37ms/step	- accuracy: 0.9970	- loss: 0.0073
93/521		15s	37ms/step	- accuracy: 0.9970	- loss: 0.0073
95/521		15s	37ms/step	- accuracy: 0.9970	- loss: 0.0073
97/521		15s	37ms/step	- accuracy: 0.9970	- loss: 0.0073
99/521		15s	37ms/step	- accuracy: 0.9970	- loss: 0.0073
101/521		15s	37ms/step	- accuracy: 0.9970	- loss: 0.0073
103/521		15s	37ms/step	- accuracy: 0.9970	- loss: 0.0074
105/521		15s	37ms/step	- accuracy: 0.9970	- loss: 0.0074
107/521		15s	37ms/step	- accuracy: 0.9971	- loss: 0.0074

109/521		15s 37ms/step	- accuracy: 0.9971	- loss: 0.0074
111/521		15s 37ms/step	- accuracy: 0.9971	- loss: 0.0074
113/521		15s 37ms/step	- accuracy: 0.9971	- loss: 0.0074
115/521		14s 37ms/step	- accuracy: 0.9971	- loss: 0.0075
117/521		14s 37ms/step	- accuracy: 0.9971	- loss: 0.0075
119/521		14s 37ms/step	- accuracy: 0.9971	- loss: 0.0075
121/521		14s 37ms/step	- accuracy: 0.9971	- loss: 0.0075
123/521		14s 37ms/step	- accuracy: 0.9971	- loss: 0.0075
125/521		14s 37ms/step	- accuracy: 0.9971	- loss: 0.0076
127/521		14s 37ms/step	- accuracy: 0.9971	- loss: 0.0076
129/521		14s 37ms/step	- accuracy: 0.9971	- loss: 0.0076
131/521		14s 37ms/step	- accuracy: 0.9971	- loss: 0.0076
133/521		14s 37ms/step	- accuracy: 0.9971	- loss: 0.0076
135/521		14s 37ms/step	- accuracy: 0.9972	- loss: 0.0077
137/521		14s 37ms/step	- accuracy: 0.9972	- loss: 0.0077
139/521		14s 37ms/step	- accuracy: 0.9972	- loss: 0.0077
141/521		14s 37ms/step	- accuracy: 0.9972	- loss: 0.0077
143/521		13s 37ms/step	- accuracy: 0.9972	- loss: 0.0077
145/521		13s 37ms/step	- accuracy: 0.9972	- loss: 0.0077
147/521		13s 37ms/step	- accuracy: 0.9972	- loss: 0.0078
149/521		13s 37ms/step	- accuracy: 0.9972	- loss: 0.0078
151/521		13s 37ms/step	- accuracy: 0.9972	- loss: 0.0078
153/521		13s 37ms/step	- accuracy: 0.9972	- loss: 0.0078
155/521		13s 37ms/step	- accuracy: 0.9972	- loss: 0.0078
157/521		13s 37ms/step	- accuracy: 0.9972	- loss: 0.0079
159/521		13s 37ms/step	- accuracy: 0.9972	- loss: 0.0079
161/521		13s 37ms/step	- accuracy: 0.9972	- loss: 0.0079
163/521		13s 37ms/step	- accuracy: 0.9972	- loss: 0.0079

165/521		13s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0079
167/521		13s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0079
169/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0080
171/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0080
173/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0080
175/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0080
177/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0080
179/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0081
181/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0081
183/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0081
185/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0081
187/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0081
189/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0081
191/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0081
193/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0082
195/521		12s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0082
197/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0082
199/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0082
201/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0082
203/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0082
205/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0083
207/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0083
209/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0083
211/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0083
213/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0083
215/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0083
217/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0083
219/521		11s	37ms/step	-	accuracy: 0.9972	-	loss: 0.0083

221/521		11s 37ms/step	- accuracy: 0.9972	- loss: 0.0083
223/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0084
225/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0084
227/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0084
229/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0084
231/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0084
233/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0084
235/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0084
237/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0084
239/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0084
241/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0084
243/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0084
245/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
247/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
249/521		10s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
251/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
253/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
255/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
257/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
259/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
261/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
263/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
265/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
267/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0085
269/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0086
271/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0086
273/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0086
275/521		9s 37ms/step	- accuracy: 0.9972	- loss: 0.0086

277/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0086
279/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0086
281/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0086
283/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0086
285/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0086
287/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0086
289/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0086
291/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0087
293/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0087
295/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0087
297/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0087
299/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0087
301/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0087
303/521		8s 37ms/step	- accuracy: 0.9972	- loss: 0.0087
305/521		7s 37ms/step	- accuracy: 0.9972	- loss: 0.0087
307/521		7s 37ms/step	- accuracy: 0.9972	- loss: 0.0087
309/521		7s 37ms/step	- accuracy: 0.9972	- loss: 0.0087
311/521		7s 37ms/step	- accuracy: 0.9972	- loss: 0.0088
313/521		7s 37ms/step	- accuracy: 0.9972	- loss: 0.0088
315/521		7s 37ms/step	- accuracy: 0.9972	- loss: 0.0088
317/521		7s 37ms/step	- accuracy: 0.9972	- loss: 0.0088
319/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0088
321/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0088
323/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0088
325/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0088
327/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0088
329/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0088
331/521		7s 37ms/step	- accuracy: 0.9971	- loss: 0.0088

333/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0088
335/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0088
337/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0088
339/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0089
341/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0089
343/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0089
345/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0089
347/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0089
349/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0089
351/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0089
353/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0089
355/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0089
357/521		6s 37ms/step	- accuracy: 0.9971	- loss: 0.0089
359/521		5s 37ms/step	- accuracy: 0.9971	- loss: 0.0089
361/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
363/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
365/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
367/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
369/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
371/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
373/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
375/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
377/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
379/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
381/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
383/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
385/521		5s 37ms/step	- accuracy: 0.9972	- loss: 0.0089
387/521		4s 37ms/step	- accuracy: 0.9972	- loss: 0.0089

389/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
391/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
393/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
395/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
397/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
399/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
401/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
403/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
405/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
407/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
409/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
411/521		4s 37ms/step - accuracy: 0.9972 - loss: 0.0089
413/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
415/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
417/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
419/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
421/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
423/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
425/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
427/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
429/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
431/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
433/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
435/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
437/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
439/521		3s 37ms/step - accuracy: 0.9972 - loss: 0.0089
441/521		2s 37ms/step - accuracy: 0.9972 - loss: 0.0089
443/521		2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

445/521  2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

447/521  2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

449/521  2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

451/521  2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

453/521  2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

455/521  2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

457/521  2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

459/521  2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

461/521  2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

463/521  2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

465/521  2s 37ms/step - accuracy: 0.9972 - loss: 0.0089

467/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

469/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

471/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

473/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

475/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

477/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

479/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

481/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

483/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

485/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

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489/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

491/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

493/521  1s 37ms/step - accuracy: 0.9972 - loss: 0.0089

495/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

497/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

499/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

501/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

503/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

505/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

507/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

509/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

511/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

513/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

515/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

517/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

519/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089

521/521  0s 37ms/step - accuracy: 0.9972 - loss: 0.0089
Epoch 13: ReduceLROnPlateau reducing learning rate to 0.0002500000118743628.

521/521  20s 39ms/step - accuracy: 0.9973 - loss: 0.0086
- val_accuracy: 0.9985 - val_loss: 0.0051 - learning_rate: 5.0000e-04
Epoch 14/25

1/521  27s 53ms/step - accuracy: 1.0000 - loss: 0.0201

3/521  19s 37ms/step - accuracy: 1.0000 - loss: 0.0129

5/521  18s 37ms/step - accuracy: 1.0000 - loss: 0.0098

7/521  18s 36ms/step - accuracy: 0.9997 - loss: 0.0087

9/521  18s 36ms/step - accuracy: 0.9991 - loss: 0.0083

11/521  18s 36ms/step - accuracy: 0.9988 - loss: 0.0081

13/521  18s 36ms/step - accuracy: 0.9986 - loss: 0.0079

15/521  18s 36ms/step - accuracy: 0.9985 - loss: 0.0077

17/521  18s 36ms/step - accuracy: 0.9982 - loss: 0.0083

19/521  18s 36ms/step - accuracy: 0.9979 - loss: 0.0089

21/521  18s 36ms/step - accuracy: 0.9977 - loss: 0.0093

23/521  18s 37ms/step - accuracy: 0.9976 - loss: 0.0096

25/521  18s 37ms/step - accuracy: 0.9976 - loss: 0.0097

27/521  18s 37ms/step - accuracy: 0.9975 - loss: 0.0099

29/521  18s 37ms/step - accuracy: 0.9975 - loss: 0.0100

31/521		18s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0100
33/521		18s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0100
35/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0100
37/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0100
39/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0100
41/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0100
43/521		17s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0100
45/521		17s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0100
47/521		17s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0100
49/521		17s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0100
51/521		17s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0100
53/521		17s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0100
55/521		17s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0100
57/521		17s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0100
59/521		17s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0100
61/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0100
63/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0100
65/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0100
67/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0099
69/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0099
71/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0099
73/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0099
75/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0099
77/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0098
79/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0098
81/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0098
83/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0098
85/521		16s	37ms/step	-	accuracy: 0.9973	-	loss: 0.0098

87/521		15s 37ms/step	- accuracy: 0.9973	- loss: 0.0097
89/521		15s 37ms/step	- accuracy: 0.9973	- loss: 0.0097
91/521		15s 37ms/step	- accuracy: 0.9973	- loss: 0.0097
93/521		15s 37ms/step	- accuracy: 0.9973	- loss: 0.0097
95/521		15s 37ms/step	- accuracy: 0.9973	- loss: 0.0097
97/521		15s 37ms/step	- accuracy: 0.9973	- loss: 0.0097
99/521		15s 37ms/step	- accuracy: 0.9973	- loss: 0.0096
101/521		15s 37ms/step	- accuracy: 0.9973	- loss: 0.0096
103/521		15s 37ms/step	- accuracy: 0.9973	- loss: 0.0096
105/521		15s 37ms/step	- accuracy: 0.9973	- loss: 0.0096
107/521		15s 37ms/step	- accuracy: 0.9973	- loss: 0.0096
109/521		15s 37ms/step	- accuracy: 0.9974	- loss: 0.0095
111/521		15s 37ms/step	- accuracy: 0.9974	- loss: 0.0095
113/521		15s 37ms/step	- accuracy: 0.9974	- loss: 0.0095
115/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0095
117/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0095
119/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0095
121/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0095
123/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0095
125/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0095
127/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0094
129/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0094
131/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0094
133/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0094
135/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0094
137/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0094
139/521		14s 37ms/step	- accuracy: 0.9974	- loss: 0.0094
141/521		13s 37ms/step	- accuracy: 0.9974	- loss: 0.0094

143/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0094
145/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0094
147/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0094
149/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0094
151/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0094
153/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0094
155/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0094
157/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0094
159/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0094
161/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0094
163/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0093
165/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0093
167/521		13s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0093
169/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0093
171/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0093
173/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0093
175/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0093
177/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0093
179/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0093
181/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0092
183/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0092
185/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0092
187/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0092
189/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0092
191/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0092
193/521		12s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0092
195/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0092
197/521		11s	37ms/step	-	accuracy: 0.9974	-	loss: 0.0091

199/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0091
201/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0091
203/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0091
205/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0091
207/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0091
209/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0090
211/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0090
213/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0090
215/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0090
217/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0090
219/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0090
221/521		11s 37ms/step	- accuracy: 0.9974	- loss: 0.0090
223/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0090
225/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0089
227/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0089
229/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0089
231/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0089
233/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0089
235/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0089
237/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0089
239/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0088
241/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0088
243/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0088
245/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0088
247/521		10s 37ms/step	- accuracy: 0.9975	- loss: 0.0088
249/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0088
251/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0088
253/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0088

255/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0088
257/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0087
259/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0087
261/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0087
263/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0087
265/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0087
267/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0087
269/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0087
271/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0087
273/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0086
275/521		9s 37ms/step	- accuracy: 0.9975	- loss: 0.0086
277/521		8s 37ms/step	- accuracy: 0.9975	- loss: 0.0086
279/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0086
281/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0086
283/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0086
285/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0086
287/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0086
289/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0085
291/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0085
293/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0085
295/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0085
297/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0085
299/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0085
301/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0085
303/521		8s 37ms/step	- accuracy: 0.9976	- loss: 0.0085
305/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0084
307/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0084
309/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0084

311/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0084
313/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0084
315/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0084
317/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0084
319/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0084
321/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0083
323/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0083
325/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0083
327/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0083
329/521		7s 37ms/step	- accuracy: 0.9976	- loss: 0.0083
331/521		6s 37ms/step	- accuracy: 0.9976	- loss: 0.0083
333/521		6s 37ms/step	- accuracy: 0.9976	- loss: 0.0083
335/521		6s 37ms/step	- accuracy: 0.9976	- loss: 0.0083
337/521		6s 37ms/step	- accuracy: 0.9977	- loss: 0.0082
339/521		6s 37ms/step	- accuracy: 0.9977	- loss: 0.0082
341/521		6s 37ms/step	- accuracy: 0.9977	- loss: 0.0082
343/521		6s 37ms/step	- accuracy: 0.9977	- loss: 0.0082
345/521		6s 37ms/step	- accuracy: 0.9977	- loss: 0.0082
347/521		6s 37ms/step	- accuracy: 0.9977	- loss: 0.0082
349/521		6s 37ms/step	- accuracy: 0.9977	- loss: 0.0082
351/521		6s 37ms/step	- accuracy: 0.9977	- loss: 0.0082
353/521		6s 37ms/step	- accuracy: 0.9977	- loss: 0.0081
355/521		6s 37ms/step	- accuracy: 0.9977	- loss: 0.0081
357/521		6s 37ms/step	- accuracy: 0.9977	- loss: 0.0081
359/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0081
361/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0081
363/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0081
365/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0081

367/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0081
369/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0081
371/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
373/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
375/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
377/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
379/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
381/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
383/521		5s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
385/521		4s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
387/521		4s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
389/521		4s 37ms/step	- accuracy: 0.9977	- loss: 0.0080
391/521		4s 37ms/step	- accuracy: 0.9977	- loss: 0.0079
393/521		4s 37ms/step	- accuracy: 0.9977	- loss: 0.0079
395/521		4s 37ms/step	- accuracy: 0.9978	- loss: 0.0079
397/521		4s 37ms/step	- accuracy: 0.9978	- loss: 0.0079
399/521		4s 37ms/step	- accuracy: 0.9978	- loss: 0.0079
401/521		4s 37ms/step	- accuracy: 0.9978	- loss: 0.0079
403/521		4s 37ms/step	- accuracy: 0.9978	- loss: 0.0079
405/521		4s 37ms/step	- accuracy: 0.9978	- loss: 0.0079
407/521		4s 37ms/step	- accuracy: 0.9978	- loss: 0.0079
409/521		4s 37ms/step	- accuracy: 0.9978	- loss: 0.0079
411/521		4s 37ms/step	- accuracy: 0.9978	- loss: 0.0079
413/521		3s 37ms/step	- accuracy: 0.9978	- loss: 0.0078
415/521		3s 37ms/step	- accuracy: 0.9978	- loss: 0.0078
417/521		3s 37ms/step	- accuracy: 0.9978	- loss: 0.0078
419/521		3s 37ms/step	- accuracy: 0.9978	- loss: 0.0078
421/521		3s 37ms/step	- accuracy: 0.9978	- loss: 0.0078

423/521  3s 37ms/step - accuracy: 0.9978 - loss: 0.0078

425/521  3s 37ms/step - accuracy: 0.9978 - loss: 0.0078

427/521  3s 37ms/step - accuracy: 0.9978 - loss: 0.0078

429/521  3s 37ms/step - accuracy: 0.9978 - loss: 0.0078

431/521  3s 37ms/step - accuracy: 0.9978 - loss: 0.0078

433/521  3s 37ms/step - accuracy: 0.9978 - loss: 0.0078

435/521  3s 37ms/step - accuracy: 0.9978 - loss: 0.0078

437/521  3s 37ms/step - accuracy: 0.9978 - loss: 0.0077

439/521  3s 37ms/step - accuracy: 0.9978 - loss: 0.0077

441/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0077

443/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0077

445/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0077

447/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0077

449/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0077

451/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0077

453/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0077

455/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0077

457/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0077

459/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0077

461/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0077

463/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0076

465/521  2s 37ms/step - accuracy: 0.9978 - loss: 0.0076

467/521  1s 37ms/step - accuracy: 0.9978 - loss: 0.0076

469/521  1s 37ms/step - accuracy: 0.9978 - loss: 0.0076

471/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0076

473/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0076

475/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0076

477/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0076

479/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0076

481/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0076

483/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0076

485/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0076

487/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0076

489/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0076

491/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0076

493/521  1s 37ms/step - accuracy: 0.9979 - loss: 0.0075

495/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

497/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

499/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

501/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

503/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

505/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

507/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

509/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

511/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

513/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

515/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

517/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

519/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0075

521/521  0s 37ms/step - accuracy: 0.9979 - loss: 0.0074

521/521  20s 38ms/step - accuracy: 0.9984 - loss: 0.0057
- val_accuracy: 0.9980 - val_loss: 0.0069 - learning_rate: 2.5000e-04

Epoch 15/25

1/521  31s 60ms/step - accuracy: 0.9844 - loss: 0.0344

3/521  22s 44ms/step - accuracy: 0.9905 - loss: 0.0223

5/521  21s 42ms/step - accuracy: 0.9929 - loss: 0.0173

7/521  21s 41ms/step - accuracy: 0.9942 - loss: 0.0148

9/521		20s	40ms/step	-	accuracy: 0.9951	-	loss: 0.0131
11/521		20s	39ms/step	-	accuracy: 0.9957	-	loss: 0.0118
13/521		19s	39ms/step	-	accuracy: 0.9962	-	loss: 0.0108
15/521		19s	38ms/step	-	accuracy: 0.9965	-	loss: 0.0100
17/521		19s	38ms/step	-	accuracy: 0.9968	-	loss: 0.0094
19/521		19s	38ms/step	-	accuracy: 0.9971	-	loss: 0.0089
21/521		18s	38ms/step	-	accuracy: 0.9973	-	loss: 0.0084
23/521		18s	37ms/step	-	accuracy: 0.9975	-	loss: 0.0080
25/521		18s	37ms/step	-	accuracy: 0.9976	-	loss: 0.0077
27/521		18s	37ms/step	-	accuracy: 0.9977	-	loss: 0.0074
29/521		18s	37ms/step	-	accuracy: 0.9979	-	loss: 0.0071
31/521		18s	37ms/step	-	accuracy: 0.9980	-	loss: 0.0068
33/521		18s	37ms/step	-	accuracy: 0.9981	-	loss: 0.0066
35/521		17s	37ms/step	-	accuracy: 0.9981	-	loss: 0.0064
37/521		17s	37ms/step	-	accuracy: 0.9982	-	loss: 0.0062
39/521		17s	37ms/step	-	accuracy: 0.9983	-	loss: 0.0060
41/521		17s	37ms/step	-	accuracy: 0.9984	-	loss: 0.0059
43/521		17s	37ms/step	-	accuracy: 0.9984	-	loss: 0.0058
45/521		17s	37ms/step	-	accuracy: 0.9985	-	loss: 0.0056
47/521		17s	37ms/step	-	accuracy: 0.9985	-	loss: 0.0055
49/521		17s	37ms/step	-	accuracy: 0.9986	-	loss: 0.0054
51/521		17s	37ms/step	-	accuracy: 0.9986	-	loss: 0.0054
53/521		17s	37ms/step	-	accuracy: 0.9986	-	loss: 0.0053
55/521		17s	37ms/step	-	accuracy: 0.9986	-	loss: 0.0052
57/521		17s	37ms/step	-	accuracy: 0.9986	-	loss: 0.0051
59/521		17s	37ms/step	-	accuracy: 0.9987	-	loss: 0.0051
61/521		16s	37ms/step	-	accuracy: 0.9987	-	loss: 0.0050
63/521		16s	37ms/step	-	accuracy: 0.9987	-	loss: 0.0049

65/521		16s 37ms/step	- accuracy: 0.9987	- loss: 0.0049
67/521		16s 37ms/step	- accuracy: 0.9987	- loss: 0.0048
69/521		16s 37ms/step	- accuracy: 0.9987	- loss: 0.0047
71/521		16s 37ms/step	- accuracy: 0.9987	- loss: 0.0047
73/521		16s 37ms/step	- accuracy: 0.9988	- loss: 0.0046
75/521		16s 37ms/step	- accuracy: 0.9988	- loss: 0.0046
77/521		16s 37ms/step	- accuracy: 0.9988	- loss: 0.0045
79/521		16s 37ms/step	- accuracy: 0.9988	- loss: 0.0045
81/521		16s 37ms/step	- accuracy: 0.9988	- loss: 0.0045
83/521		16s 37ms/step	- accuracy: 0.9988	- loss: 0.0044
85/521		16s 37ms/step	- accuracy: 0.9988	- loss: 0.0044
87/521		15s 37ms/step	- accuracy: 0.9988	- loss: 0.0043
89/521		15s 37ms/step	- accuracy: 0.9988	- loss: 0.0043
91/521		15s 37ms/step	- accuracy: 0.9988	- loss: 0.0043
93/521		15s 37ms/step	- accuracy: 0.9988	- loss: 0.0042
95/521		15s 37ms/step	- accuracy: 0.9988	- loss: 0.0042
97/521		15s 37ms/step	- accuracy: 0.9988	- loss: 0.0042
99/521		15s 37ms/step	- accuracy: 0.9988	- loss: 0.0042
101/521		15s 37ms/step	- accuracy: 0.9988	- loss: 0.0041
103/521		15s 37ms/step	- accuracy: 0.9989	- loss: 0.0041
105/521		15s 37ms/step	- accuracy: 0.9989	- loss: 0.0041
107/521		15s 37ms/step	- accuracy: 0.9989	- loss: 0.0041
109/521		15s 37ms/step	- accuracy: 0.9989	- loss: 0.0040
111/521		15s 37ms/step	- accuracy: 0.9989	- loss: 0.0040
113/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0040
115/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0040
117/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0039
119/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0039

121/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0039
123/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0039
125/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0039
127/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0038
129/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0038
131/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0038
133/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0038
135/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0038
137/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0038
139/521		14s 37ms/step	- accuracy: 0.9989	- loss: 0.0038
141/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0037
143/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0037
145/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0037
147/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0037
149/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0037
151/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0037
153/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0037
155/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0037
157/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0037
159/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0037
161/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0036
163/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0036
165/521		13s 37ms/step	- accuracy: 0.9989	- loss: 0.0036
167/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0036
169/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0036
171/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0036
173/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0036
175/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0036

177/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0036
179/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0036
181/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0036
183/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0035
185/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0035
187/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0035
189/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0035
191/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0035
193/521		12s 37ms/step	- accuracy: 0.9989	- loss: 0.0035
195/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0035
197/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0035
199/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0035
201/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0035
203/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0035
205/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
207/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
209/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
211/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
213/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
215/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
217/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
219/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
221/521		11s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
223/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
225/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
227/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
229/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0034
231/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0033

233/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
235/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
237/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
239/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
241/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
243/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
245/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
247/521		10s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
249/521		9s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
251/521		9s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
253/521		9s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
255/521		9s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
257/521		9s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
259/521		9s 37ms/step	- accuracy: 0.9989	- loss: 0.0033
261/521		9s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
263/521		9s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
265/521		9s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
267/521		9s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
269/521		9s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
271/521		9s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
273/521		9s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
275/521		9s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
277/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
279/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
281/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
283/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
285/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
287/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0032

289/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
291/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0032
293/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
295/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
297/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
299/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
301/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
303/521		8s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
305/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
307/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
309/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
311/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
313/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
315/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
317/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
319/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
321/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
323/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
325/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
327/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
329/521		7s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
331/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
333/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
335/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
337/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
339/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
341/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
343/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031

345/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
347/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
349/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
351/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
353/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
355/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
357/521		6s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
359/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0031
361/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
363/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
365/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
367/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
369/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
371/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
373/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
375/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
377/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
379/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
381/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
383/521		5s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
385/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
387/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
389/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
391/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
393/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
395/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
397/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
399/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030

401/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
403/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
405/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
407/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
409/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
411/521		4s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
413/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
415/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
417/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
419/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
421/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
423/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
425/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
427/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
429/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
431/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
433/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
435/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
437/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
439/521		3s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
441/521		2s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
443/521		2s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
445/521		2s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
447/521		2s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
449/521		2s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
451/521		2s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
453/521		2s 37ms/step	- accuracy: 0.9990	- loss: 0.0030
455/521		2s 37ms/step	- accuracy: 0.9990	- loss: 0.0030

457/521  2s 37ms/step - accuracy: 0.9990 - loss: 0.0030

459/521  2s 37ms/step - accuracy: 0.9990 - loss: 0.0030

461/521  2s 37ms/step - accuracy: 0.9990 - loss: 0.0030

463/521  2s 37ms/step - accuracy: 0.9990 - loss: 0.0030

465/521  2s 37ms/step - accuracy: 0.9990 - loss: 0.0030

467/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

469/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

471/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

473/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

475/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

477/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

479/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

481/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

483/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

485/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

487/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

489/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

491/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

493/521  1s 37ms/step - accuracy: 0.9990 - loss: 0.0030

495/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

497/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

499/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

501/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

503/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

505/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

507/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

509/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

511/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

513/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

515/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

517/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

519/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

521/521  0s 37ms/step - accuracy: 0.9990 - loss: 0.0030

521/521  20s 38ms/step - accuracy: 0.9992 - loss: 0.0025
- val_accuracy: 0.9986 - val_loss: 0.0045 - learning_rate: 2.5000e-04
Epoch 16/25

1/521  31s 60ms/step - accuracy: 1.0000 - loss: 0.0042

3/521  22s 43ms/step - accuracy: 1.0000 - loss: 0.0026

5/521  21s 42ms/step - accuracy: 1.0000 - loss: 0.0021

7/521  20s 41ms/step - accuracy: 1.0000 - loss: 0.0018

9/521  20s 40ms/step - accuracy: 1.0000 - loss: 0.0017

11/521  20s 39ms/step - accuracy: 1.0000 - loss: 0.0018

13/521  19s 39ms/step - accuracy: 1.0000 - loss: 0.0018

15/521  19s 39ms/step - accuracy: 1.0000 - loss: 0.0018

17/521  19s 38ms/step - accuracy: 1.0000 - loss: 0.0018

19/521  19s 38ms/step - accuracy: 1.0000 - loss: 0.0017

21/521  18s 38ms/step - accuracy: 1.0000 - loss: 0.0017

23/521  18s 38ms/step - accuracy: 1.0000 - loss: 0.0017

25/521  18s 37ms/step - accuracy: 1.0000 - loss: 0.0017

27/521  18s 37ms/step - accuracy: 1.0000 - loss: 0.0017

29/521  18s 37ms/step - accuracy: 0.9999 - loss: 0.0017

31/521  18s 37ms/step - accuracy: 0.9999 - loss: 0.0017

33/521  18s 37ms/step - accuracy: 0.9999 - loss: 0.0017

35/521  18s 37ms/step - accuracy: 0.9999 - loss: 0.0017

37/521  18s 37ms/step - accuracy: 0.9999 - loss: 0.0017

39/521  17s 37ms/step - accuracy: 0.9998 - loss: 0.0017

41/521  17s 37ms/step - accuracy: 0.9998 - loss: 0.0017

43/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0017
45/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0017
47/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0017
49/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0017
51/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0017
53/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0017
55/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0017
57/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
59/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
61/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
63/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
65/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
67/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
69/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
71/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
73/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
75/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
77/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
79/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
81/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0016
83/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
85/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
87/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
89/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
91/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
93/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
95/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
97/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015

99/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
101/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
103/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
105/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
107/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
109/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
111/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
113/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
115/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
117/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
119/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
121/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
123/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
125/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
127/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
129/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
131/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
133/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
135/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
137/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
139/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
141/521		14s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
143/521		13s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
145/521		13s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
147/521		13s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
149/521		13s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
151/521		13s	37ms/step	- accuracy: 0.9998	- loss: 0.0015
153/521		13s	37ms/step	- accuracy: 0.9998	- loss: 0.0015

155/521		13s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
157/521		13s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
159/521		13s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
161/521		13s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
163/521		13s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
165/521		13s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
167/521		13s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
169/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
171/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
173/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
175/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
177/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
179/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
181/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
183/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
185/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
187/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
189/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
191/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
193/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
195/521		12s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
197/521		11s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
199/521		11s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
201/521		11s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
203/521		11s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
205/521		11s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
207/521		11s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015
209/521		11s	37ms/step	-	accuracy: 0.9998	-	loss: 0.0015

211/521		11s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
213/521		11s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
215/521		11s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
217/521		11s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
219/521		11s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
221/521		11s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
223/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
225/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
227/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
229/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
231/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
233/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
235/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
237/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
239/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
241/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
243/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
245/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
247/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
249/521		10s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
251/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
253/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
255/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
257/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
259/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
261/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
263/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
265/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015

267/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
269/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
271/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
273/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
275/521		9s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
277/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
279/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
281/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
283/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
285/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
287/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
289/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
291/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
293/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
295/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
297/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
299/521		8s 37ms/step	- accuracy: 0.9998	- loss: 0.0015
301/521		8s 37ms/step	- accuracy: 0.9997	- loss: 0.0015
303/521		8s 37ms/step	- accuracy: 0.9997	- loss: 0.0015
305/521		7s 37ms/step	- accuracy: 0.9997	- loss: 0.0015
307/521		7s 37ms/step	- accuracy: 0.9997	- loss: 0.0015
309/521		7s 37ms/step	- accuracy: 0.9997	- loss: 0.0015
311/521		7s 37ms/step	- accuracy: 0.9997	- loss: 0.0015
313/521		7s 37ms/step	- accuracy: 0.9997	- loss: 0.0015
315/521		7s 37ms/step	- accuracy: 0.9997	- loss: 0.0015
317/521		7s 37ms/step	- accuracy: 0.9997	- loss: 0.0015
319/521		7s 37ms/step	- accuracy: 0.9997	- loss: 0.0015
321/521		7s 37ms/step	- accuracy: 0.9997	- loss: 0.0015

323/521		7s 37ms/step - accuracy: 0.9997 - loss: 0.0015
325/521		7s 37ms/step - accuracy: 0.9997 - loss: 0.0015
327/521		7s 37ms/step - accuracy: 0.9997 - loss: 0.0015
329/521		7s 37ms/step - accuracy: 0.9997 - loss: 0.0015
331/521		6s 37ms/step - accuracy: 0.9997 - loss: 0.0016
333/521		6s 37ms/step - accuracy: 0.9997 - loss: 0.0016
335/521		6s 37ms/step - accuracy: 0.9997 - loss: 0.0016
337/521		6s 37ms/step - accuracy: 0.9997 - loss: 0.0016
339/521		6s 37ms/step - accuracy: 0.9997 - loss: 0.0016
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349/521		6s 37ms/step - accuracy: 0.9997 - loss: 0.0016
351/521		6s 37ms/step - accuracy: 0.9997 - loss: 0.0016
353/521		6s 37ms/step - accuracy: 0.9997 - loss: 0.0016
355/521		6s 37ms/step - accuracy: 0.9997 - loss: 0.0016
357/521		6s 37ms/step - accuracy: 0.9997 - loss: 0.0016
359/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
361/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
363/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
365/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
367/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
369/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
371/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
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375/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
377/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016

379/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
381/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
383/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
385/521		5s 37ms/step - accuracy: 0.9997 - loss: 0.0016
387/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
389/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
391/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
393/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
395/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
397/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
399/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
401/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
403/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
405/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
407/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
409/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
411/521		4s 37ms/step - accuracy: 0.9997 - loss: 0.0016
413/521		3s 37ms/step - accuracy: 0.9997 - loss: 0.0016
415/521		3s 37ms/step - accuracy: 0.9997 - loss: 0.0016
417/521		3s 37ms/step - accuracy: 0.9997 - loss: 0.0016
419/521		3s 37ms/step - accuracy: 0.9997 - loss: 0.0016
421/521		3s 37ms/step - accuracy: 0.9997 - loss: 0.0016
423/521		3s 37ms/step - accuracy: 0.9997 - loss: 0.0016
425/521		3s 37ms/step - accuracy: 0.9997 - loss: 0.0016
427/521		3s 37ms/step - accuracy: 0.9997 - loss: 0.0016
429/521		3s 37ms/step - accuracy: 0.9997 - loss: 0.0016
431/521		3s 37ms/step - accuracy: 0.9997 - loss: 0.0016
433/521		3s 37ms/step - accuracy: 0.9997 - loss: 0.0016

435/521  3s 37ms/step - accuracy: 0.9997 - loss: 0.0016

437/521  3s 37ms/step - accuracy: 0.9997 - loss: 0.0016

439/521  3s 37ms/step - accuracy: 0.9997 - loss: 0.0016

441/521  2s 37ms/step - accuracy: 0.9997 - loss: 0.0016

443/521  2s 37ms/step - accuracy: 0.9997 - loss: 0.0016

445/521  2s 37ms/step - accuracy: 0.9997 - loss: 0.0016

447/521  2s 37ms/step - accuracy: 0.9997 - loss: 0.0016

449/521  2s 37ms/step - accuracy: 0.9997 - loss: 0.0016

451/521  2s 37ms/step - accuracy: 0.9997 - loss: 0.0016

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457/521  2s 37ms/step - accuracy: 0.9997 - loss: 0.0016

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461/521  2s 37ms/step - accuracy: 0.9997 - loss: 0.0016

463/521  2s 37ms/step - accuracy: 0.9997 - loss: 0.0016

465/521  2s 37ms/step - accuracy: 0.9997 - loss: 0.0016

467/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

469/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

471/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

473/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

475/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

477/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

479/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

481/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

483/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

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489/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

491/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

493/521  1s 37ms/step - accuracy: 0.9997 - loss: 0.0016

495/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

497/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

499/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

501/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

503/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

505/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

507/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

509/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

511/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

513/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

515/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

517/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

519/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

521/521  0s 37ms/step - accuracy: 0.9997 - loss: 0.0016

521/521  20s 39ms/step - accuracy: 0.9996 - loss: 0.0015
- val_accuracy: 0.9991 - val_loss: 0.0026 - learning_rate: 2.5000e-04

Epoch 17/25

1/521  32s 62ms/step - accuracy: 1.0000 - loss: 0.0061

3/521  22s 43ms/step - accuracy: 1.0000 - loss: 0.0044

5/521  21s 41ms/step - accuracy: 1.0000 - loss: 0.0034

7/521  20s 40ms/step - accuracy: 1.0000 - loss: 0.0029

9/521  19s 39ms/step - accuracy: 0.9996 - loss: 0.0037

11/521  19s 38ms/step - accuracy: 0.9994 - loss: 0.0042

13/521  19s 38ms/step - accuracy: 0.9993 - loss: 0.0045

15/521  19s 38ms/step - accuracy: 0.9992 - loss: 0.0046

17/521  18s 37ms/step - accuracy: 0.9992 - loss: 0.0046

19/521  18s 37ms/step - accuracy: 0.9992 - loss: 0.0045

21/521		18s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0045
23/521		18s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0044
25/521		18s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0043
27/521		18s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0042
29/521		18s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0041
31/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0040
33/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0039
35/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0039
37/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0038
39/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0038
41/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0037
43/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0037
45/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0036
47/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0036
49/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0036
51/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0035
53/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0035
55/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0035
57/521		17s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0035
59/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0034
61/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0034
63/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0034
65/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0033
67/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0033
69/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0033
71/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0033
73/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0032
75/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0032

77/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0032
79/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0032
81/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0032
83/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0031
85/521		16s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0031
87/521		15s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0031
89/521		15s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0031
91/521		15s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0031
93/521		15s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0031
95/521		15s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0030
97/521		15s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0030
99/521		15s	37ms/step	-	accuracy: 0.9993	-	loss: 0.0030
101/521		15s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
103/521		15s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
105/521		15s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
107/521		15s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
109/521		15s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
111/521		15s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
113/521		14s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
115/521		14s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
117/521		14s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
119/521		14s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
121/521		14s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
123/521		14s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
125/521		14s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
127/521		14s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0030
129/521		14s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0029
131/521		14s	37ms/step	-	accuracy: 0.9992	-	loss: 0.0029

133/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
135/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
137/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
139/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
141/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
143/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
145/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
147/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
149/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
151/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
153/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
155/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
157/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
159/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0029
161/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
163/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
165/521		13s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
167/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
169/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
171/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
173/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
175/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
177/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
179/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
181/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
183/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
185/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
187/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028

189/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
191/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
193/521		12s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
195/521		11s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
197/521		11s 37ms/step	- accuracy: 0.9992	- loss: 0.0028
199/521		11s 37ms/step	- accuracy: 0.9992	- loss: 0.0027
201/521		11s 37ms/step	- accuracy: 0.9992	- loss: 0.0027
203/521		11s 37ms/step	- accuracy: 0.9992	- loss: 0.0027
205/521		11s 37ms/step	- accuracy: 0.9992	- loss: 0.0027
207/521		11s 37ms/step	- accuracy: 0.9992	- loss: 0.0027
209/521		11s 37ms/step	- accuracy: 0.9992	- loss: 0.0027
211/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
213/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
215/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
217/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
219/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
221/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
223/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
225/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
227/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
229/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
231/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
233/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0027
235/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
237/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
239/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
241/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
243/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0026

245/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
247/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
249/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
251/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
253/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
255/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
257/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
259/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
261/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
263/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
265/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
267/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
269/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
271/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
273/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
275/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
277/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
279/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
281/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
283/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0026
285/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
287/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
289/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
291/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
293/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
295/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
297/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
299/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0025

301/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
303/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
305/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
307/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
309/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
311/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
313/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
315/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
317/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
319/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
321/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
323/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
325/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
327/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
329/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
331/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
333/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
335/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
337/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
339/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0025
341/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
343/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
345/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
347/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
349/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
351/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
353/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
355/521		6s 37ms/step	- accuracy: 0.9993	- loss: 0.0024

357/521		6s 37ms/step - accuracy: 0.9993 - loss: 0.0024
359/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0024
361/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0024
363/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0024
365/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0024
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371/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0024
373/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0024
375/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0024
377/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0024
379/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0024
381/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0024
383/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0024
385/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
387/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
389/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
391/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
393/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
395/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
397/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
399/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
401/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
403/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
405/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
407/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
409/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024
411/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0024

413/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
415/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
417/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
419/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
421/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
423/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
425/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
427/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
429/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
431/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
433/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
435/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0024
437/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
439/521		3s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
441/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
443/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
445/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
447/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
449/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
451/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
453/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
455/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
457/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
459/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
461/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
463/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
465/521		2s 37ms/step	- accuracy: 0.9993	- loss: 0.0023
467/521		1s 37ms/step	- accuracy: 0.9993	- loss: 0.0023

469/521  1s 37ms/step - accuracy: 0.9993 - loss: 0.0023

471/521  1s 37ms/step - accuracy: 0.9993 - loss: 0.0023

473/521  1s 37ms/step - accuracy: 0.9993 - loss: 0.0023

475/521  1s 37ms/step - accuracy: 0.9993 - loss: 0.0024

477/521  1s 37ms/step - accuracy: 0.9993 - loss: 0.0024

479/521  1s 37ms/step - accuracy: 0.9993 - loss: 0.0024

481/521  1s 37ms/step - accuracy: 0.9993 - loss: 0.0024

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489/521  1s 37ms/step - accuracy: 0.9993 - loss: 0.0024

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493/521  1s 37ms/step - accuracy: 0.9993 - loss: 0.0024

495/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0024

497/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0024

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511/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0024

513/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0024

515/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0024

517/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0024

519/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0024

521/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0024

521/521  20s 38ms/step - accuracy: 0.9991 - loss: 0.0026
- val_accuracy: 0.9985 - val_loss: 0.0042 - learning_rate: 2.5000e-04
Epoch 18/25

1/521		27s 54ms/step - accuracy: 1.0000 - loss: 0.0011
3/521		19s 37ms/step - accuracy: 1.0000 - loss: 7.3102e-04
5/521		18s 37ms/step - accuracy: 1.0000 - loss: 7.2915e-04
7/521		18s 37ms/step - accuracy: 1.0000 - loss: 9.9114e-04
9/521		18s 37ms/step - accuracy: 1.0000 - loss: 0.0011
11/521		18s 37ms/step - accuracy: 1.0000 - loss: 0.0011
13/521		18s 37ms/step - accuracy: 1.0000 - loss: 0.0012
15/521		18s 37ms/step - accuracy: 1.0000 - loss: 0.0011
17/521		18s 37ms/step - accuracy: 1.0000 - loss: 0.0011
19/521		18s 37ms/step - accuracy: 1.0000 - loss: 0.0011
21/521		18s 37ms/step - accuracy: 1.0000 - loss: 0.0011
23/521		18s 37ms/step - accuracy: 1.0000 - loss: 0.0012
25/521		18s 37ms/step - accuracy: 1.0000 - loss: 0.0012
27/521	 	18s 37ms/step - accuracy: 1.0000 - loss: 0.0012
29/521	 	18s 37ms/step - accuracy: 1.0000 - loss: 0.0013
31/521	 	17s 37ms/step - accuracy: 1.0000 - loss: 0.0013
33/521	 	17s 37ms/step - accuracy: 1.0000 - loss: 0.0013
35/521	 	17s 37ms/step - accuracy: 1.0000 - loss: 0.0014
37/521	 	17s 37ms/step - accuracy: 0.9999 - loss: 0.0015
39/521	 	17s 37ms/step - accuracy: 0.9999 - loss: 0.0015
41/521	 	17s 37ms/step - accuracy: 0.9999 - loss: 0.0016
43/521	 	17s 37ms/step - accuracy: 0.9999 - loss: 0.0016
45/521	 	17s 37ms/step - accuracy: 0.9999 - loss: 0.0017
47/521	 	17s 37ms/step - accuracy: 0.9999 - loss: 0.0017
49/521	 	17s 37ms/step - accuracy: 0.9999 - loss: 0.0018

51/521		17s	37ms/step	- accuracy: 0.9999	- loss: 0.0018
53/521		17s	37ms/step	- accuracy: 0.9999	- loss: 0.0018
55/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0018
57/521		17s	37ms/step	- accuracy: 0.9998	- loss: 0.0018
59/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
61/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
63/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
65/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
67/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
69/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
71/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
73/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
75/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
77/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
79/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
81/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
83/521		16s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
85/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
87/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
89/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0019
91/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0020
93/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0020
95/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0020
97/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0020
99/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0020
101/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0020
103/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0020
105/521		15s	37ms/step	- accuracy: 0.9998	- loss: 0.0020

107/521		15s 37ms/step	- accuracy: 0.9998	- loss: 0.0020
109/521		15s 36ms/step	- accuracy: 0.9998	- loss: 0.0020
111/521		14s 36ms/step	- accuracy: 0.9998	- loss: 0.0020
113/521		14s 36ms/step	- accuracy: 0.9998	- loss: 0.0021
115/521		14s 36ms/step	- accuracy: 0.9998	- loss: 0.0021
117/521		14s 36ms/step	- accuracy: 0.9998	- loss: 0.0021
119/521		14s 36ms/step	- accuracy: 0.9998	- loss: 0.0021
121/521		14s 36ms/step	- accuracy: 0.9998	- loss: 0.0021
123/521		14s 36ms/step	- accuracy: 0.9998	- loss: 0.0021
125/521		14s 37ms/step	- accuracy: 0.9998	- loss: 0.0021
127/521		14s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
129/521		14s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
131/521		14s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
133/521		14s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
135/521		14s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
137/521		14s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
139/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
141/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
143/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
145/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
147/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
149/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
151/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
153/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
155/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
157/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
159/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021
161/521		13s 37ms/step	- accuracy: 0.9997	- loss: 0.0021

163/521		13s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
165/521		13s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
167/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
169/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
171/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
173/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
175/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
177/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
179/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
181/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
183/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
185/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
187/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
189/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
191/521		12s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
193/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
195/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
197/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
199/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
201/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
203/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
205/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
207/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
209/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
211/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
213/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
215/521		11s	37ms/step	-	accuracy: 0.9997	-	loss: 0.0022
217/521		11s	37ms/step	-	accuracy: 0.9996	-	loss: 0.0022

219/521		11s 37ms/step	- accuracy: 0.9996	- loss: 0.0022
221/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0022
223/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0022
225/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0022
227/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0022
229/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0022
231/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
233/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
235/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
237/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
239/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
241/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
243/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
245/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
247/521		10s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
249/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
251/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
253/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
255/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
257/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
259/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
261/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
263/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
265/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
267/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
269/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
271/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
273/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023

275/521		9s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
277/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
279/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
281/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
283/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
285/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
287/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
289/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
291/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
293/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
295/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
297/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
299/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
301/521		8s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
303/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
305/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0023
307/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0024
309/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0024
311/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0024
313/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0024
315/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0024
317/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0024
319/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0024
321/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0024
323/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0024
325/521		7s 37ms/step	- accuracy: 0.9996	- loss: 0.0024
327/521		7s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
329/521		7s 37ms/step	- accuracy: 0.9995	- loss: 0.0024

331/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
333/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
335/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
337/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
339/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
341/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
343/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
345/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
347/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
349/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
351/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
353/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
355/521		6s 37ms/step - accuracy: 0.9995 - loss: 0.0024
357/521		5s 37ms/step - accuracy: 0.9995 - loss: 0.0024
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375/521		5s 37ms/step - accuracy: 0.9995 - loss: 0.0024
377/521		5s 37ms/step - accuracy: 0.9995 - loss: 0.0024
379/521		5s 37ms/step - accuracy: 0.9995 - loss: 0.0024
381/521		5s 37ms/step - accuracy: 0.9995 - loss: 0.0024
383/521		5s 37ms/step - accuracy: 0.9995 - loss: 0.0024
385/521		4s 37ms/step - accuracy: 0.9995 - loss: 0.0024

387/521		4s 37ms/step - accuracy: 0.9995 - loss: 0.0024
389/521		4s 37ms/step - accuracy: 0.9995 - loss: 0.0024
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407/521		4s 37ms/step - accuracy: 0.9995 - loss: 0.0024
409/521		4s 37ms/step - accuracy: 0.9995 - loss: 0.0024
411/521		4s 37ms/step - accuracy: 0.9995 - loss: 0.0024
413/521		3s 37ms/step - accuracy: 0.9995 - loss: 0.0024
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417/521		3s 37ms/step - accuracy: 0.9995 - loss: 0.0024
419/521		3s 37ms/step - accuracy: 0.9995 - loss: 0.0024
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423/521		3s 37ms/step - accuracy: 0.9995 - loss: 0.0024
425/521		3s 37ms/step - accuracy: 0.9995 - loss: 0.0024
427/521		3s 37ms/step - accuracy: 0.9995 - loss: 0.0024
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437/521		3s 37ms/step - accuracy: 0.9995 - loss: 0.0024
439/521		3s 37ms/step - accuracy: 0.9995 - loss: 0.0024
441/521		2s 37ms/step - accuracy: 0.9995 - loss: 0.0024

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465/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0024

467/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0024

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495/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0024

497/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0024

499/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
501/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
503/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
505/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
507/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
509/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
511/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
513/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
515/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
517/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
519/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
521/521		0s 37ms/step	- accuracy: 0.9995	- loss: 0.0024
521/521  20s 38ms/step - accuracy: 0.9994 - loss: 0.0021				
- val_accuracy: 0.9990 - val_loss: 0.0034 - learning_rate: 2.5000e-04				
Epoch 19/25				
1/521		30s 60ms/step	- accuracy: 1.0000	- loss: 4.1770e-04
3/521		22s 43ms/step	- accuracy: 0.9983	- loss: 0.0017
5/521		21s 41ms/step	- accuracy: 0.9976	- loss: 0.0026
7/521		20s 40ms/step	- accuracy: 0.9976	- loss: 0.0028
9/521		19s 39ms/step	- accuracy: 0.9977	- loss: 0.0028
11/521		19s 38ms/step	- accuracy: 0.9978	- loss: 0.0028
13/521		19s 38ms/step	- accuracy: 0.9980	- loss: 0.0027
15/521		19s 38ms/step	- accuracy: 0.9981	- loss: 0.0026
17/521		18s 37ms/step	- accuracy: 0.9982	- loss: 0.0025
19/521		18s 37ms/step	- accuracy: 0.9983	- loss: 0.0025
21/521		18s 37ms/step	- accuracy: 0.9984	- loss: 0.0024
23/521		18s 37ms/step	- accuracy: 0.9985	- loss: 0.0023
25/521		18s 37ms/step	- accuracy: 0.9985	- loss: 0.0023
27/521		18s 37ms/step	- accuracy: 0.9985	- loss: 0.0023

29/521		18s	37ms/step	- accuracy: 0.9986	- loss: 0.0022
31/521		17s	37ms/step	- accuracy: 0.9986	- loss: 0.0022
33/521		17s	37ms/step	- accuracy: 0.9986	- loss: 0.0022
35/521		17s	37ms/step	- accuracy: 0.9987	- loss: 0.0022
37/521		17s	37ms/step	- accuracy: 0.9987	- loss: 0.0021
39/521		17s	37ms/step	- accuracy: 0.9987	- loss: 0.0021
41/521		17s	37ms/step	- accuracy: 0.9987	- loss: 0.0021
43/521		17s	37ms/step	- accuracy: 0.9988	- loss: 0.0020
45/521		17s	37ms/step	- accuracy: 0.9988	- loss: 0.0020
47/521		17s	37ms/step	- accuracy: 0.9988	- loss: 0.0020
49/521		17s	37ms/step	- accuracy: 0.9988	- loss: 0.0019
51/521		17s	37ms/step	- accuracy: 0.9988	- loss: 0.0019
53/521		17s	37ms/step	- accuracy: 0.9989	- loss: 0.0019
55/521		17s	37ms/step	- accuracy: 0.9989	- loss: 0.0019
57/521		16s	37ms/step	- accuracy: 0.9989	- loss: 0.0019
59/521		16s	37ms/step	- accuracy: 0.9989	- loss: 0.0019
61/521		16s	37ms/step	- accuracy: 0.9989	- loss: 0.0018
63/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0018
65/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0018
67/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0018
69/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0018
71/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0018
73/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0017
75/521		16s	37ms/step	- accuracy: 0.9991	- loss: 0.0017
77/521		16s	37ms/step	- accuracy: 0.9991	- loss: 0.0017
79/521		16s	37ms/step	- accuracy: 0.9991	- loss: 0.0017
81/521		16s	37ms/step	- accuracy: 0.9991	- loss: 0.0017
83/521		16s	37ms/step	- accuracy: 0.9991	- loss: 0.0017

85/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0017
87/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0017
89/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0017
91/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0017
93/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0017
95/521		15s 37ms/step	- accuracy: 0.9992	- loss: 0.0017
97/521		15s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
99/521		15s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
101/521		15s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
103/521		15s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
105/521		15s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
107/521		15s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
109/521		15s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
111/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
113/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
115/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
117/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
119/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
121/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
123/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
125/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
127/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
129/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
131/521		14s 37ms/step	- accuracy: 0.9992	- loss: 0.0016
133/521		14s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
135/521		14s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
137/521		14s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
139/521		13s 37ms/step	- accuracy: 0.9993	- loss: 0.0016

141/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
143/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
145/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
147/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
149/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
151/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
153/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
155/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
157/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
159/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
161/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
163/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
165/521		13s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
167/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
169/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
171/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
173/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
175/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
177/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
179/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
181/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
183/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
185/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
187/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
189/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
191/521		12s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
193/521		11s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
195/521		11s	37ms/step	- accuracy: 0.9993	- loss: 0.0016

197/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
199/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
201/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
203/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
205/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
207/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
209/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
211/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
213/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
215/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
217/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
219/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
221/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
223/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
225/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
227/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
229/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
231/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
233/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
235/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
237/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
239/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
241/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
243/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
245/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
247/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
249/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
251/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016

253/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
255/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
257/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
259/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
261/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
263/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
265/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
267/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
269/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
271/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
273/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
275/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
277/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
279/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
281/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
283/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
285/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
287/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
289/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
291/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
293/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
295/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
297/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
299/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
301/521		8s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
303/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
305/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0016
307/521		7s 37ms/step	- accuracy: 0.9993	- loss: 0.0016

309/521		7s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
311/521		7s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
313/521		7s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
315/521		7s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
317/521		7s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
319/521		7s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
321/521		7s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
323/521		7s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
325/521		7s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
327/521		7s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
329/521		7s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
331/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
333/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
335/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
337/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
339/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
341/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
343/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
345/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
347/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
349/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
351/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
353/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
355/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
357/521		6s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
359/521		5s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
361/521		5s	37ms/step	- accuracy: 0.9993	- loss: 0.0016
363/521		5s	37ms/step	- accuracy: 0.9993	- loss: 0.0016

365/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0016
367/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0016
369/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0016
371/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0016
373/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0016
375/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0016
377/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0016
379/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0016
381/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0016
383/521		5s 37ms/step - accuracy: 0.9993 - loss: 0.0016
385/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
387/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
389/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
391/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
393/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
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397/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
399/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
401/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
403/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
405/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
407/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
409/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
411/521		4s 37ms/step - accuracy: 0.9993 - loss: 0.0016
413/521		3s 37ms/step - accuracy: 0.9993 - loss: 0.0016
415/521		3s 37ms/step - accuracy: 0.9993 - loss: 0.0016
417/521		3s 37ms/step - accuracy: 0.9993 - loss: 0.0016
419/521		3s 37ms/step - accuracy: 0.9993 - loss: 0.0016

421/521  3s 37ms/step - accuracy: 0.9993 - loss: 0.0016

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439/521  3s 37ms/step - accuracy: 0.9993 - loss: 0.0016

441/521  2s 37ms/step - accuracy: 0.9993 - loss: 0.0016

443/521  2s 37ms/step - accuracy: 0.9993 - loss: 0.0016

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463/521  2s 37ms/step - accuracy: 0.9993 - loss: 0.0017

465/521  2s 37ms/step - accuracy: 0.9993 - loss: 0.0017

467/521  1s 37ms/step - accuracy: 0.9993 - loss: 0.0017

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495/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0017

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515/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0017

517/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0017

519/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0017

521/521  0s 37ms/step - accuracy: 0.9993 - loss: 0.0017

Epoch 19: ReduceLROnPlateau reducing learning rate to 0.0001250000059371814.

521/521  20s 38ms/step - accuracy: 0.9992 - loss: 0.0025
- val_accuracy: 0.9990 - val_loss: 0.0050 - learning_rate: 2.5000e-04

Epoch 20/25

1/521  28s 54ms/step - accuracy: 1.0000 - loss: 4.4964e-04

3/521  19s 37ms/step - accuracy: 1.0000 - loss: 2.9849e-04

5/521		18s 37ms/step	accuracy: 1.0000	loss: 2.5554e-04
7/521		18s 37ms/step	accuracy: 1.0000	loss: 2.9901e-04
9/521		18s 37ms/step	accuracy: 1.0000	loss: 3.7437e-04
11/521		18s 37ms/step	accuracy: 1.0000	loss: 4.1998e-04
13/521		18s 37ms/step	accuracy: 1.0000	loss: 4.9724e-04
15/521		18s 37ms/step	accuracy: 1.0000	loss: 5.7528e-04
17/521		18s 37ms/step	accuracy: 1.0000	loss: 6.8771e-04
19/521		18s 37ms/step	accuracy: 1.0000	loss: 7.7535e-04
21/521		18s 37ms/step	accuracy: 1.0000	loss: 9.0295e-04
23/521		18s 37ms/step	accuracy: 0.9999	loss: 0.0011
25/521		18s 37ms/step	accuracy: 0.9999	loss: 0.0012
27/521		18s 37ms/step	accuracy: 0.9998	loss: 0.0013
29/521		18s 37ms/step	accuracy: 0.9998	loss: 0.0013
31/521		18s 37ms/step	accuracy: 0.9998	loss: 0.0014
33/521		18s 37ms/step	accuracy: 0.9998	loss: 0.0014
35/521		17s 37ms/step	accuracy: 0.9997	loss: 0.0015
37/521		17s 37ms/step	accuracy: 0.9997	loss: 0.0015
39/521		17s 37ms/step	accuracy: 0.9997	loss: 0.0016
41/521		17s 37ms/step	accuracy: 0.9997	loss: 0.0016
43/521		17s 37ms/step	accuracy: 0.9996	loss: 0.0017
45/521		17s 37ms/step	accuracy: 0.9996	loss: 0.0017
47/521		17s 37ms/step	accuracy: 0.9996	loss: 0.0017
49/521		17s 37ms/step	accuracy: 0.9996	loss: 0.0017

51/521		17s	37ms/step	- accuracy: 0.9996	- loss: 0.0017
53/521		17s	37ms/step	- accuracy: 0.9996	- loss: 0.0018
55/521		17s	37ms/step	- accuracy: 0.9996	- loss: 0.0018
57/521		17s	37ms/step	- accuracy: 0.9996	- loss: 0.0018
59/521		16s	37ms/step	- accuracy: 0.9996	- loss: 0.0018
61/521		16s	37ms/step	- accuracy: 0.9996	- loss: 0.0018
63/521		16s	37ms/step	- accuracy: 0.9996	- loss: 0.0018
65/521		16s	37ms/step	- accuracy: 0.9996	- loss: 0.0018
67/521		16s	37ms/step	- accuracy: 0.9996	- loss: 0.0018
69/521		16s	37ms/step	- accuracy: 0.9996	- loss: 0.0018
71/521		16s	37ms/step	- accuracy: 0.9996	- loss: 0.0018
73/521		16s	37ms/step	- accuracy: 0.9995	- loss: 0.0018
75/521		16s	37ms/step	- accuracy: 0.9995	- loss: 0.0018
77/521		16s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
79/521		16s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
81/521		16s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
83/521		16s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
85/521		16s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
87/521		15s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
89/521		15s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
91/521		15s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
93/521		15s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
95/521		15s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
97/521		15s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
99/521		15s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
101/521		15s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
103/521		15s	37ms/step	- accuracy: 0.9995	- loss: 0.0019
105/521		15s	37ms/step	- accuracy: 0.9995	- loss: 0.0019

107/521		15s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
109/521		15s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
111/521		15s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
113/521		15s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
115/521		15s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
117/521		14s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
119/521		14s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
121/521		14s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
123/521		14s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
125/521		14s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
127/521		14s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
129/521		14s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
131/521		14s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
133/521		14s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
135/521		14s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
137/521		14s 37ms/step	- accuracy: 0.9995	- loss: 0.0020
139/521		14s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
141/521		14s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
143/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
145/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
147/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
149/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
151/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
153/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
155/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
157/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
159/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
161/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020

163/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
165/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
167/521		13s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
169/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
171/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
173/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
175/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
177/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
179/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
181/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
183/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
185/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
187/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
189/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
191/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
193/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
195/521		12s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
197/521		11s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
199/521		11s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
201/521		11s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
203/521		11s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
205/521		11s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
207/521		11s 37ms/step	- accuracy: 0.9994	- loss: 0.0020
209/521		11s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
211/521		11s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
213/521		11s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
215/521		11s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
217/521		11s 37ms/step	- accuracy: 0.9994	- loss: 0.0019

219/521		11s 37ms/step - accuracy: 0.9994 - loss: 0.0019
221/521		11s 37ms/step - accuracy: 0.9994 - loss: 0.0019
223/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
225/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
227/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
229/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
231/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
233/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
235/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
237/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
239/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
241/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
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245/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
247/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
249/521		10s 37ms/step - accuracy: 0.9994 - loss: 0.0019
251/521		9s 37ms/step - accuracy: 0.9994 - loss: 0.0019
253/521		9s 37ms/step - accuracy: 0.9994 - loss: 0.0019
255/521		9s 37ms/step - accuracy: 0.9994 - loss: 0.0019
257/521		9s 37ms/step - accuracy: 0.9994 - loss: 0.0019
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261/521		9s 37ms/step - accuracy: 0.9994 - loss: 0.0019
263/521		9s 37ms/step - accuracy: 0.9994 - loss: 0.0019
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267/521		9s 37ms/step - accuracy: 0.9994 - loss: 0.0019
269/521		9s 37ms/step - accuracy: 0.9994 - loss: 0.0019
271/521		9s 37ms/step - accuracy: 0.9994 - loss: 0.0019
273/521		9s 37ms/step - accuracy: 0.9994 - loss: 0.0019

275/521		9s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
277/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
279/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
281/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
283/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
285/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
287/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
289/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
291/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
293/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
295/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
297/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
299/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
301/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
303/521		8s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
305/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
307/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
309/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
311/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
313/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
315/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
317/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
319/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
321/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
323/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
325/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
327/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
329/521		7s 37ms/step	- accuracy: 0.9994	- loss: 0.0019

331/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
333/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
335/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
337/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
339/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
341/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
343/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
345/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
347/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
349/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
351/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
353/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
355/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
357/521		6s 37ms/step - accuracy: 0.9994 - loss: 0.0019
359/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
361/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
363/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
365/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
367/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
369/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
371/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
373/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
375/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
377/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
379/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
381/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
383/521		5s 37ms/step - accuracy: 0.9994 - loss: 0.0019
385/521		4s 37ms/step - accuracy: 0.9994 - loss: 0.0019

387/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
389/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
391/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
393/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
395/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
397/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
399/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
401/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
403/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
405/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
407/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
409/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
411/521		4s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
413/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
415/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
417/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
419/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
421/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
423/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
425/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
427/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
429/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
431/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
433/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
435/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
437/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
439/521		3s 37ms/step	- accuracy: 0.9994	- loss: 0.0019
441/521		2s 37ms/step	- accuracy: 0.9995	- loss: 0.0019

443/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

445/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

447/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

449/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

451/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

453/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

455/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

457/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

459/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

461/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

463/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

465/521  2s 37ms/step - accuracy: 0.9995 - loss: 0.0019

467/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

469/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

471/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

473/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

475/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

477/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

479/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

481/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

483/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

485/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

487/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

489/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

491/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

493/521  1s 37ms/step - accuracy: 0.9995 - loss: 0.0019

495/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0019

497/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0019

499/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0019

501/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0019

503/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0019

505/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0019

507/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0018

509/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0018

511/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0018

513/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0018

515/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0018

517/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0018

519/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0018

521/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0018

521/521  20s 38ms/step - accuracy: 0.9996 - loss: 0.0016
- val_accuracy: 0.9991 - val_loss: 0.0045 - learning_rate: 1.2500e-04

Epoch 21/25

1/521  31s 60ms/step - accuracy: 1.0000 - loss: 0.0029

3/521  22s 43ms/step - accuracy: 0.9957 - loss: 0.0076

5/521  21s 41ms/step - accuracy: 0.9960 - loss: 0.0073

7/521  20s 39ms/step - accuracy: 0.9964 - loss: 0.0066

9/521  19s 39ms/step - accuracy: 0.9968 - loss: 0.0060

11/521  19s 38ms/step - accuracy: 0.9971 - loss: 0.0055

13/521  19s 38ms/step - accuracy: 0.9974 - loss: 0.0050

15/521  19s 38ms/step - accuracy: 0.9976 - loss: 0.0047

17/521  18s 37ms/step - accuracy: 0.9978 - loss: 0.0044

19/521  18s 37ms/step - accuracy: 0.9979 - loss: 0.0041

21/521  18s 37ms/step - accuracy: 0.9980 - loss: 0.0039

23/521  18s 37ms/step - accuracy: 0.9981 - loss: 0.0037

25/521  18s 37ms/step - accuracy: 0.9982 - loss: 0.0035

27/521  18s 37ms/step - accuracy: 0.9983 - loss: 0.0034

29/521		18s	37ms/step	- accuracy: 0.9984	- loss: 0.0032
31/521		18s	37ms/step	- accuracy: 0.9985	- loss: 0.0031
33/521		17s	37ms/step	- accuracy: 0.9985	- loss: 0.0030
35/521		17s	37ms/step	- accuracy: 0.9986	- loss: 0.0029
37/521		17s	37ms/step	- accuracy: 0.9986	- loss: 0.0028
39/521		17s	37ms/step	- accuracy: 0.9987	- loss: 0.0027
41/521		17s	37ms/step	- accuracy: 0.9987	- loss: 0.0026
43/521		17s	37ms/step	- accuracy: 0.9988	- loss: 0.0026
45/521		17s	37ms/step	- accuracy: 0.9988	- loss: 0.0025
47/521		17s	37ms/step	- accuracy: 0.9989	- loss: 0.0024
49/521		17s	37ms/step	- accuracy: 0.9989	- loss: 0.0024
51/521		17s	37ms/step	- accuracy: 0.9989	- loss: 0.0024
53/521		17s	37ms/step	- accuracy: 0.9989	- loss: 0.0023
55/521		17s	37ms/step	- accuracy: 0.9989	- loss: 0.0023
57/521		17s	37ms/step	- accuracy: 0.9989	- loss: 0.0023
59/521		17s	37ms/step	- accuracy: 0.9990	- loss: 0.0023
61/521		17s	37ms/step	- accuracy: 0.9990	- loss: 0.0023
63/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0022
65/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0022
67/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0022
69/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0022
71/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0022
73/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0022
75/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0022
77/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0021
79/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0021
81/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0021
83/521		16s	37ms/step	- accuracy: 0.9990	- loss: 0.0021

85/521		16s 37ms/step	- accuracy: 0.9990	- loss: 0.0021
87/521		16s 37ms/step	- accuracy: 0.9990	- loss: 0.0021
89/521		15s 37ms/step	- accuracy: 0.9990	- loss: 0.0021
91/521		15s 37ms/step	- accuracy: 0.9990	- loss: 0.0021
93/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0021
95/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0021
97/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0021
99/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0021
101/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0021
103/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0021
105/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0021
107/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0021
109/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0021
111/521		15s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
113/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
115/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
117/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
119/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
121/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
123/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
125/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
127/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
129/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
131/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
133/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
135/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
137/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020
139/521		14s 37ms/step	- accuracy: 0.9991	- loss: 0.0020

141/521		13s	37ms/step	- accuracy: 0.9991	- loss: 0.0020
143/521		13s	37ms/step	- accuracy: 0.9991	- loss: 0.0020
145/521		13s	37ms/step	- accuracy: 0.9992	- loss: 0.0020
147/521		13s	37ms/step	- accuracy: 0.9992	- loss: 0.0020
149/521		13s	37ms/step	- accuracy: 0.9992	- loss: 0.0020
151/521		13s	37ms/step	- accuracy: 0.9992	- loss: 0.0020
153/521		13s	37ms/step	- accuracy: 0.9992	- loss: 0.0020
155/521		13s	37ms/step	- accuracy: 0.9992	- loss: 0.0020
157/521		13s	37ms/step	- accuracy: 0.9992	- loss: 0.0020
159/521		13s	37ms/step	- accuracy: 0.9992	- loss: 0.0020
161/521		13s	37ms/step	- accuracy: 0.9992	- loss: 0.0020
163/521		13s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
165/521		13s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
167/521		12s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
169/521		12s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
171/521		12s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
173/521		12s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
175/521		12s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
177/521		12s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
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185/521		12s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
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189/521		12s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
191/521		12s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
193/521		12s	37ms/step	- accuracy: 0.9992	- loss: 0.0019
195/521		11s	37ms/step	- accuracy: 0.9992	- loss: 0.0019

197/521		11s 37ms/step	- accuracy: 0.9992	- loss: 0.0019
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209/521		11s 37ms/step	- accuracy: 0.9992	- loss: 0.0019
211/521		11s 37ms/step	- accuracy: 0.9992	- loss: 0.0019
213/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0019
215/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0019
217/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0019
219/521		11s 37ms/step	- accuracy: 0.9993	- loss: 0.0019
221/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0019
223/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0019
225/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0019
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237/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0019
239/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0019
241/521		10s 37ms/step	- accuracy: 0.9993	- loss: 0.0018
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249/521		9s 37ms/step	- accuracy: 0.9993	- loss: 0.0018
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343/521		6s	37ms/step	- accuracy: 0.9994	- loss: 0.0018
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359/521		5s	37ms/step	- accuracy: 0.9994	- loss: 0.0017
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519/521  0s 37ms/step - accuracy: 0.9994 - loss: 0.0016

521/521  0s 37ms/step - accuracy: 0.9995 - loss: 0.0016

521/521  20s 38ms/step - accuracy: 0.9997 - loss: 0.0013
 - val_accuracy: 0.9991 - val_loss: 0.0045 - learning_rate: 1.2500e-04
 Epoch 21: early stopping
 Restoring model weights from the end of the best epoch: 16.
 Tiempo total de entrenamiento: 7.1 min

8. Curvas de aprendizaje

Lo que queremos ver:

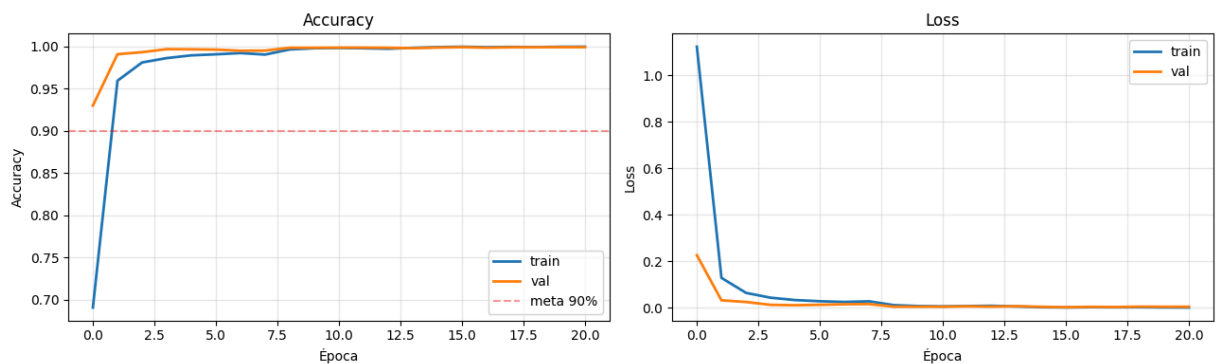
- La curva de **training** y **validation** convergen sin separarse demasiado → sin sobreajuste.
- La curva de **validation accuracy** termina por encima de **0.95**.
- La curva de **loss** baja rápido al inicio y se estabiliza.

```
In [11]: fig, axes = plt.subplots(1, 2, figsize=(13, 4))

axes[0].plot(history.history["accuracy"], label="train", linewidth=2)
axes[0].plot(history.history["val_accuracy"], label="val", linewidth=2)
axes[0].axhline(0.90, color="red", linestyle="--", alpha=0.4, label="meta 90")
axes[0].set_xlabel("Época"); axes[0].set_ylabel("Accuracy"); axes[0].legend()
axes[0].set_title("Accuracy")

axes[1].plot(history.history["loss"], label="train", linewidth=2)
axes[1].plot(history.history["val_loss"], label="val", linewidth=2)
axes[1].set_xlabel("Época"); axes[1].set_ylabel("Loss"); axes[1].legend(); axes[1].set_title("Loss")

plt.tight_layout(); plt.show()
```



9. Evaluación sobre el set de test oficial del GTSRB

El test set oficial son 12 630 imágenes que **nunca** vio el modelo durante el entrenamiento (ni siquiera para validación). Si la accuracy en este set está cerca de la de validación, el modelo generaliza bien.

```
In [12]: test_csv = pd.read_csv(DATASET_DIR / "Test.csv")
print(f"Cargando {len(test_csv):,} imágenes de test...")

X_test_list, y_test_list = [], []
for _, row in tqdm(test_csv.iterrows(), total=len(test_csv), desc="Test"):
    img_path = DATASET_DIR / row["Path"]
    X_test_list.append(load_image(img_path))
    y_test_list.append(int(row["ClassId"]))

X_test = np.array(X_test_list, dtype=np.float32)
y_test = np.array(y_test_list, dtype=np.int32)
y_test_oh = to_categorical(y_test, NUM_CLASSES)
```

```
print(f"X_test.shape: {X_test.shape}")

loss, accuracy = model.evaluate(X_test, y_test_oh, verbose=1)
print(f"\n>>> Accuracy en test oficial: {accuracy*100:.2f}% <<<")
print(f">>> Loss en test oficial      : {loss:.4f} <<<")
```

Cargando 12,630 imágenes de test...

```
Test:  0%|          | 0/12630 [00:00<?, ?it/s]
Test:  3%||         | 440/12630 [00:00<00:02, 4394.47it/s]
Test:  8%|█         | 948/12630 [00:00<00:02, 4795.68it/s]
Test: 12%|██        | 1497/12630 [00:00<00:02, 5109.61it/s]
Test: 16%|███       | 2059/12630 [00:00<00:01, 5303.71it/s]
Test: 21%|████      | 2637/12630 [00:00<00:01, 5475.01it/s]
Test: 25%|█████     | 3193/12630 [00:00<00:01, 5502.95it/s]
Test: 30%|██████    | 3744/12630 [00:00<00:01, 5410.81it/s]
Test: 34%|███████   | 4333/12630 [00:00<00:01, 5559.76it/s]
Test: 39%|████████  | 4925/12630 [00:00<00:01, 5664.00it/s]
Test: 43%|█████████ | 5492/12630 [00:01<00:01, 5347.42it/s]
Test: 48%|█████████ | 6048/12630 [00:01<00:01, 5407.70it/s]
Test: 52%|█████████ | 6598/12630 [00:01<00:01, 5428.19it/s]
Test: 57%|█████████ | 7143/12630 [00:01<00:01, 5304.27it/s]
Test: 61%|█████████ | 7676/12630 [00:01<00:00, 5175.57it/s]
Test: 65%|█████████ | 8196/12630 [00:01<00:00, 5164.33it/s]
Test: 69%|█████████ | 8746/12630 [00:01<00:00, 5253.87it/s]
Test: 73%|█████████ | 9273/12630 [00:01<00:00, 5123.37it/s]
Test: 78%|█████████ | 9824/12630 [00:01<00:00, 5234.19it/s]
Test: 82%|█████████ | 10389/12630 [00:01<00:00, 5354.67it/s]
Test: 87%|█████████ | 10967/12630 [00:02<00:00, 5478.61it/s]
Test: 91%|█████████ | 11544/12630 [00:02<00:00, 5563.32it/s]
Test: 96%|█████████ | 12127/12630 [00:02<00:00, 5641.83it/s]
Test: 100%|█████████| 12630/12630 [00:02<00:00, 5380.77it/s]
```

X_test.shape: (12630, 32, 32, 3)

1/395 ————— 30s 79ms/step - accuracy: 1.0000 - loss: 2.7147e-04

7/395 ————— 3s 9ms/step - accuracy: 0.9770 - loss: 0.0847

14/395 ————— 3s 8ms/step - accuracy: 0.9747 - loss: 0.0904

21/395 ————— 3s 8ms/step - accuracy: 0.9747 - loss: 0.0925

28/395 ————— 3s 8ms/step - accuracy: 0.9759 - loss: 0.0902

35/395 ————— 2s 8ms/step - accuracy: 0.9772 - loss: 0.0865

42/395 ————— 2s 8ms/step - accuracy: 0.9783 - loss: 0.0823

49/395 ————— 2s 8ms/step - accuracy: 0.9792 - loss: 0.0788

56/395		2s 8ms/step	- accuracy: 0.9799	- loss: 0.0764
63/395		2s 8ms/step	- accuracy: 0.9804	- loss: 0.0744
70/395		2s 8ms/step	- accuracy: 0.9808	- loss: 0.0727
77/395		2s 8ms/step	- accuracy: 0.9811	- loss: 0.0715
84/395		2s 8ms/step	- accuracy: 0.9813	- loss: 0.0708
91/395		2s 8ms/step	- accuracy: 0.9815	- loss: 0.0700
98/395		2s 8ms/step	- accuracy: 0.9817	- loss: 0.0693
105/395		2s 8ms/step	- accuracy: 0.9819	- loss: 0.0685
112/395		2s 8ms/step	- accuracy: 0.9820	- loss: 0.0677
119/395		2s 8ms/step	- accuracy: 0.9821	- loss: 0.0672
126/395		2s 8ms/step	- accuracy: 0.9822	- loss: 0.0667
132/395		2s 8ms/step	- accuracy: 0.9823	- loss: 0.0663
139/395		2s 8ms/step	- accuracy: 0.9824	- loss: 0.0659
145/395		2s 8ms/step	- accuracy: 0.9824	- loss: 0.0656
152/395		1s 8ms/step	- accuracy: 0.9825	- loss: 0.0652
159/395		1s 8ms/step	- accuracy: 0.9826	- loss: 0.0648
166/395		1s 8ms/step	- accuracy: 0.9826	- loss: 0.0646
173/395		1s 8ms/step	- accuracy: 0.9827	- loss: 0.0643
180/395		1s 8ms/step	- accuracy: 0.9827	- loss: 0.0641
186/395		1s 8ms/step	- accuracy: 0.9827	- loss: 0.0640
193/395		1s 8ms/step	- accuracy: 0.9827	- loss: 0.0638
200/395		1s 8ms/step	- accuracy: 0.9828	- loss: 0.0637
207/395		1s 8ms/step	- accuracy: 0.9828	- loss: 0.0635
214/395		1s 8ms/step	- accuracy: 0.9828	- loss: 0.0633
221/395		1s 8ms/step	- accuracy: 0.9828	- loss: 0.0632
228/395		1s 8ms/step	- accuracy: 0.9828	- loss: 0.0631
235/395		1s 8ms/step	- accuracy: 0.9828	- loss: 0.0630
242/395		1s 8ms/step	- accuracy: 0.9829	- loss: 0.0629

```

249/395  1s 8ms/step - accuracy: 0.9829 - loss: 0.0629
256/395  1s 8ms/step - accuracy: 0.9829 - loss: 0.0628
263/395  1s 8ms/step - accuracy: 0.9829 - loss: 0.0628
270/395  1s 8ms/step - accuracy: 0.9829 - loss: 0.0627
277/395  0s 8ms/step - accuracy: 0.9829 - loss: 0.0626
284/395  0s 8ms/step - accuracy: 0.9829 - loss: 0.0625
291/395  0s 8ms/step - accuracy: 0.9829 - loss: 0.0625
298/395  0s 8ms/step - accuracy: 0.9830 - loss: 0.0624
305/395  0s 8ms/step - accuracy: 0.9830 - loss: 0.0623
312/395  0s 8ms/step - accuracy: 0.9830 - loss: 0.0621
319/395  0s 8ms/step - accuracy: 0.9830 - loss: 0.0620
326/395  0s 8ms/step - accuracy: 0.9831 - loss: 0.0619
333/395  0s 8ms/step - accuracy: 0.9831 - loss: 0.0618
340/395  0s 8ms/step - accuracy: 0.9831 - loss: 0.0617
347/395  0s 8ms/step - accuracy: 0.9831 - loss: 0.0616
354/395  0s 8ms/step - accuracy: 0.9832 - loss: 0.0615
361/395  0s 8ms/step - accuracy: 0.9832 - loss: 0.0614
368/395  0s 8ms/step - accuracy: 0.9832 - loss: 0.0613
375/395  0s 8ms/step - accuracy: 0.9832 - loss: 0.0612
382/395  0s 8ms/step - accuracy: 0.9832 - loss: 0.0611
389/395  0s 8ms/step - accuracy: 0.9832 - loss: 0.0610
395/395  0s 8ms/step - accuracy: 0.9833 - loss: 0.0609

395/395  3s 8ms/step - accuracy: 0.9838 - loss: 0.0573
>>> Accuracy en test oficial: 98.38% <<<
>>> Loss en test oficial    : 0.0573 <<<

```

10. Matriz de confusión y classification report

La matriz de confusión nos dice **qué clases se confunden con cuáles**. Esperamos confusiones razonables como "Speed limit 50" con "Speed limit 80" (números similares) y no confusiones entre clases visualmente muy distintas.

```
In [13]: y_pred_probs = model.predict(X_test, batch_size=128, verbose=1)
y_pred = np.argmax(y_pred_probs, axis=1)

print(classification_report(
    y_test, y_pred,
    target_names=[f"{i}-{SIGN_NAMES[i][:18]}" for i in range(NUM_CLASSES)],
    digits=3,
))
```

1/99 ————— 14s 148ms/step

6/99 ————— 1s 13ms/step

11/99 ————— 1s 12ms/step

17/99 ————— 0s 11ms/step

23/99 ————— 0s 10ms/step

29/99 ————— 0s 10ms/step

35/99 ————— 0s 10ms/step

41/99 ————— 0s 10ms/step

47/99 ————— 0s 10ms/step

53/99 ————— 0s 10ms/step

59/99 ————— 0s 10ms/step

65/99 ————— 0s 10ms/step

70/99 ————— 0s 10ms/step

75/99 ————— 0s 10ms/step

80/99 ————— 0s 10ms/step

85/99 ————— 0s 10ms/step

90/99 ————— 0s 10ms/step

95/99 ————— 0s 10ms/step

99/99 ————— 0s 12ms/step

99/99	1s 12ms/step			
	precision	recall	f1-score	support
0-Speed limit 20	1.000	1.000	1.000	60
1-Speed limit 30	0.988	1.000	0.994	720
2-Speed limit 50	0.987	0.999	0.993	750
3-Speed limit 60	0.975	0.962	0.969	450
4-Speed limit 70	1.000	0.988	0.994	660
5-Speed limit 80	0.953	0.995	0.974	630
6-End speed limit 80	1.000	0.827	0.905	150
7-Speed limit 100	0.998	0.991	0.994	450
8-Speed limit 120	0.991	0.991	0.991	450
9-No passing	0.990	1.000	0.995	480
10-No passing trucks	0.997	0.997	0.997	660
11-Priority intersect	0.974	0.983	0.979	420
12-Priority road	0.983	0.988	0.986	690
13-Yield	0.992	0.997	0.994	720
14-Stop	1.000	1.000	1.000	270
15-No vehicles	0.963	0.995	0.979	210
16-No trucks	1.000	1.000	1.000	150
17-No entry	1.000	0.992	0.996	360
18-General caution	0.997	0.913	0.953	390
19-Dangerous curve le	0.968	1.000	0.984	60
20-Dangerous curve ri	0.938	1.000	0.968	90
21-Double curve	0.924	0.811	0.864	90
22-Bumpy road	1.000	0.958	0.979	120
23-Slippery road	0.962	1.000	0.980	150
24-Road narrows	1.000	0.989	0.994	90
25-Road work	0.977	0.990	0.983	480
26-Traffic signals	0.933	1.000	0.965	180
27-Pedestrians	0.717	0.550	0.623	60
28-Children crossing	0.974	0.993	0.983	150
29-Bicycles crossing	0.968	1.000	0.984	90
30-Beware ice	0.971	0.887	0.927	150
31-Wild animals	0.971	1.000	0.985	270
32-End restrictions	1.000	1.000	1.000	60
33-Turn right	0.991	1.000	0.995	210
34-Turn left	1.000	0.992	0.996	120
35-Ahead only	0.995	1.000	0.997	390
36-Go straight/right	0.992	1.000	0.996	120
37-Go straight/left	1.000	1.000	1.000	60
38-Keep right	0.999	1.000	0.999	690
39-Keep left	0.967	0.978	0.972	90
40-Roundabout	1.000	0.967	0.983	90
41-End no passing	1.000	0.917	0.957	60
42-End no passing tru	0.918	1.000	0.957	90
accuracy			0.984	12630
macro avg	0.976	0.969	0.971	12630
weighted avg	0.984	0.984	0.983	12630

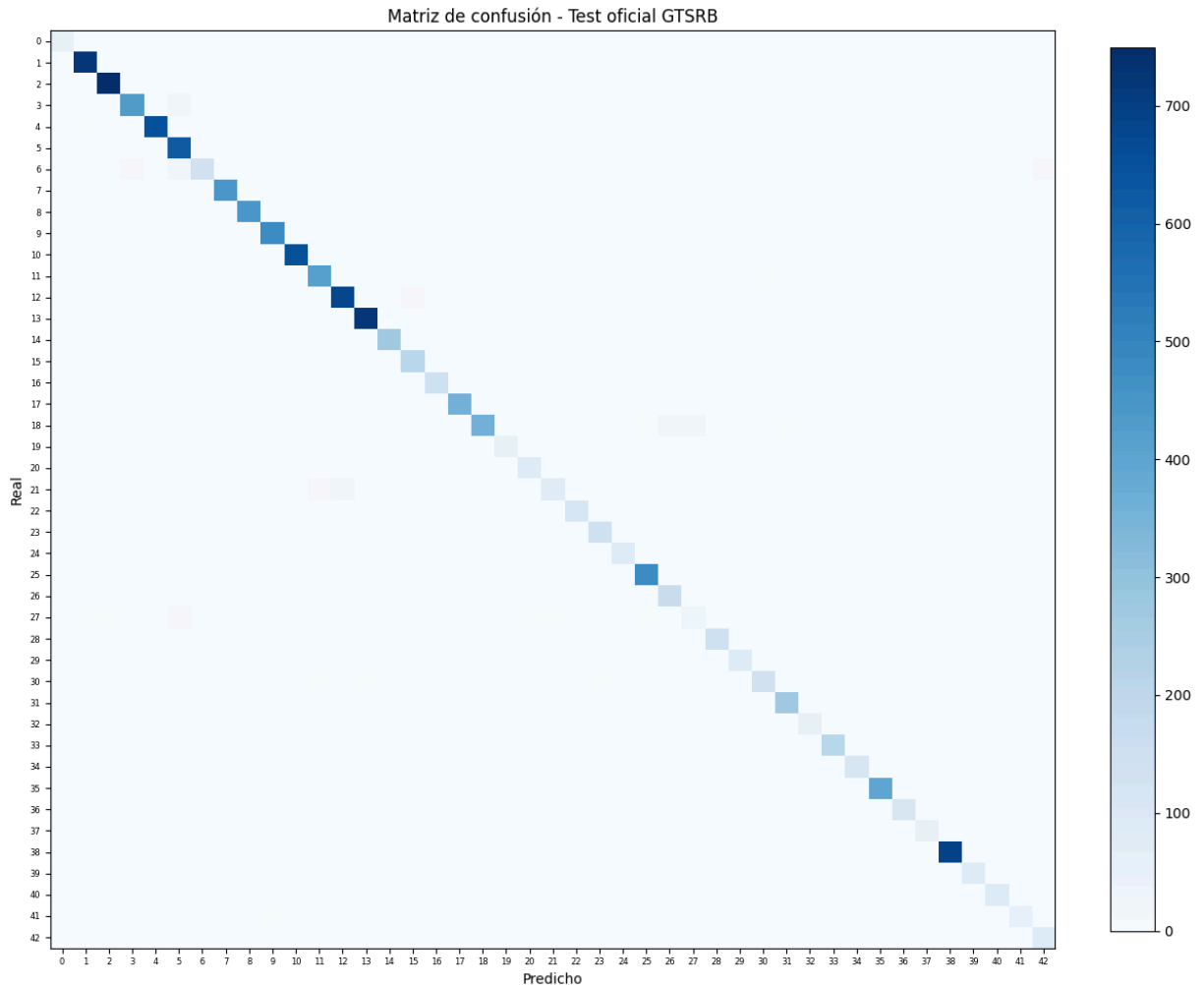
```
In [14]: cm = confusion_matrix(y_test, y_pred)

fig, ax = plt.subplots(figsize=(12, 10))
```

```

im = ax.imshow(cm, cmap="Blues", aspect="auto")
ax.set_title("Matriz de confusión - Test oficial GTSRB", fontsize=12)
ax.set_xlabel("Predicho"); ax.set_ylabel("Real")
ax.set_xticks(range(NUM_CLASSES)); ax.set_yticks(range(NUM_CLASSES))
ax.set_xticklabels(range(NUM_CLASSES), fontsize=6)
ax.set_yticklabels(range(NUM_CLASSES), fontsize=6)
plt.colorbar(im, ax=ax, fraction=0.04)
plt.tight_layout(); plt.show()

```



11. Visualización de predicciones

Mostramos 8 aciertos y 8 errores con su probabilidad asignada. Los errores son los más informativos: permiten detectar si la red está sesgada por alguna clase específica o si las confusiones tienen sentido visual.

```

In [15]: correct_idx = np.where(y_pred == y_test)[0]
wrong_idx = np.where(y_pred != y_test)[0]
print(f"Aciertos: {len(correct_idx):,} - Errores: {len(wrong_idx):,}")

def show_predictions(indices, title, n=8):
    fig, axes = plt.subplots(2, 4, figsize=(13, 6))
    chosen = np.random.choice(indices, size=min(n, len(indices)), replace=False)
    for ax, i in zip(axes.flat, chosen):

```

```

ax.imshow(X_test[i])
prob = y_pred_probs[i, y_pred[i]]
ax.set_title(
    f"real: {y_test[i]}--{SIGN_NAMES[y_test[i]][:14]}\n"
    f"pred: {y_pred[i]}--{SIGN_NAMES[y_pred[i]][:14]} ({prob:.2f})",
    fontsize=8,
    color="green" if y_pred[i] == y_test[i] else "red",
)
ax.axis("off")
fig.suptitle(title, fontsize=12)
plt.tight_layout(); plt.show()

show_predictions(correct_idx, "Aciertos (verde)")
show_predictions(wrong_idx, "Errores (rojo)")

```

Aciertos: 12,426 - Errores: 204

Aciertos (verde)



Errores (rojo)



12. Exportación del modelo

Guardamos el modelo en formato `.keras` (formato nativo recomendado de Keras 3) en `models/`. Este archivo es lo que el controlador de Webots cargará vía `tensorflow.keras.models.load_model()`.

```
In [16]: MODEL_OUT = Path("../models/modelo_gtsrb.keras").resolve()
MODEL_OUT.parent.mkdir(parents=True, exist_ok=True)
model.save(str(MODEL_OUT))

size_mb = os.path.getsize(MODEL_OUT) / 1e6
print(f"Modelo guardado en: {MODEL_OUT}")
print(f"Tamaño: {size_mb:.1f} MB")
print(f"Test accuracy final: {accuracy*100:.2f}%")
```

Modelo guardado en: /Users/rafaelsanchezmarmolejo/Desktop/Tec de Monterrey/NAvegación Autónoma/Teams/Actividad 4.1 – Detección de Señales de Tránsito/models/modelo_gtsrb.keras
Tamaño: 16.4 MB
Test accuracy final: 98.38%

13. Notas para la integración con Webots

El controlador del vehículo (`controllers/vehicle/vehicle.py`) carga este modelo y, en cada paso de simulación:

1. Captura el frame de la cámara onboard (`BGRA` 256×128).
2. Convierte a RGB y extrae una **Region of Interest** del lado superior-derecho (donde aparecen las señales del lado del conductor).
3. Redimensiona la ROI a 32×32, normaliza a `[0,1]`.
4. Llama a `model.predict()` y toma `np.argmax` + `np.max` para clase y confianza.
5. Si `confianza > 0.85`, muestra el nombre de la señal y dibuja una caja en el display.

Importante: las señales del mundo Webots NO son fotos del GTSRB. Por eso el data augmentation agresivo (rotaciones, brillo, zoom) que aplicamos durante el entrenamiento es lo que le permite al modelo generalizar al render sintético del simulador.

Referencias

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